

Comparative Forecasting of Malaysia's Electrical and Electronics Exports Using
Machine-Learning Models and Arima

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LIST OF ABBREVIATIONS

AIC	-	Akaike Information Criterion
ANN	-	Artificial Neural Network
ARDL	-	Autoregressive Distributed Lag
ARIMA	-	Autoregressive Integrated Moving Average
COVID-19	-	Coronavirus Disease 2019
CPI	-	Consumer Price Index
CRISP-DM	-	Cross-Industry Standard Process for Data Mining
DOSM	-	Department of Statistics Malaysia
E&E	-	Electrical and Electronic
FDI	-	Foreign Direct Investment
FRED	-	Federal Reserve Economic Data
GARCH	-	Generalised Autoregressive Conditional Heteroskedasticity
GDP	-	Gross Domestic Product
GFC	-	Global Financial Crisis
IPI	-	Industrial Production Index
KPI	-	Key Performance Indicator
LASSO	-	Least Absolute Shrinkage and Selection Operator
LSTM	-	Long Short-Term Memory
MAE	-	Mean Absolute Error
MATRADE	-	Malaysia External Trade Development Corporation
ML	-	Machine Learning
MLP	-	Multilayer Perceptron
MSE	-	Mean Squared Error
MSME	-	Micro, Small and Medium Enterprises
MYR	-	Malaysian Ringgit
R ²	-	Coefficient of Determination

RBF	-	Radial Basis Function
RF	-	Random Forest
RMSE	-	Root Mean Squared Error
SARIMA	-	Seasonal Autoregressive Integrated Moving Average
SHAP	-	SHapley Additive exPlanations
SITC	-	Standard International Trade Classification
SVM	-	Support Vector Machine
SVR	-	Support Vector Regression
TBATS	-	Trigonometric, Box-Cox Transform, ARMA Errors, Trend and Seasonal Components
USD	-	United States Dollar
UTM	-	Universiti Teknologi Malaysia
VAR	-	Vector Autoregression
WSTS	-	World Semiconductor Trade Statistics
XGBoost	-	Extreme Gradient Boosting
ZAR	-	South African Rand

LIST OF SYMBOLS

ε	-	Epsilon (loss margin parameter in SVR)
γ	-	Gamma (kernel coefficient in SVR)
C	-	Regularisation parameter in SVR
R^2	-	Coefficient of determination (model fit)

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CHAPTER 1

INTRODUCTION

1.1 Overview

Malaysia's Electrical and Electronics (E&E) industry plays an important role in the country's manufacturing and export sector and is closely connected to international markets and global electronics supply chains. However, fluctuations in domestic economic conditions, exchange rates, inflation and external industrial activity can create challenges in forecasting Malaysia's E&E export performance, particularly over short forecasting horizons.

This study addresses this problem by developing a machine-learning-based forecasting system for Malaysia's E&E exports using monthly data from January 1990 to March 2025. The study examines the relationships between E&E export values and three selected macroeconomic indicators, namely CPI inflation, the MYR/USD exchange rate and the U.S. Industrial Production Index. Random Forest, Support Vector Regression (SVR) and XGBoost are developed and evaluated to generate forecasts at one-, two-, and three-month-ahead horizons of Malaysia's E&E export values. The resulting historical trends, model performance and forecast results are presented through an interactive Power BI dashboard to support evidence-based decision-making.

This chapter presents the problem background, problem statement, research questions, research objectives, research scope and significance of the study, followed by the organisation of the thesis.

1.2 Problem Background

The Electrical and Electronics (E&E) industry remains one of Malaysia's most important economic sectors, contributing approximately 6.9% to national GDP in 2021 and around 32% of total manufacturing sales value in 2022 (Sheng and Ismail, 2025). Malaysia's overall exports reached a record RM1.607 trillion in 2025, with technology-driven products, including E&E, playing a substantial role (MATRADE, 2025). Because the sector is deeply embedded in global electronics supply chains, its export performance is highly sensitive to shifts in domestic economic conditions and external demand, and these shifts can occur within short time windows that are difficult to anticipate using conventional planning tools.

Accurate one-, two- and three-month-ahead forecasts of E&E export performance are valuable to several groups of decision-makers. Policymakers and government agencies rely on timely forecasts to anticipate emerging risks and calibrate trade and economic policy responses before conditions deteriorate further. Trade analysts and financial institutions use short-term forecasts to assess market conditions and support risk assessment and lending decisions. E&E exporters and businesses depend on near-term visibility into export performance to plan production, manage inventory and allocate resources efficiently. Because these decisions typically operate on monthly or quarterly planning cycles, forecasts that are both short-horizon and available at monthly resolution carry more practical value than longer-horizon or lower-frequency projections.

Existing forecasting approaches for Malaysia's E&E sector present two related limitations for this practical problem. First, prior Malaysia-specific forecasting work has tended to rely on quarterly rather than monthly data. One study, for example, collected monthly E&E sales data but aggregated it into quarterly observations to align with quarterly GDP figures, reducing the dataset to only 64 quarterly observations (Sheng and Ismail, 2025). Such aggregation smooths short-term fluctuations and reduces the number of observations available for model training, limiting the ability of forecasting models to capture how export performance responds to changes in exchange rates, prices and external demand within a given quarter. Second,

econometric studies examining the determinants of Malaysia's electronics and manufacturing exports including exchange rate, inflation and related macroeconomic conditions have generally used cointegration and ARDL frameworks designed to estimate long-run relationships rather than generate short-term operational forecasts (Wong, 2008; Baharin, Shahperi and Guangqin, 2023; Hakim, 2024). These methods are well suited to explaining long-run structural relationships but are not designed to produce the short-horizon, monthly forecasts that policymakers, analysts and E&E businesses require for near-term planning. A fuller review of these studies, and of machine-learning approaches to export forecasting more broadly, is provided in Chapter 2.

This study addresses these limitations by retaining monthly E&E export data across a 35-year period (January 1990 to March 2025) and applying machine-learning models capable of capturing nonlinear relationships between E&E exports and selected macroeconomic conditions, namely CPI inflation, the MYR/USD exchange rate and the U.S. Industrial Production Index. These indicators were selected for their established association with Malaysia's export competitiveness and external demand conditions, as detailed in Chapter 2; other potentially relevant variables, such as GDP and semiconductor-demand indices, were considered but excluded from the predictor set for the reasons explained in the Research Scope (Section 1.6).

1.3 Problem Statement

Malaysia's E&E export values fluctuate in response to short-term shifts in both domestic and external economic conditions, yet forecasting these fluctuations accurately remains difficult. Because the sector is closely tied to global electronics demand and international supply chains, its export performance can shift within short time windows in ways that are not always captured by forecasting approaches built on lower-frequency or narrowly scoped data.

Existing research has approached this problem from several directions, but each carries a specific limitation. Malaysia-specific forecasting work has concentrated on E&E sales values and their contribution to GDP rather than export values directly,

and where monthly data existed, it was aggregated into quarterly observations to match quarterly GDP, reducing both the resolution and the number of observations available for modelling (Sheng and Ismail, 2025). Separately, econometric studies have established that exchange rate and related macroeconomic conditions influence Malaysia's electronics and manufacturing exports (Wong, 2008; Baharin, Shahperi and Guangqin, 2023; Hakim, 2024), but these studies rely on cointegration and ARDL frameworks designed to estimate long-run relationships rather than to generate short-term forecasts or to compare how different modelling approaches perform against one another. Meanwhile, machine-learning export forecasting has advanced in other contexts including China, Indonesia and MSME exports more broadly but none of this work is applied to Malaysia's E&E export values specifically, nor does it incorporate the combination of domestic and external monthly indicators relevant to this sector (NICKELSON, NOORAENI and EFLIZA, 2022; Qiu, 2022; R, M.S. and Pachava, 2025). Furthermore, none of the reviewed Malaysia-specific or machine-learning studies directly benchmark a machine-learning forecasting framework against a traditional statistical baseline such as ARIMA for this specific forecasting problem, leaving it unclear whether machine-learning approaches offer any practical improvement over conventional time-series methods for Malaysia's E&E exports.

This gap the absence of a monthly, E&E-export-specific forecasting framework that both incorporates relevant domestic and external macroeconomic indicators and is benchmarked against a traditional statistical method motivates the present study. To address it, this study develops and compares Random Forest, Support Vector Regression (SVR) and XGBoost against ARIMA as a statistical baseline, using monthly E&E export data from January 1990 to March 2025 integrated with three selected macroeconomic indicators, to determine whether machine-learning approaches can meaningfully improve short-term forecasting of Malaysia's E&E exports.

1.3.1 Research Questions

This following research questions in this study are:

- a) RQ1 - What patterns, trends and relationships exist between Malaysia's monthly E&E export values, historical time-series characteristics and the selected macroeconomic indicators?
- b) RQ2 - How effectively can Random Forest, Support Vector Regression (SVR), and XGBoost forecast Malaysia's E&E export values over 1-month, 2-month and 3-month ahead forecasting horizon?
- c) RQ3 - How can an interactive Power BI dashboard effectively present historical E&E export trends, model performance, and 1-month, 2-month and 3-month ahead forecast results to support evidence-based decision-making?

1.4 Project Goal and Objectives

This project aims to create a machine learning-based forecasting model for Malaysia's E&E export using monthly historical export data and selected domestic and external macroeconomic indicators. The proposed system will apply Random Forest, Support Vector Regression (SVR), and XGBoost to generate three-month-ahead forecasts of Malaysia's E&E exports. The forecasting results will be evaluated and compared to identify the model that provides the most effective forecasting performance. The resulting historical trends, model performance and pre-generated forecast outputs will be presented through an interactive Power BI dashboard to support policymakers, trade analysts and business decision-makers in understanding E&E export trends and making evidence-based decisions.

The objectives of this project are:

- a) RO1 – To analyse the patterns, trends and relationships between Malaysia's E&E export values, historical time-series patterns and selected macroeconomic indicators, namely inflation, exchange rate, and the U.S. Industrial Production Index.
- b) RO2 – To develop and evaluate Random Forest, Support Vector Regression (SVR), and XGBoost models for 1-month, 2-month and 3-month-ahead forecasting of Malaysia's E&E export values using monthly

historical export data, engineered time-series features, and selected macroeconomic indicators.

- c) RO3 – To develop an interactive Power BI dashboard for visualising historical E&E export trends, model performance and 1-month, 2-month and 3-month ahead forecast results to support evidence-based decision-making.

1.5 Research Significance

This study contributes to Malaysia's E&E export forecasting by providing empirical evidence on the comparative forecasting performance of Random Forest, SVR and XGBoost against ARIMA using monthly E&E export data. It addresses the data-resolution limitation of previous Malaysian E&E forecasting research by retaining monthly observations rather than aggregating to quarterly data; however, monthly resolution alone does not guarantee improved forecast accuracy, and this study's findings where ARIMA outperformed all three machine-learning models — provide direct evidence that traditional statistical methods remain a competitive and necessary benchmark rather than a method that should be excluded from forecasting studies of this kind. The study also provides new evidence on the combined use of CPI inflation, the MYR/USD exchange rate and the U.S. Industrial Production Index for short-term E&E export forecasting, and develops three-month-ahead forecasts that can support policymakers in calibrating trade policy responses, trade analysts and financial institutions in assessing near-term market conditions, and E&E exporters and businesses in production and inventory planning. More broadly, the study offers a reproducible framework for evaluating machine-learning approaches against statistical benchmarks that future Malaysian trade-forecasting research can build on.

1.6 Research Scope

The following scopes are included in this research:

- (a) This study focuses exclusively on forecasting Malaysia's Electrical and Electronics (E&E) export values at the aggregate national level; product-level or firm-level forecasting is not covered.
- (b) The study uses monthly historical E&E export data covering the period from January 1990 to March 2025; no real-time or post-March-2025 data integration is included.
- (c) The analysis uses three selected macroeconomic indicators, namely CPI inflation, MYR/USD exchange rate, and the U.S. Industrial Production Index (IPI); other potentially relevant indicators such as GDP and semiconductor-demand indices were excluded due to data-frequency and availability constraints, as discussed in Section 1.2.
- (d) Random Forest, Support Vector Regression (SVR), and XGBoost are evaluated at one-, two- and three-month-ahead forecasting horizons, with ARIMA used as a statistical benchmark; deep-learning approaches (e.g. LSTM) are outside the scope of this study given the relatively limited number of monthly observations available.
- (e) The forecasting results, model performance, and historical E&E export trends are presented through an interactive Power BI dashboard populated with pre-generated Python outputs, rather than a live/real-time forecasting system, to support evidence-based decision-making.

1.7 Project Stakeholders

The forecasting system developed in this study is expected to provide potential benefits to several stakeholders within Malaysia's trade and economic ecosystem.

- a) **Government Agencies:** Government agencies such as the Ministry of Economy Malaysia, Department of Statistics Malaysia (DOSM), and Bank Negara Malaysia (BNM) may use the forecasting results and dashboard visualisations to monitor Malaysia's E&E export trends and

changing economic conditions. The information can support the preparation and assessment of evidence-based trade and economic policies.

- b) Policy Makers: Policy makers may use the three-month-ahead E&E export forecasts to anticipate potential changes in export performance, identify emerging risks and support the development of strategies aimed at strengthening Malaysia's E&E export competitiveness.
- c) E&E Exporters and Businesses: Companies operating within Malaysia's Electrical and Electronics sector may use the forecasting information to support production planning, inventory management, resource allocation and business strategy development. Short-term forecasts can provide additional information for businesses when planning their activities in response to expected changes in export performance and economic conditions.
- d) Financial Institutions: Banks and other financial institutions may use E&E export forecasts and related macroeconomic information to better understand economic and trade conditions. The information may support risk assessment, lending decisions and financial planning for businesses that are exposed to the E&E and international trade sectors.
- e) Researchers and Academicians: The forecasting framework, model comparison results and findings from the study may provide a reference for researchers and academic institutions interested in machine-learning-based trade forecasting, economic forecasting and Malaysia's E&E sector. The study may also provide a basis for future research using alternative variables, forecasting methods or economic sectors.
- f) Investors and Market Analysts: Investors and market analysts may use the historical trends and forecast outputs to gain insights into the expected direction of Malaysia's E&E export performance. Such information can support the assessment of market conditions; identification of potential opportunities and analysis of risks associated with the E&E sector and international trade.

1.8 Thesis Organisation

The thesis has seven chapters each covering a different part of the research.

Chapter 1 – Introduction: Introduces the research background and problem related to forecasting Malaysia’s E&E exports. It presents the problem statement, research questions, research objectives, research significance and research scope. The chapter also provides an overview of the thesis organisation.

Chapter 2 – Literature Review: Reviews existing literature on Malaysia’s E&E industry, E&E trade and export forecasting, selected macroeconomic indicators, and machine-learning approaches to forecasting. The chapter also examines previous applications of Random Forest, Support Vector Regression (SVR) and XGBoost and establishes the research gap addressed by this study.

Chapter 3 – Methodology: Describes the methodology adopted for the study, including data collection, data preprocessing, exploratory data analysis, feature engineering, time-series data splitting, model development, model training and evaluation. The chapter also describes the development of the Power BI dashboard for presenting the forecasting results.

Chapter 4 – Exploratory Data Analysis: Presents the results of the data preprocessing, exploratory data analysis and feature engineering. The chapter examines the patterns, trends and relationships between Malaysia’s E&E export values and the selected macroeconomic indicators and presents the characteristics of the prepared dataset before model development.

Chapter 5 – Implementation: The chapter describes the implementation of the three selected machine-learning models and the ARIMA statistical baseline and the proposed machine-learning forecasting system using Random Forest, Support Vector Regression (SVR) and XGBoost. The chapter presents the model training process, forecasting implementation and development of the interactive Power BI dashboard.

Chapter 6 – Results and Discussion: Presents and discusses the forecasting results obtained from the three machine-learning models. The performance of Random Forest, SVR and XGBoost is compared using the selected evaluation metrics, followed by the three-month-ahead E&E export forecasts and interpretation of the findings. The chapter also discusses the results in relation to the research questions and findings from previous studies.

Chapter 7 – Conclusion: Summarises the main findings of the study and discusses how the research questions and objectives have been addressed. The chapter also presents the contributions, limitations of the study and recommendations for future research.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter critically reviews existing literature related to forecasting Malaysia's Electrical and Electronics (E&E) export values. The review examines the characteristics of Malaysia's E&E export sector, macroeconomic factors that influence export performance, traditional forecasting approaches, and the application of machine-learning techniques to economic and trade forecasting. Particular attention is given to Random Forest (RF), Support Vector Regression (SVR), and Extreme Gradient Boosting (XGBoost), which are the three models adopted in this study. The literature is further compared in terms of data frequency, forecasting target, predictor variables, geographical context, forecasting horizon, and model performance. This comparison is used to identify limitations in existing research and establish the research gap addressed by the present study.

2.2 Problem Domain

The problem domain of this study is the forecasting of Malaysia's Electrical and Electronics (E&E) export values. The E&E industry is an important component of Malaysia's manufacturing and export sector and includes a broad range of products such as semiconductors, electronic components, telecommunications equipment and other electronic products. Malaysia's participation in global E&E value chains has made the sector an important part of the country's international trade activities. The industry's performance is also closely connected to developments in international markets and changes in global demand for electronic and semiconductor-related products (Sheng and Ismail, 2025).

The E&E sector operates within a highly interconnected international environment, where changes in domestic economic conditions and external market activity can influence export performance. Malaysia's E&E exports are exposed to international demand because the industry is integrated into global production and supply chains. Previous research on Malaysia's E&E exports has identified major trading markets, including the United States, Singapore, Japan and Hong Kong, demonstrating the sector's dependence on international markets (Muhammad and Yaacob, 2009). This international exposure makes short-term changes in economic and industrial conditions relevant to the forecasting of Malaysia's E&E export values.

These characteristics make E&E export performance potentially sensitive to changes in both domestic macroeconomic conditions and external industrial demand. Consequently, forecasting Malaysia's E&E exports requires consideration of economic indicators that capture both domestic conditions and external demand rather than relying solely on historical export values.

2.2.1 Malaysia's E&E Industry

The Electrical and Electronics (E&E) industry in Malaysia includes semiconductors, electronic components, telecommunication equipment, electrical machine and other advanced electronic products (Sheng and Ismail, 2025). The industry has become an integral part of the manufacturing-based economy of Malaysia in the last 30 years since the country shifted from being a commodity-based economy to a manufacturing-based export economy through its industrialisation policies such as import-substitution industrialisation in the 1960s, export-oriented industrialisation in the 1970s, and various successive Industrial Master Plans in the 1980s and 1990s (Wong, 2008). Foreign direct investment inflows, especially from the United States, Japan, Singapore and the European Union, played a significant part in the growth and development of the sector, as shown in subsequent research which identified the exchange rate, income tax, financial development and economic risk as important determinants of FDI in the Malaysia E&E sector specifically (Tang, Yip and Ozturk, 2014).

Within the broader manufacturing sector, the E&E industry has become Malaysia's leading non-resource-based export-oriented industry, representing the largest single component of manufactured exports in terms of output, employment, and export value (Muhammad and Yaacob, 2009). Malaysia's participation in the global E&E value chain has positioned the country as a significant international player, with the industry contributing approximately 6.9% to national GDP in 2021 and accounting for around 32% of total manufacturing sales value by 2022 (Sheng and Ismail, 2025). The industry's deep integration into international supply chains means that Malaysia's E&E producers do not operate in isolation from external economic conditions rather, the sector's performance is directly linked to global demand cycles, particularly rising demand for semiconductor-related products driven by electric vehicles, artificial intelligence, and other emerging technologies (Sheng and Ismail, 2025).

2.2.2 Malaysia's E&E Export Performance

Malaysia's E&E exports have historically been concentrated in a small number of major international markets. Early disaggregated analysis identified the United States, Singapore, Japan, and Hong Kong as Malaysia's principal E&E export destinations (Muhammad and Yaacob, 2009), reflecting the sector's exposure to demand conditions in these economies. Comparative competitiveness analysis using Constant Market Share and Revealed Comparative Advantage frameworks found that in the early 1990s, Malaysia's E&E exports were genuinely competitive in these four markets; however, in later periods, continued export growth was found to be driven more by rising global import demand (a structural effect) than by improvements in Malaysia's underlying competitiveness (Muhammad and Yaacob, 2009).

More recent trade data confirms the sector's continued significance: Malaysia's exports reached a record RM1.607 trillion in 2025, with growth underpinned by record shipments to major trading partners including ASEAN, the United States, Taiwan, and the European Union, reflecting continued integration into technology-driven global supply chains (MATRADE, 2025). At the same time, export performance has proven sensitive to external shocks (Thorbecke, 2024) documents that Malaysia's semiconductor-related firms were among the sectors most negatively affected during

the recent period of overlapping global crises, including the COVID-19 pandemic and associated supply chain disruptions, underscoring the sector's vulnerability to conditions outside Malaysia's direct control.

Despite this recognised sensitivity to both domestic and external economic conditions, Malaysia-specific research forecasting E&E export performance directly rather than sales value or GDP contribution remains limited, a gap discussed further in Section 2.6.

2.3 Macroeconomic Factors Influencing E&E Exports

2.3.1 Exchange Rate

The exchange rate has been consistently identified as a significant determinant of Malaysia's electronics and manufacturing export performance. Wong, in a disaggregated study of Malaysia's top five electronics export categories by SITC product group, found a long-run cointegrating relationship between export demand and relative prices, with estimated price elasticities exceeding one for most product categories with the notable exception of semiconductor exports, where price appeared less influential than for other electronics categories. This suggests that exchange-rate-driven price competitiveness affects different E&E sub-sectors unevenly (Wong, 2008).

Similarly it confirms the exchange rate's role using Autoregressive Distributed Lag (ARDL) analysis of Malaysia's manufacturing exports (of which E&E products contributed 30.2% in 2022) over 1990–2021, finding a long-run cointegrating relationship between exchange rate movements and manufacturing export value (Muhammad and Yaacob, 2009). It also reinforces this finding specifically in terms of volatility, concluding that exchange rate volatility negatively affects Malaysia's export volumes and revenues, and that stable exchange rate conditions are important for maintaining export competitiveness (Hakim, 2024). Taken together, this literature establishes exchange rate not merely as a theoretically relevant variable but as an

empirically confirmed determinant of Malaysia's E&E and manufacturing export performance specifically.

2.3.2 Inflation

Inflation is theorised to affect export performance primarily through its effect on production costs: rising domestic prices increase the cost of producing export goods, which can reduce price competitiveness relative to foreign producers (Baharin et al., 2024). While less extensively tested in the Malaysia-specific E&E literature than exchange rate, inflation has been incorporated as a control variable in broader manufacturing export studies (Baharin et al., 2024).

Recent machine-learning research on inflation dynamics provides additional justification for how inflation should be modelled in a forecasting context. Pratap, Pawar and Sengupta, applying explainable machine learning to inflation forecasting in an emerging-market context, demonstrate that inflation relationships are frequently non-linear, involving thresholds and interaction effects between variables that traditional linear models struggle to capture. This finding supports the use of non-linear machine-learning approaches such as those applied in the present study over linear econometric specifications when inflation is included as a predictor of export behaviour (Sengupta, Pratap and Pawar, 2025).

2.3.3 US Industrial Production Index

The U.S. Industrial Production Index (IPI) has been used in prior research as a proxy for external industrial demand affecting a trading partner's exports. It was also directly incorporated U.S. IPI, alongside exchange rate and GDP, as one of six variables used to forecast China's export value to the United States, finding that models incorporating this indicator produced satisfactory forecasting performance across multiple modelling approaches. This precedent supports the use of a similar external-demand proxy for Malaysia's E&E exports, given the sector's substantial trade exposure to U.S. demand (Qiu, 2022).

Further support comes from Kumar and Ahmad, who examine the forecasting relationship between U.S. and Chinese industrial production and industrial production across G20 economies. Their findings indicate that incorporating U.S. industrial production data measurably improves the forecast accuracy of other countries' industrial production, particularly for developed and trade-linked economies, with the effect concentrated in economies most integrated into U.S.-linked supply chains (Kumar and Ahmad, 2024).

2.4 Traditional E&E Export Forecasting Methods and Their Limitations

Traditional approaches to export and E&E-related forecasting in Malaysia have relied primarily on statistical time-series and econometric techniques. Sheng and Ismail applied six forecasting techniques Decomposition, Holt-Winters Exponential Smoothing, Seasonal ARIMA (SARIMA), Time Series Regression, Artificial Neural Networks, and Support Vector Machines to forecast Malaysia's E&E industry sales value and its contribution to GDP through 2030, finding that Time Series Regression produced the most robust long-term projection (Sheng and Ismail, 2025). However, this study forecasts sales value and GDP contribution rather than export value directly, and critically for the present study's design, the researchers aggregated their originally monthly E&E sales data into quarterly observations to align with quarterly GDP figures, reducing the usable dataset to only 64 quarterly data points. This aggregation illustrates a structural limitation common to forecasting studies that anchor E&E data to lower-frequency macroeconomic series: it discards short-term variation and reduces the sample available for model training.

Separately, Malaysia-specific export determinant research has relied on cointegration and ARDL-based econometric frameworks. (Wong, 2008) applied cointegration analysis to estimate long-run export demand elasticities for disaggregated electronics product categories, while another study used ARDL bounds-testing to establish long-run relationships between manufacturing exports and exchange rate, FDI, and trade openness (Baharin, Shahperi and Guangqin, 2023). These econometric approaches are primarily used to examine long-run structural relationships between variables, while relatively less attention has been given to their

use for short-term monthly forecasting of Malaysia's E&E export values. Outside Malaysia, export forecasting research using classical statistical methods has similarly relied on ARIMA and related time-series techniques. Another study also compared BP neural networks, ARIMA, and AR-GARCH models for forecasting China's export value, finding ARIMA to have a slight performance advantage overall despite its simpler linear structure (Qiu, 2022). However, such comparisons suggest that no single traditional method dominates uniformly, and that machine-learning alternatives may offer complementary strengths particularly in capturing the non-linear relationships noted in Section 2.3 motivating the shift toward machine-learning approaches examined in Section 2.5.

Overall, the reviewed studies indicate that Malaysia's export performance is influenced by both domestic and external economic conditions. Exchange rate movements have been repeatedly associated with export competitiveness, while inflation may affect export performance through production costs and price competitiveness. The use of U.S. Industrial Production as an external demand indicator is supported by international trade forecasting research, particularly for economies integrated into global manufacturing supply chains. However, these variables have generally been examined individually or within econometric frameworks rather than being jointly evaluated as predictors of Malaysia's monthly E&E export values using machine-learning methods. This provides a basis for investigating their combined predictive usefulness within the present study.

2.5 Machine Learning Models

2.5.1 Random Forest

Random Forest (RF) is an ensemble learning technique, which is used to integrate the predictions of several decision trees (Balasubramanian, 2026). The algorithm is trained on a bootstrapped sample of the training set for each tree and uses a random subset of features at each split. The combination of multiple trees reduces the dependence on any individual tree and allows the model to capture nonlinear relationships and interactions between predictors. The model can also provide feature-

importance measures, which are useful for examining the relative contribution of explanatory variables in economic forecasting.

Empirical evidence supports the application of Random Forest to economic forecasting. Silva, Wilhelm and Amancio compared several supervised machine-learning regressors for economic-growth forecasting across more than 200 countries. Their results ranked Random Forest above the other models examined, followed by XGBoost and LightGBM, with Random Forest, XGBoost and k-nearest neighbours identified among the strongest overall performers (Silva, Wilhelm and Amancio, 2024).

However, the findings of Silva, Wilhelm and Amancio cannot be directly generalised to Malaysia's E&E export forecasting because their target was economic growth rather than monthly E&E export values. This distinction is important because model performance depends on the characteristics of the forecasting problem. Therefore, although previous research supports Random Forest's ability to model nonlinear economic relationships, its performance for Malaysia's E&E exports must be established empirically (Silva, Wilhelm and Amancio, 2024).

In the present study, Random Forest is therefore evaluated using historical E&E export information, engineered time-series features and selected macroeconomic indicators. Its forecasting performance is compared with SVR and XGBoost over the same one-, two- and three-month forecasting horizons.

2.5.2 Support Vector Regression

Support Vector Regression (SVR) is a variant of the Support Vector Machine (SVM) algorithm, first introduced by Vapnik (1995). Support Vector Regression (SVR) is the regression form of Support Vector Machine (SVM). It is designed to estimate a function while controlling model complexity and can use kernel functions to represent nonlinear relationships between predictors and the target variable. Common kernel functions include linear, polynomial and radial basis function kernels.

This flexibility makes SVR applicable to forecasting problems where the relationship between explanatory variables and the target is not necessarily linear.

Yang, Lee and Chang provide relevant evidence from international trade forecasting. Their study compared SVR with several statistical and machine-learning approaches, including ARIMA, Exponential Smoothing, TBATS, Random Forest Regression, LASSO, Multilayer Perceptron and XGBoost, across ten major countries. Their results showed that SVR achieved competitive forecasting performance in several country-specific cases, particularly when economic structural variables and lagged trade values were incorporated as predictors (Yang, Lee and Chang, 2023).

The use of economic predictors in the study by Yang, Lee and Chang is relevant to the present research because the proposed SVR model also incorporates macroeconomic information alongside historical export features. However, the previous study examined international trade forecasting across several countries rather than specifically forecasting Malaysia's E&E export values. Therefore, its findings provide methodological support for SVR but do not establish its performance for the Malaysian E&E export series (Yang, Lee and Chang, 2023).

The present study evaluates SVR alongside Random Forest and XGBoost using the same predictor set, forecasting horizons and evaluation framework. This allows its performance to be assessed relative to both other machine-learning approaches and the traditional statistical benchmark used in the study.

2.5.3 Extreme Gradient Boosting (XGBoost)

Extreme Gradient Boosting (XGBoost) is a tree-based ensemble learning method that builds a sequence of decision trees, with subsequent trees designed to improve the errors produced by earlier trees. The approach includes regularisation mechanisms intended to control model complexity and reduce overfitting. Its ability to represent nonlinear relationships makes it suitable for forecasting problems involving multiple economic predictors.

Recent comparative forecasting research provides evidence supporting the use of XGBoost. Santoro, Ciano and Ferrara reported that XGBoost outperformed simpler machine-learning methods and deep-learning models within their forecasting context. However, the result was specific to the characteristics of their dataset and forecasting problem and therefore cannot be assumed to apply directly to Malaysia's E&E export forecasting (Santoro, Ciano and Ferrara, 2024).

A study provides a further comparison between XGBoost and SVR in currency forecasting using macroeconomic predictors. Their findings indicated relatively small differences in forecasting performance between the two approaches. This suggests that the relative performance of XGBoost and SVR cannot necessarily be determined in advance and supports the use of empirical model comparison (Fundama *et al.*, 2026).

Kim also reported strong performance from tree-based ensemble models compared with linear approaches in monthly macroeconomic forecasting using economic predictors. Together, these findings provide support for including XGBoost in the present study while also reinforcing the importance of evaluating its performance using the specific Malaysian E&E export dataset rather than assuming superiority based on results from other forecasting contexts (Kim, 2025).

2.5.4 Comparative Studies of RF, SVR, and XGBoost

Previous research demonstrates that Random Forest, SVR and XGBoost can all be applied to economic, trade and financial forecasting, but their relative performance differs across forecasting problems. Silva, Wilhelm and Amancio reported strong performance from Random Forest in economic-growth forecasting, while Santoro, Ciano and Ferrara reported strong performance from XGBoost in their forecasting application (Santoro, Ciano and Ferrara, 2024; Silva, Wilhelm and Amancio, 2024). Another study, meanwhile, found relatively small differences between SVR and XGBoost (Fundama *et al.*, 2026).

These findings indicate that the performance of a machine-learning model is context-dependent. Differences in the target variable, predictor variables, economic

environment and forecasting horizon can affect model performance. Consequently, selecting a forecasting algorithm solely on the basis of its performance in another domain may not produce the same results when applied to Malaysia's E&E exports.

The present study therefore compares Random Forest, SVR and XGBoost using the same historical E&E export dataset, engineered time-series features and selected macroeconomic indicators. This controlled comparison allows differences in forecasting performance to be attributed more directly to the modelling approaches rather than differences in datasets or experimental settings.

An important consideration for the present study is that the available dataset, after chronological train-test splitting, provides a relatively limited number of monthly observations for model training compared with the large-sample settings typical of many machine-learning applications. This consideration favours Random Forest, SVR and XGBoost over more data-intensive approaches such as deep-learning models. Random Forest and XGBoost are ensemble tree-based methods that can learn effectively from moderate-sized datasets because they build on relatively simple base learners (decision trees) combined through bagging or boosting, and both incorporate mechanisms such as bootstrap sampling, random feature selection and regularisation that reduce the risk of overfitting when the number of training observations is limited. SVR similarly performs well on smaller datasets because its complexity is controlled through the margin and regularisation parameters (C and epsilon) rather than requiring large volumes of data to estimate a high-dimensional parameter space, as would be the case for deep neural networks. This supports the selection of RF, SVR and XGBoost, rather than deep-learning alternatives, as appropriate machine-learning approaches for the dataset size and monthly time-series characteristics of Malaysia's E&E export series.

Table 2:1 Summary and Comparison of Previous Studies

Study	Forecasting context	Target / Application	Model(s)	Main finding	Relevance to present study
Qiu (2022)	Export forecasting	Export value	Neural network and time-series methods	Examined data-driven and time-series approaches for export forecasting	Supports the use of forecasting models for export values, but does not specifically address Malaysia's E&E exports
Nooraeni et al. (2022)	Export-import forecasting	Indonesian export-import values	ANN, ARIMA	Compared statistical and neural-network approaches for export-import forecasting	Supports the use of ARIMA as a traditional forecasting benchmark

Yang, Lee & Chang (2023)	International trade forecasting	Foreign trade	ARIMA, ETS, TBATS, SVR, RF, XGBoost, LASSO, MLP and other approaches	SVR achieved competitive performance in several country-specific forecasting cases	Provides direct evidence for comparing SVR with statistical and other ML approaches
Silva, Wilhelm & Amancio (2024)	Economic forecasting	Economic growth	RF, XGBoost, LightGBM and other ML models	RF ranked strongly among the models examined, with XGBoost also performing strongly	Supports RF and XGBoost for nonlinear economic forecasting, but the target differs from E&E exports
Santoro, Ciano & Ferrara (2024)	Time-series forecasting	Economic forecasting	XGBoost and other ML/deep-learning	XGBoost performed strongly in the forecasting	Supports inclusion of XGBoost, but performance

			approaches	context examined	e cannot be assumed to transfer to Malaysian E&E exports
Kim (2025)	Macroeconomic forecasting	Monthly economic variables	Tree-based ensemble and linear approaches	Tree-based ensemble models performed strongly compared with linear approaches	Supports tree-based ML models when economic predictors are available
Fundam a et al. (2026)	Financial forecasting	Currency forecasting	SVR, XGBoost	Reported relatively small differences between SVR and XGBoost	Supports direct empirical comparison between SVR and XGBoost

2.6 Research Gap

Although previous studies have examined Malaysia's E&E sector and the application of machine-learning methods to economic and trade forecasting, several limitations remain. First, existing Malaysia-specific E&E forecasting research has primarily focused on quarterly data or E&E sales rather than monthly E&E export values. For example, Sheng and Ismail aggregated monthly E&E sales data into quarterly observations to align with the availability of quarterly GDP data, resulting in a smaller number of observations and potentially reducing the ability to capture short-term fluctuations in E&E performance (Sheng and Ismail, 2025). Therefore, there is limited evidence on whether monthly E&E export data can provide more detailed information for short-term forecasting.

Second, previous studies on the determinants of Malaysia's electronics and manufacturing exports have mainly examined individual economic relationships using econometric approaches. Studies such as Wong, Baharin, and Hakim provide evidence of the importance of exchange rate and other macroeconomic conditions (Wong, 2008b; Baharin, Shahperi and Guangqin, 2023; Hakim, 2024), but they do not examine the combined predictive contribution of inflation, MYR/USD exchange rate and U.S. Industrial Production Index within a machine-learning forecasting framework for Malaysia's E&E exports. Meanwhile, international machine-learning export forecasting studies have used different combinations of economic indicators and have mainly focused on other countries or broader export categories (Nooraeni *et al.*, 2022; Qiu, 2022; R, M.S. and Pachava, 2025). Consequently, the predictive usefulness of this particular combination of domestic and external indicators for Malaysia's E&E exports remains insufficiently established.

Third, there is limited empirical evidence on short-term forecasting of Malaysia's E&E export values using machine-learning methods. Existing studies provide evidence that Random Forest, SVR and XGBoost can perform effectively in different economic and forecasting contexts (Yang, Lee and Chang, 2023; Silva, Wilhelm and Amancio, 2024; Kim, 2025; Fundama *et al.*, 2026). However, model performance is dependent on the characteristics of the dataset, forecasting target and

predictor variables. Therefore, findings from other countries, economic sectors or forecasting applications cannot be assumed to represent the performance of these models for Malaysia's E&E exports.

Overall, previous studies demonstrate the usefulness of both traditional statistical and machine-learning approaches for economic and trade forecasting. However, the reported performance of individual machine-learning models varies across forecasting contexts. Existing Malaysian research has not established whether Random Forest, SVR or XGBoost can provide accurate short-term forecasts of Malaysia's E&E export values when historical export features are combined with domestic and external macroeconomic indicators. Furthermore, limited evidence directly compares these machine-learning approaches with a traditional statistical benchmark for this specific forecasting problem. Therefore, the present study compares Random Forest, SVR and XGBoost against ARIMA as a traditional statistical benchmark for one-, two- and three-month-ahead forecasting of Malaysia's E&E exports.

2.7 Chapter Summary

In the literature reviewed, this chapter concentrated on the special features of the Electrical and Electronics (E&E) sector, macroeconomic conditions impacting export performance, traditional forecasting methods and machine learning techniques. The review revealed that existing E&E forecasting research in Malaysia has some limitations, such as the use of lower-frequency data, the fact that the majority of the forecast models focus on export values only, and the lack of evidence for short-term monthly forecasting by combining domestic and external economic indicators. Previous studies also demonstrated that Random Forest, Support Vector Regression and XGBoost can perform effectively across different economic and forecasting applications, although their relative performance varies according to the dataset, predictors and forecasting context. To address these limitations, the present study develops a monthly comparative machine-learning framework for forecasting Malaysia's E&E export values over a three-month horizon using selected macroeconomic and historical time-series features.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

In this chapter, a methodology for the development of forecasting system using machine learning for Electrical and Electronics (E&E) exports in Malaysia was described. The methodology for the study is the CRISP-DM (Cross-Industry Standard Process for Data Mining) that has a structured approach to the study with phases: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation and Deployment. Three machine-learning models: Support Vector Regression (SVR), Random Forest and XGBoost are built and tested with the historical monthly data of E&E exports and chosen macroeconomic indicators. ARIMA is additionally employed as a statistical baseline for benchmarking the machine-learning models.

3.2 Research Design

This study uses quantitative, empirical and data driven research design to create the machine-learning based forecasting system for Electrical and Electronics (E&E) export values for Malaysia. The study employs monthly export data of E&E commodities from 1975 to 2009 and selected macroeconomic and external economic indicators for forecasting the value of E&E exports in Malaysia for short term forecasting period.

The study employs a time series forecasting method that employs monthly data from January 1990 to March 2025. Monthly frequency permits the study to reflect changes in the performance of E&E exports, seasonal trends, economic sensitivity and short term variations of selected explanatory variables. The forecasting framework is

intended to produce 1, 2 and 3-month ahead forecasts, focusing on the three-month forecasting horizon.

Monthly observations are made instead of quarterly or annual observations, in order to keep the short-term information that would be averaged away at the higher level of aggregation that occurs when observations are quarterly or annual, and because the forecast horizons of interest (one, two and three months ahead) are also short-term. The dataset contains 423 monthly observations from January 1990 to March 2025, offering a satisfactory length of time to observe multiple full economic cycles, as needed for the selected machine-learning models to learn robust patterns instead of fitting to a specific economic regime. After chronological splitting and the removal of observations required for lag and rolling feature construction, the remaining training and testing samples (326 and 82 observations respectively) fall within the range typically used in comparable applied machine-learning forecasting studies using classical (non-deep-learning) models, further supporting the choice of Random Forest, SVR and XGBoost over more data-intensive approaches.

The study examines the predictive contribution of historical E&E export patterns and selected external and domestic economic indicators, namely inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index. Historical export values are incorporated through time-series feature engineering, while the selected economic indicators are used to provide information on domestic price conditions, exchange-rate movements, and external industrial activity that may be associated with Malaysia's E&E export performance.

A comparative research design is adopted to evaluate three machine-learning models: Random Forest (RF), Support Vector Regression (SVR), and XGBoost. The three models are trained and evaluated using the same prepared dataset, forecasting horizons, and evaluation framework. This enables a systematic comparison of their forecasting performance and allows the best-performing model to be identified for short-term forecasting of Malaysia's E&E exports.

3.3 CRISP-DM Framework

The primary method for creating a machine-learning based forecasting system in this study is based on the Cross-Industry Standard Process for Data Mining (CRISP-DM). The CRISP-DM process is an iterative and structured research process that consists of the following phases: problem specification for the forecasting task, data preparation, model building, model evaluation and model deployment. There are six phases: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation and Deployment.

Because it was explicitly designed around an iterative feedback between the business goals and technical development, CRISP-DM was chosen over other process frameworks, like SEMMA (Sample, Explore, Modify, Model, Assess) and KDD (Knowledge Discovery in Databases), which did not explicitly include this type of feedback. This is appropriate for a forecasting study like this where model evaluation results (forecast accuracy at various horizons) may warrant repeating decisions made earlier in the study for feature engineering or data preparation. CRISP-DM is also the most widely adopted and vendor-neutral standard for structuring applied data-science and machine-learning projects, providing a transparent and replicable structure for reporting the present study's methodology (Chapman *et al.*, 2000).

In this study, the Business Understanding phase defines the need to forecast Malaysia's E&E export values over a short-term horizon and establishes the research objectives and success criteria. The Data Understanding phase involves collecting and examining monthly E&E export data together with the selected economic indicators, including inflation, the MYR/USD exchange rate and the U.S. Industrial Production Index. The Data Preparation phase covers data cleaning, integration, transformation and the construction of time-series features required for forecasting.

During the Modelling phase, Random Forest, Support Vector Regression (SVR), and XGBoost models are developed using the prepared monthly dataset. The Evaluation phase compares the forecasting performance of the models using MSE, RMSE, MAE and R^2 across the selected forecasting horizons. Finally, the Deployment

phase involves presenting the forecasting results through an interactive Power BI dashboard to support the interpretation and communication of historical trends, model performance and three-month-ahead forecasts.

CRISP-DM is iterative rather than strictly sequential, allowing findings from later stages to inform earlier stages. For example, model evaluation may identify the need to revise the feature engineering or data preparation process before the models are evaluated again. This iterative characteristic is particularly relevant to the present time-series forecasting study because feature selection, preprocessing and model configuration may require refinement based on forecasting performance.

3.4 Research Methodology Workflow

In this study, the overall research process follows the CRISP-DM framework and covers the stages from business understanding and data collection to model development, evaluation and dashboard deployment. The workflow is designed to provide a systematic and repeatable process for developing a machine-learning-based forecasting system for Malaysia's E&E exports.

The workflow begins with Business Understanding, where the forecasting problem, research objectives and success criteria are defined. This is followed by Data Understanding, which involves collecting and examining monthly Malaysia E&E export data and the selected economic indicators. Data Preparation then involves cleaning, integrating and transforming the datasets into a consistent monthly dataset. Feature Engineering is subsequently performed to construct historical time-series features, such as lagged and rolling features, that can provide information about previous E&E export behaviour.

The Modelling stage involves developing Random Forest, Support Vector Regression (SVR), and XGBoost models using the prepared dataset. The models are evaluated for 1-month-, 2-month-, and 3-month-ahead forecasting using MSE, RMSE, MAE and R^2 . The best-performing models are then used to generate the required short-term E&E export forecasts. Finally, the Deployment stage presents the historical

trends, model performance and three-month-ahead forecast results through an interactive Power BI dashboard.

The stages are not treated as strictly sequential, as feedback may be applied between stages when necessary. For example, the results obtained during model evaluation may indicate that changes are required in feature engineering, preprocessing or model configuration. This iterative process allows the forecasting models to be refined before the final results are presented and deployed through the dashboard.

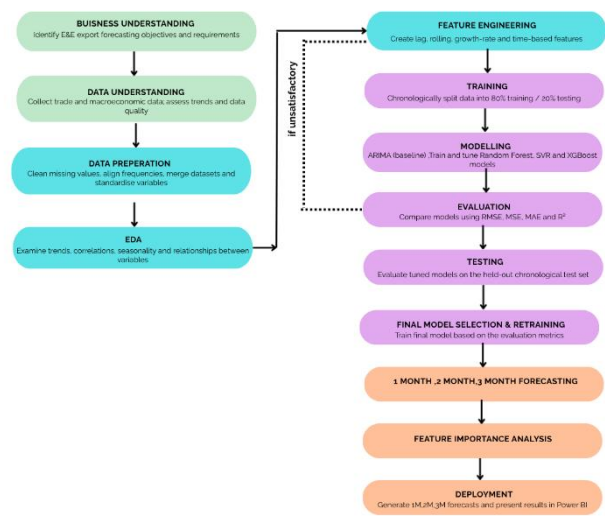


Figure 3:1 Research Workflow Diagram

3.4.1 Business Understanding

3.4.1.1 Project Objectives

The project pursues three primary objectives:

- a) To analyse the patterns, trends and relationships between Malaysia's E&E export values, historical time-series patterns and

selected macroeconomic indicators, namely inflation, exchange rate, and the U.S. Industrial Production Index.

- b) To develop and evaluate Random Forest, Support Vector Regression (SVR), and XGBoost models for 1-month-, 2-month-, and 3-month-ahead forecasting of Malaysia's E&E export values using monthly historical export data, engineered time-series features, and selected macroeconomic indicators.
- c) To develop an interactive Power BI dashboard for visualising historical E&E export trends, model performance and three-month-ahead forecast results to support evidence-based decision-making.

The stakeholders identified in Section 1.4 are interested in near-term forecasts, as their planning and decision-making processes have short time horizons: policymakers and government agencies rely on short-term forecasts to calibrate trade responses; E&E exporters are interested in 1, 2 and 3 month forecasts for planning production and inventory; financial institutions are interested in short-term forecasts for making lending and risk decisions. ARIMA is also considered as a benchmark for statistical models in order to test if the extra complexity of machine-learning models that use macroeconomic indicators and engineered time-series features offers a useful forecasting advantage over a univariate statistical model for these same short time horizons.

3.4.1.2 Success Criteria

The success of this study is primarily assessed based on the forecasting performance of the Random Forest, Support Vector Regression (SVR), and XGBoost models for Malaysia's E&E exports. Model performance is evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R^2 . Lower MSE, RMSE and MAE values indicate lower prediction errors, while a higher R^2 value indicates a greater proportion of variation in the target variable being explained by the model.

RMSE, MAE, MSE and R^2 were selected as the evaluation metrics because together they provide complementary information about forecasting error: MAE gives an easily interpretable average error magnitude, MSE and RMSE penalise larger errors more heavily (which is important for a forecasting problem where large misses have greater practical consequences for planning), and R^2 indicates the proportion of variation in E&E export values explained by the model, allowing forecasting performance to be assessed both in absolute error terms and relative to the variability of the target series itself.

The best-performing model will be identified based primarily on RMSE, with MAE, MSE and R^2 used as supporting evaluation measures. The evaluation will be performed across the forecasting horizons considered in the study, with particular emphasis on the three-month-ahead forecasting objective. The final model should demonstrate reliable predictive performance and produce forecasts that can be effectively communicated through the proposed Power BI dashboard.

3.4.1.3 Project Planning

Table 3:1 Project Execution Plan

Category	Description
Project Plan	The project follows the CRISP-DM methodology, beginning with problem definition and business understanding, followed by data collection, data understanding, data preparation, feature engineering, model development, model evaluation and deployment through an interactive forecasting dashboard.
Data Resources	Historical monthly Malaysia E&E export data and selected economic indicators, namely CPI/inflation, MYR/USD exchange rate and the U.S. Industrial Production Index. E&E export data are obtained from the Ministry of Economy Malaysia's External Trade Statistics portal, CPI/inflation data from the Department of Statistics Malaysia (DOSM), and exchange rate and U.S. Industrial Production Index data from the Federal Reserve Bank of St. Louis (FRED).

Computing Resources	A typical personal computer with sufficient processing capacity for data preprocessing, feature engineering, machine-learning model training, hyperparameter tuning and model evaluation.
Human Resources	The project is implemented by the researcher with academic supervision and regular monitoring and guidance throughout the project development process.
Tools and Software	Python is used as the primary programming language. The data preprocessing and manipulation are done using pandas and NumPy, while scikit-learn is employed for machine learning modelling and evaluation, and XGBoost is used for gradient-boosting modelling. Power BI is used to develop the dashboard and visualise the data.
Timeline	The project is divided into 8 weeks as follows: Weeks 1–2: Data Collection and Understanding; Weeks 3–4: Data Cleaning and Feature Engineering; Weeks 5–6: Model Development and Evaluation; Weeks 7–8: Deployment, Visualisation and Final Report.

Table 3:1 maps directly onto the CRISP-DM phases and research objectives: the Weeks 1–2 activities (Data Collection and Understanding) correspond to the Business Understanding and Data Understanding phases and support RO1; Weeks 3–4 (Data Cleaning and Feature Engineering) correspond to the Data Preparation phase and support RO1 and RO2; Weeks 5–6 (Model Development and Evaluation) correspond to the Modelling and Evaluation phases and support RO2; and Weeks 7–8 (Deployment, Visualisation and Final Report) correspond to the Deployment phase and support RO3.

3.4.2 Data Understanding

3.4.2.1 Data Sources and Acquisition

The data used in this study were collected from official and established data sources to provide consistent and credible information for forecasting Malaysia's Electrical and Electronics (E&E) exports. The study uses monthly observations

covering the period from January 1990 to March 2025, resulting in 423 monthly observations. The long observation period provides sufficient historical variation for examining changes in E&E export performance across different economic conditions and supports the development of machine-learning forecasting models.

The data used in this study were collected from four official and established sources, summarised in the table below.

Table 3:2 Data Sources

Variable	Description	Source Organisation	Access Method	Study Period	Access Date	Frequency	Unit	Role
E&E Exports	Malaysia's E&E export value	Ministry of Economy Malaysia – External Trade Statistics	Public portal download	Jan 1990 – Mar 2025	March 2025	Monthly	RM million	Target
CPI Inflation	Malaysia's inflation rate	Department of Statistics Malaysia (DOSM)	Public portal download (OpenDOSM)	Jan 1990 – Mar 2025	March 2025	Monthly	% change	Predictor
MYR/USD Exchange Rate	MYR to USD exchange rate	Federal Reserve Bank of St. Louis (FRED)	Public download via FRED	Jan 1990 – Mar 2025	March 2025	Monthly	MYR per USD	Predictor
U.S. Industrial Production Index	U.S. industrial output index	Federal Reserve Bank of St. Louis (FRED)	Public download via FRED	Jan 1990 – Mar 2025	March 2025	Monthly	Index (2017=100)	Predictor

All four datasets were sourced from official government or central-bank-affiliated institutions rather than third-party aggregators, which supports their reliability and consistency for academic research. The Ministry of Economy Malaysia and DOSM are the official Malaysian government sources for trade and price statistics respectively, while FRED is a widely used and academically recognised repository maintained by the Federal Reserve Bank of St. Louis for macroeconomic time-series

data. All datasets were accessed and downloaded in March 2025 to align with the end of the study's observation period.

The forecasting target is Malaysia's monthly E&E export value, obtained from the Ministry of Economy Malaysia's External Trade Statistics portal. The predictor variables consist of CPI/inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index. CPI/inflation data are obtained from the Department of Statistics Malaysia (DOSM), while the exchange rate and U.S. Industrial Production Index are obtained from the Federal Reserve Bank of St. Louis (FRED).

Monthly frequency is adopted because the study focuses on short-term forecasting of E&E exports up to three months ahead. Using monthly observations allows the model to capture more frequent changes in export performance and movements in the selected economic indicators compared with an annual dataset. The monthly frequency therefore provides an appropriate temporal resolution for the forecasting objective.

The datasets were standardised before integration to ensure consistency in their date formats, units and variable names. The Date field was used as the common time index for combining the datasets. After standardisation, the datasets were integrated into a single multivariate time-series dataset containing the E&E export target and the selected predictor variables. The resulting dataset provides a consistent monthly structure for subsequent exploratory analysis, feature engineering and machine-learning modelling.

3.4.3 Data Preparation

Data preparation was conducted to transform the collected monthly datasets into a consistent format suitable for machine-learning-based forecasting. The preparation process included data selection, data cleaning, data integration and formatting, exploratory analysis, and feature construction and transformation. These steps were performed to ensure that the E&E export target and selected economic

predictors were appropriately structured while preserving the chronological nature of the time-series data.

3.4.3.1 Data Selection

The variables were selected based on their relevance to Malaysia's E&E export performance, availability throughout the study period, and suitability for short-term machine-learning forecasting. The selected datasets consist of monthly Malaysia E&E exports as the target variable and three predictors: inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index.

Inflation was included because changes in domestic price levels can affect production costs and export price competitiveness, which may influence Malaysia's E&E export performance. The MYR/USD exchange rate was selected because exchange-rate movements can affect the relative price competitiveness of Malaysian exports and the cost of imported inputs. The U.S. Industrial Production Index was selected as an external indicator of industrial activity and demand conditions that may provide information relevant to Malaysia's E&E exports, given the sector's integration with international markets and supply chains.

Together, these variables represent domestic price conditions, exchange-rate movements and external industrial activity, providing complementary information for forecasting Malaysia's E&E export values.

The exclusion of other potentially relevant variables was a deliberate methodological choice rather than an oversight. GDP was excluded because it is only available at quarterly frequency and disaggregating it to monthly resolution would require interpolation introducing estimated rather than the directly observed values is inconsistent with this study's monthly design (Sheng and Ismail, 2025). A dedicated semiconductor-demand index (e.g. WSTS) was excluded because no consistent, freely available monthly series exists across the full 1990–2025 study period; its influence is considered to be partly captured indirectly through the U.S. Industrial Production Index, given the significant role of U.S. demand in global semiconductor and

electronics cycles (Qiu, 2022; Kumar and Ahmad, 2024). The three retained indicators inflation, exchange rate and U.S. IPI are each independently supported by prior Malaysian and international export-forecasting literature as relevant to electronics and manufacturing export performance (Wong, 2008; Qiu, 2022; Baharin, Shahperi and Guangqin, 2023; Hakim, 2024), providing a literature-supported and data-feasible predictor set for this study's monthly forecasting framework.

Records that were not relevant to the monthly forecasting analysis, such as annual summary records, empty rows and invalid records, were removed during data cleaning. Only observations and variables relevant to Malaysia's E&E export forecasting were retained.

3.4.3.2 Data Cleaning

The source datasets were cleaned to improve consistency and ensure that the observations were suitable for subsequent integration and modelling. Non-relevant rows, including subtotal rows, metadata and empty records, were removed, while only observations relevant to the monthly E&E export forecasting analysis were retained.

Missing-value and duplicate checks were performed on each of the four analytical variables (E&E Exports, CPI Inflation, Exchange Rate and U.S. IPI) across all 423 monthly observations. No missing values and no duplicate records were identified in any of the four variables (detailed results are presented in Table 4:2, Chapter 4), and therefore no imputation or deduplication was required. Each variable was also visually inspected for outliers (boxplots and time-series plots) and the interquartile range (IQR) method was used. High and low points of the E&E export and inflation series could be attributed to genuine and identifiable economic events such as the 2008 Global Financial Crisis, the 2022 inflation surge and the COVID-19 pandemic, not to data entry errors. Such observations were therefore not deleted or capped from the data, but were instead kept so as not to lose the true historical variability that is important for the forecasting problem.

The cleaned datasets were then prepared for integration using the common monthly date index. The resulting data were checked for inconsistencies in data types, duplicate observations and chronological ordering before proceeding to data integration and feature construction.

3.4.3.3 Data Integration and Formatting

Before integration, each of the four source datasets contained 423 monthly observations. A temporary Date_YM key in YYYY-MM format was created for each dataset to ensure that observations were merged on the same calendar month rather than exact date, since the original datasets used differing date conventions and reporting frequencies. The E&E export dataset was used as the base table and progressively left-joined with CPI inflation, exchange rate and U.S. IPI data on this common monthly key. After integration, the temporary key was removed, the date field was converted to a proper datetime index, and the observations were sorted chronologically. The resulting integrated dataset retained all 423 monthly observations with no rows lost or duplicated during merging, and post-integration checks confirmed zero missing values and zero duplicate records across the combined dataset (Table 4:3, Chapter 4). The integrated dataset was subsequently prepared for feature construction and model development. The chronological ordering was also maintained when separating the data into training and testing periods to avoid using future observations to predict earlier periods.

3.4.3.4 Data Exploration

Exploratory Data Analysis (EDA) was conducted to understand the structure, trends and relationships within the integrated monthly dataset before feature engineering and model development. The analysis focused primarily on the historical behaviour of Malaysia's E&E exports and its relationship with the selected economic indicators. Four types of exploratory analysis were conducted to support the subsequent feature-engineering and modelling decisions.

Time-series visualisations were used to examine the historical trends, fluctuations and potential seasonal patterns in Malaysia's E&E export values. Major economic events, including the 2008 global financial crisis and the COVID-19 pandemic, were also considered when interpreting substantial changes in export performance. These events provide contextual information for understanding unusual movements in the E&E export series.

Correlation analysis was used to examine the direction and strength of the linear relationships between E&E exports and the selected economic indicators. The analysis provides an initial understanding of the associations between E&E exports, inflation, the MYR/USD exchange rate and the U.S. Industrial Production Index. However, correlation results were treated as exploratory evidence rather than proof of causality or predictive significance.

Seasonality analysis, using monthly boxplots, was used to assess whether particular calendar months systematically exhibit higher or lower export values, informing the inclusion of month and quarter as calendar-based features.

Descriptive statistics (mean, median, standard deviation, minimum and maximum) were used to characterise the scale and variability of each variable, informing the decision to apply feature scaling for SVR. Detailed results of each of these analyses are presented in Chapter 4.

The findings from the exploratory analysis were subsequently used to support the feature engineering and modelling stages of the study.

3.4.3.5 Feature Construction and Transformation

Feature construction and transformation were performed to convert the integrated monthly dataset into a supervised learning structure suitable for machine-learning-based forecasting. Since the study uses time-series data, historical information from previous periods was incorporated to capture temporal patterns and changes in Malaysia's E&E export performance.

Lag features were created using previous E&E export observations at 1-month, 3-month, and 12-month intervals. The 1-month lag captures recent export behaviour, while the 3-month and 12-month lags provide information on medium-term and annual patterns.

$$Lag_k(t) = E\&E_{t-k} \quad (3.1)$$

where $k = 1, 3$, and 12 months. This represents the E&E export value from k months before the current observation.

Rolling statistical features were also constructed using 3-month and 12-month windows. Rolling means were used to represent recent and longer-term export trends, while rolling standard deviations were used to capture the variability of E&E export values over the corresponding periods. The rolling features were calculated using only historical observations to prevent information from future periods from being incorporated into the predictors.

$$RollingMean_w(t) = \frac{1}{w} \sum_{i=1}^w E\&E_{t-i} \quad (3.2)$$

where $w = 3$ and 12 months.

$$RollingStd_w(t) = Std(E\&E_{t-1}, E\&E_{t-2}, \dots, E\&E_{t-w}) \quad (3.3)$$

where $w = 3$ and 12 months.

Growth rate features were calculated to capture changes in E&E export performance over time.

3.4)

$$Growth(t) = \left(\frac{E\&E_t - E\&E_{t-1}}{E\&E_{t-1}} \right) \times 100$$

This measures the percentage change in E&E exports relative to the previous month.

In addition, month and quarter variables were constructed as time-based features to represent potential seasonal and calendar-related patterns in monthly E&E exports.

3.5)

$$Month(t) \in \{1, 2, \dots, 12\}$$

(3.6)

$$Quarter(t) \in \{1, 2, 3, 4\}$$

where $Month(t)$ represents the calendar month and $Quarter(t)$ represents the calendar quarter of observation t .

The selected economic indicators, namely inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index, were incorporated into the feature set. Where lagged predictor variables were constructed, only information available from previous periods was used to maintain the chronological nature of the forecasting problem.

Feature transformation was also applied according to the requirements of the machine-learning algorithms. Standardisation was applied to the features used by Support Vector Regression (SVR) because the model is sensitive to differences in feature scales. Random Forest and XGBoost were not subjected to standardisation because their tree-based structures are not dependent on the scale of the input variables.

Because every lag, rolling and growth-rate feature at time t uses only values from $t-1$ and earlier, none of these features incorporate information from the target period or any future observation, preventing information leakage. Feature scaling (standardisation) applied for SVR was fitted using only the training partition (the

earliest 326 observations); the fitted scaler parameters (mean and standard deviation) were then applied to transform the test partition, ensuring that no information from the held-out test set influenced the scaling process.

The feature construction process resulted in the removal of observations for which the required historical information was unavailable, particularly at the beginning of the dataset where lagged and rolling features could not be calculated. The resulting dataset was subsequently structured for supervised learning and model development. Future E&E export values were used only as target values corresponding to the relevant forecasting periods and were not included as predictor variables. This approach helps prevent information from future periods from being used as input during model training.

3.4.4 Modelling

The modelling stage focuses on developing and comparing three machine-learning regression models for forecasting Malaysia's E&E export values: Random Forest (RF), Support Vector Regression (SVR), and XGBoost. The models were selected because they represent different machine-learning approaches and can capture nonlinear relationships between the engineered time-series features, selected economic indicators, and E&E export values.

Random Forest was selected as a bagging-based ensemble method that combines multiple decision trees and can capture nonlinear relationships and interactions between predictors. Support Vector Regression was selected as a kernel-based method that can model nonlinear relationships through the use of an appropriate kernel function. XGBoost was selected as a gradient-boosting ensemble method capable of learning complex nonlinear patterns while incorporating regularisation to control model complexity.

The use of these three models provides a comparative evaluation of different machine-learning approaches for short-term E&E export forecasting. All three models

are trained and evaluated using the same prepared dataset, forecasting horizons and evaluation framework to ensure a fair comparison of their predictive performance.

3.4.4.1 Model Selection

To ensure diversity of learning approaches and to capture the range of relationships, both linear and nonlinear, in the economic time-series data, three machine learning algorithms will be chosen.

- Support Vector Regression (SVR) was added as a kernel-based method that can be used for complex, nonlinear relationships in high dimensional feature space. SVR is especially useful in noisy economic data where the relationship between the macroeconomic indicators and the trade values is not necessarily linear.
- Random Forest Regressor was chosen as it is an ensemble model using the bootstrap aggregation (bagging) method, which has the following advantages:
 - It can model nonlinear interactions,
 - it is less prone to overfitting, and
 - it can provide feature importance scores, which can be used to interpret the key economic drivers of trade.
- XGBoost (Extreme Gradient Boosting), an iterative error correction-based ensemble learning method, was added because of its high predictive performance and efficiency in learning complex patterns, and its use of a sequential boosting approach.

The three models were carefully selected from different methodological families (kernel-based, bagging ensemble, boosting ensemble, respectively) to avoid any methodological bias, and to allow a thorough comparison of the performance of the models with respect to different learning strategies.

The three models can be used together to examine the various ways that economic relationships can be learned. Random Forest is robust against nonlinear

interactions and can be used for feature importance analysis, SVR is flexible to model the nonlinear interactions in a limited time-series data set in a reasonable time, and XGBoost is an optimized boosting framework that offers both good predictive power and regularization. These models can be combined to make a complete comparison of accuracy and suitability for the task of trade forecasting in Malaysia.

ARIMA was included as a statistical baseline model to provide a benchmark against which the machine-learning models could be compared. The ARIMA baseline used only the historical E&E export series and did not use the engineered machine-learning predictors or macroeconomic features. This allowed the study to determine whether the machine-learning approaches provided improvements over a univariate statistical forecasting approach. Deep-learning models such as LSTM were also not considered because the study focuses on a relatively limited number of monthly observations and aims to maintain a focused comparison of classical machine-learning approaches with different learning mechanisms. Therefore, the study concentrates on SVR, Random Forest and XGBoost as three complementary machine-learning approaches suitable for the available dataset and research objectives.

ARIMA forecasts were generated separately for the 1-month-, 2-month-, and 3-month-ahead forecasting horizons using only the historical E&E export series. The ARIMA model did not use the engineered lag, rolling, growth or macroeconomic predictor variables used by the machine-learning models. The resulting ARIMA forecasts were evaluated against the same corresponding test observations used for evaluating the machine-learning models, allowing the statistical baseline and machine-learning models to be compared under the same forecasting horizons and evaluation criteria.

Beyond their methodological diversity, RF, SVR and XGBoost were also chosen with the dataset's size and time-series characteristics specifically in mind. After chronological splitting and feature construction, the training set contains 326 monthly observations a sample size well suited to these three classical machine-learning approaches, which do not require the very large training samples typically needed for deep-learning models to avoid overfitting. Each of the three models also has

established mechanisms for controlling model complexity on moderate-sized datasets (bootstrap aggregation and feature randomness for Random Forest; margin-based regularisation for SVR; boosting-round and depth regularisation for XGBoost), making them appropriate for this study's data volume, while ARIMA provides a univariate benchmark that requires no exogenous predictors and is well established for monthly economic time series of this length.

3.4.4.2 Data Splitting Strategy

Since this study is of a time-series forecasting nature, a random train-test split was not employed as this might include information from future periods in the training set. Rather, an 80:20 split of the train and test sets was used, with the 80 subjecting the 20. 80% of observations were used for model training and validation, and 20% for the most recent observations were retained as an unseen test set.

This chronological splitting approach maintains the temporal sequence of the observations and more closely mimics a real-world forecasting problem, in which future values are to be forecast based upon the information from the previous time periods. The same chronological evaluation period was maintained for Random Forest, SVR and XGBoost to ensure a fair comparison between the models.

Concretely, the training partition covers the earliest 326 monthly observations, and the testing partition covers the most recent 82 monthly observations, held out entirely from training and hyperparameter tuning. An 80:20 split was selected as it is a widely used convention in applied forecasting studies that balances providing sufficient historical data for the models (including five-fold time-series cross-validation within the training set) against retaining a test period long enough (82 months, approximately seven years) to evaluate performance across varied economic conditions rather than a narrow or unrepresentative window.

Table 3:3 Data Splitting Strategy

Partition	Description
80%	Earliest 326 observations. Will be used to train models and for hyperparameter optimization using five-fold time-series cross validation. Identical to SVR, Random Forest, and XGBoost.
20%	Most recent 82 observations. Will be completely withheld from training. Employed as a sort of "blind test" for all three models to determine accuracy in real world deployment.

3.4.4.3 Model Training Process

Forecasting model was created for Electrical and Electronics (E&E) Exports. All the preprocessing, feature engineering, training, validation and evaluation procedures will be followed for to make all the experiments consistent and comparable.

To have a fair comparison between the three algorithms, the same preprocessed dataset with the same data split configuration will be used in their training. The training procedure, information on hyperparameters and information about each algorithm are given below.

Table 3:4 Model-Specific Training Procedures and Hyperparameter Optimization

Model	Training Approach	Hyperparameters Tuned
SVR	The SVR model was trained using standardised features because SVR is sensitive to differences in feature scale. An RBF kernel was used to capture nonlinear relationships between the	C, epsilon (ϵ), gamma (γ) and kernel

	engineered time-series features and E&E export values. Hyperparameter tuning was performed using 5-fold cross-validation on the training data. Separate models were developed for the 1-month, 2-month and 3-month forecasting horizons.	
Random Forest	The Random Forest Regressor was trained as an ensemble of decision trees using the same training data and forecasting features. Tree-based models don't care about the magnitude of features, and didn't require feature scaling. 5-fold cross validation was used for hyper parameter tuning. Separate models were developed for the 1-month, 2-month and 3-month forecasting horizons.	n_estimators, max_depth, min_samples_split, min_samples_leaf
XGBoost	The XGBoost model was trained with the gradient-boosted decision tree, which learned the nonlinearity and	n_estimators, learning_rate, max_depth, subsample, colsample_bytree

	interaction between the engineered features. The Hyperparameter tuning was done by 5-fold cross validation on the training data.	
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Hyperparameter search ranges were as follows: for Random Forest, the number of trees (200, 300, 500), maximum depth (None, 10, 20), minimum samples to split (2, 5) and minimum samples per leaf (1, 2) were searched; for SVR, C (10, 100, 500, 1000), gamma ('scale', 0.001, 0.01, 0.1) and epsilon (0.01, 0.1, 0.5) were searched; for XGBoost, the number of boosting rounds (200, 300, 500), maximum depth (3, 5, 7), learning rate (0.03, 0.05, 0.1), row subsampling (0.8, 1.0) and column subsampling (0.8, 1.0) were searched. The `random_state = 42` argument was included in all models to ensure their repeatability. Hyper parameter optimization was done with five-fold time series cross validation to select the configuration with the lowest cross validated RMSE. For the ARIMA baseline, the order (p, d, q) was determined by fitting six specifications of the model to the training series, and choosing the one with the lowest Akaike Information Criterion (AIC); (2,1,2) was the chosen order.

Hyperparameter optimization was performed separately for each forecasting horizon to select the hyperparameter configuration that gave the lowest cross-validation RMSE. For Random Forest, 36 parameter combinations were evaluated for each forecasting horizon, while 48 combinations were evaluated for SVR and 108 combinations were evaluated for XGBoost. A total of 180, 240 and 540 model fits were therefore performed for each forecasting horizon for Random Forest, SVR and XGBoost, respectively. The best-performing configurations identified through cross-validation were subsequently used to train the final models before evaluation on the held-out chronological test set.

3.4.4.4 Model Evaluation Metrics

Once the models have been trained, the most recent 20% of observations will be used as the unseen test set for evaluating the forecasting performance of ARIMA (Baseline Model), Support Vector Regression (SVR), Random Forest (RF), and XGBoost. The same chronological test period will be used for all models to ensure a fair comparison.

Model performance will be evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

Table 3:5 Performance Evaluation Criteria

Metric	What it measures	Interpretation	Equation
	Average absolute difference between actual and forecast values.	Lower values indicate smaller average forecasting errors.	$\frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $
MSE	Average squared difference between actual and forecast values.	Lower values indicate smaller errors, while larger errors receive greater penalty.	$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$
RMSE	Square root of MSE, expressed in the same unit as the E&E export target.	Lower values indicate better forecasting accuracy and provide an interpretable	$\sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$

		measure of forecast error.	
R ²	Proportion of variation in E&E export values explained by the model.	Higher values indicate better explanatory performance, with values closer to 1 indicating stronger fit.	$1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2}$

Model performance will be evaluated separately for the 1-month-, 2-month-, and 3-month-ahead forecasting horizons. The same evaluation metrics will be applied across all models and forecasting horizons to support consistent comparison.

In addition to the quantitative metrics, actual-versus-predicted plots and residual analysis will be used to visually examine forecasting performance and identify potential systematic errors, underfitting or overfitting. The consistency between validation and test performance will also be considered when assessing model generalisation.

When the evaluation metrics produce different model rankings, greater emphasis will be placed on RMSE because it penalises larger forecasting errors and expresses the error in the same units as the E&E export target. MAE, MSE and R² will be considered as supporting measures. The model demonstrating the most suitable overall performance will subsequently be selected to generate the three-month-ahead E&E export forecasts.

RMSE was designated as the primary metric for final model selection because it is expressed in the same unit as the E&E export target (RM million), making it directly interpretable, and because it penalises large forecasting errors more heavily than MAE a desirable property given the practical cost of large planning errors. Unlike MAE, MSE and RMSE, R² is not bounded below by zero: a negative R² occurs when

a model's sum of squared errors exceeds the sum of squared deviations of the actual values from their mean, meaning the model performs worse than simply predicting the historical average for every observation. This can arise when a model's errors are large or systematically biased in a particular direction over the test period, and is reported and interpreted for each model in Chapter 6.

3.4.5 Evaluation

The evaluation stage assesses the forecasting performance and practical relevance of the developed models. The evaluation considers both quantitative forecasting accuracy and the ability of the models to represent the overall movement of Malaysia's E&E export values. The results from the evaluation stage are used to identify the most suitable model for generating the final three-month-ahead forecasts.

3.4.5.1 Technical Performance Evaluation

The technical performance of ARIMA, Support Vector Regression (SVR), Random Forest (RF), and XGBoost will be evaluated using MAE, MSE, RMSE and R^2 . These metrics provide complementary information about forecasting error and model performance. MAE represents the average magnitude of forecasting errors, while MSE and RMSE give greater emphasis to larger errors. RMSE is particularly useful for practical interpretation because it is expressed in the same unit as the E&E export values. R^2 provides an additional measure of the proportion of variation in E&E exports explained by the model.

The models will be compared separately across the 1-month-, 2-month-, and 3-month-ahead forecasting horizons. The model with the strongest overall performance will be identified based on the evaluation results, with greater emphasis placed on RMSE when differences occur between the evaluation metrics.

3.4.5.2 Business Relevance Assessment

The business relevance of the forecasting models will be assessed based on their potential usefulness in monitoring and anticipating Malaysia's E&E export performance. In addition to the quantitative evaluation metrics, actual and predicted E&E export values will be compared using time-series visualisations to assess whether the models can reasonably capture the overall trends and fluctuations in export performance.

The selected model will then be used to generate the three-month-ahead E&E export forecasts based on the most recent available information. The forecast results will subsequently be presented through the Power BI dashboard to support the interpretation and communication of Malaysia's expected short-term E&E export performance.

As no formal stakeholder consultation or expert evaluation was conducted as part of this study, the following should be understood as an interpretation of the forecasts' potential relevance to each stakeholder group rather than a formally validated business assessment. For government agencies and policymakers, the forecasts may help flag anticipated declines or surges in E&E export performance early enough to inform policy discussion. For E&E exporters and businesses, the one-to-three-month horizon aligns with typical production and inventory planning cycles, potentially supporting more informed short-term operational decisions. For financial institutions and market analysts, the forecasts and associated model-performance information may provide supplementary context when assessing sector-level risk and trade exposure, alongside their own analysis.

3.4.5.3 Process Review and Iteration

Model development followed an iterative process in which initial model performance was evaluated and hyperparameters were subsequently tuned using time-series cross-validation. Feature engineering and model configuration were reviewed based on validation and test performance. The final best performing models were then

retrained using the complete historical modelling dataset before generating the future three-month forecast.

3.4.5.4 Determining Next Steps

Based on the comparative evaluation, the best-performing model was identified separately for each forecasting horizon. The selected final model was subsequently retrained using the complete historical modelling dataset and used to generate forecasts for April, May and June 2025 based on information available up to March 2025.

The final model-selection rule adopted in this study distinguishes between the best-performing model overall and the best-performing machine-learning model. Although ARIMA achieved the strongest accuracy overall and is reported as such, the operational three-month-ahead forecasts presented through the dashboard are generated using XGBoost, the best-performing machine-learning model among the three candidates evaluated. This choice reflects the study's core objective (RO2) of developing and evaluating a machine-learning forecasting system specifically, with ARIMA serving as the statistical benchmark against which that system's performance is assessed, rather than as a candidate for operational deployment within the proposed machine-learning framework.

3.4.6 Deployment

3.4.6.1 Dashboard Development and Visualisation

The forecasts were presented through an interactive Power BI dashboard that serves two main purposes: (1) visualisation of historical Electrical and Electronics (E&E) export performance, and (2) presentation and interactive exploration of the generated forecast results. The completed dashboard, presented in Section 5.8, incorporates summary key performance indicator (KPI) cards, an interactive forecast-horizon slicer, model-performance visualisations, an XGBoost feature-importance

chart, and supporting macroeconomic trend charts. The main dashboard functionalities are summarised in Table 3:5.

Table 3:6 Dashboard Functionalities for Historical and Forecast Data Visualisation

Dashboard Function	Description
Historical Visualisation	Presents the historical monthly E&E export values through a time-series visualisation, allowing users to observe changes and trends in Malaysia's E&E export performance over the historical period. Supporting trend charts for CPI inflation, MYR/USD exchange rate, and the U.S. Industrial Production Index are also provided to provide additional economic context.
Forecast Result Presentation	Presents the pre-generated E&E export forecasts produced by the trained Python machine learning models. The XGBoost forecast values for the 1-month, 2-month, and 3-month horizons are imported into Power BI and displayed interactively.
Model Performance Comparison	Displays the RMSE and R^2 values for Random Forest, SVR and XGBoost, together with ARIMA as the statistical baseline, across the different forecasting horizons, allowing users to compare the predictive performance of the three machine-learning models against the traditional statistical benchmark. (Note: the actual ARIMA performance results are presented and discussed in Chapter 6.)
Feature Importance Analysis	Presents the top five features contributing to the XGBoost predictions. The feature-importance visualisation can be examined according to the selected forecasting horizon, allowing users to identify the most influential historical and macroeconomic-related features for each horizon.
Interactive Forecast-	Provides a forecast-horizon slicer that allows users to select the 1-month, 2-month, or 3-month forecasting horizon. The

Horizon	selected horizon dynamically updates the relevant forecast,
Analysis	model-performance, and feature-importance visualisations.

The dashboard therefore functions as an interactive analytical interface for presenting and exploring the forecasting results, rather than as the environment in which the machine learning models are trained. The forecasting process was conducted entirely in Python, where the data preparation, feature engineering, model training, tuning, evaluation, and forecast generation were performed. The resulting forecast values, model-performance metrics, and feature-importance results were subsequently extracted and imported into Power BI for visualisation.

The interactive capabilities of Power BI allow users to examine the generated results according to the selected forecasting horizon without directly interacting with the underlying Python models or code. In particular, the forecast-horizon slicer enables users to switch between the 1-month, 2-month, and 3-month horizons, while the corresponding dashboard visualisations update to present the relevant forecast results, model-performance measures, and XGBoost feature-importance values. Thus, Power BI serves primarily as the visualisation and interactive reporting layer, while Python remains responsible for the underlying forecasting process.

3.4.6.2 Monitoring and Maintenance Strategy

A monitoring and maintenance plan was devised for the sustainability of the prediction system in terms of its continued efficacy after implementation. Due to the ever-changing nature of the data being used by the machine learning models, there will be some decline in their performances because of changes in the economy and other unforeseen situations around the world. The performance of each model will therefore be evaluated periodically according to key performance indicators such as MSE, MAE, RMSE, and R. If the system were deployed in a real-world environment, model performance could be monitored periodically and retraining could be considered when substantial deterioration in forecasting performance or changes in the underlying data patterns are observed.

3.4.6.3 Documentation and Reporting

The documentation phase will integrate all the methodology, experimentation, and findings in a reproducible framework. The whole workflow ranging from data acquisition and cleaning to feature engineering, modeling, evaluation, and deployment is systematically documented. Performance measures of the three models viz., SVR, Random Forest, and XGBoost models are quantitatively evaluated and compared through visualization. Finally, conclusions are made based on the findings, and suggestions are made regarding improvements in terms of feature engineering, adding more macroeconomic factors, and a periodic retraining policy.

3.5 Chapter Summary

This chapter presented the research methodology adopted for developing a machine-learning-based forecasting system for Malaysia's E&E exports, structured according to the six phases of the CRISP-DM framework. The chapter described the research design and rationale for using monthly data over the January 1990 – March 2025 period; the data sources, cleaning, integration and feature-engineering procedures used to prepare the dataset; the selection and training of Random Forest, SVR and XGBoost, benchmarked against ARIMA as a statistical baseline; the chronological data-splitting and hyperparameter-optimisation strategy; and the evaluation metrics (MAE, MSE, RMSE and R^2) used to compare model performance. Finally, the chapter described the deployment of results through an interactive Power BI dashboard. The next chapter presents the results of the exploratory data analysis and data-preparation stages described here.

CHAPTER 4

EXPLORATORY DATA ANALYSIS

4.1 Introduction

This chapter presents the results of the exploratory data analysis and data preparation conducted for the forecasting of Malaysia's Electrical and Electronics (E&E) exports. The analysis examines the characteristics, trends, patterns and relationships within the integrated dataset before the development and evaluation of the forecasting models.

The chapter first presents an overview of the original and final datasets, followed by the results of the data quality assessment and preprocessing procedures. Exploratory analysis is then performed to analyse the historical patterns of E&E exports and the selected macroeconomic indicators such as the MYR/USD exchange rate, U.S. Industrial Production Index and inflation. Key economic upheavals such as the Global Financial Crisis of 2008 and the COVID-19 pandemic also have been taken into account.

In addition, the chapter presents the results of feature construction, including lag features, rolling statistics, growth rates and time-based features. The relationships between E&E exports and the selected predictor variables are examined using correlation analysis. Finally, the resulting feature set and feature importance analysis are presented to provide an understanding of the information used by the forecasting models in the subsequent chapters.

4.2 Dataset Overview

4.2.1 Data Sources

The dataset used in this study was constructed by integrating monthly Electrical and Electronics (E&E) export data with selected macroeconomic indicators. The data were obtained from official and publicly available sources to ensure the reliability and consistency of the information used for forecasting.

The monthly E&E export data were obtained from the Ministry of Economy Malaysia External Trade Statistics portal. The dataset represents the value of Malaysia's Electrical and Electronics (E&E) exports and serves as the target variable for the forecasting models. The E&E export data covered the period from January 1990 to March 2025.

The macroeconomic variables used were Malaysia's CPI inflation, MYR/USD exchange rate and U.S. Industrial Production Index (IPI). CPI inflation data are from the Department of Statistics Malaysia (DOSM) and the exchange rate MYR/USD and U.S. Industrial Production Index are from the Federal Reserve Bank of St. Louis (FRED). These variables were selected because they represent domestic price conditions, exchange-rate movements and external industrial activity that may be associated with changes in Malaysia's E&E export performance.

The datasets were obtained at monthly frequency and subsequently standardized using a common monthly date index before integration. The resulting integrated dataset provided the basis for the exploratory analysis, feature construction and subsequent machine-learning forecasting process.

Table 4:1 Data Sources

Variable	Description	Source Organisation	Access Method	Study Period	Access Date	Frequency	Unit	Role
----------	-------------	---------------------	---------------	--------------	-------------	-----------	------	------

E&E Exports	Malaysia's E&E export value	Ministry of Economy Malaysia – External Trade Statistics	Public portal download	Jan 1990 – Mar 2025	March 2025	Monthly	RM million	Target
CPI Inflation	Malaysia's inflation rate	Department of Statistics Malaysia (DOSM)	Public portal download (OpenDOSM)	Jan 1990 – Mar 2025	March 2025	Monthly	% change	Predictor
MYR/USD Exchange Rate	MYR to USD exchange rate	Federal Reserve Bank of St. Louis (FRED)	Public download via FRED	Jan 1990 – Mar 2025	March 2025	Monthly	MYR per USD	Predictor
U.S. Industrial Production Index	U.S. industrial output index	Federal Reserve Bank of St. Louis (FRED)	Public download via FRED	Jan 1990 – Mar 2025	March 2025	Monthly	Index (2017=100)	Predictor

A sample of the E&E export dataset in its original form is shown below, illustrating its structure prior to integration:

SITC 3		1990											
DIGIT	DESCRI	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
77	Electric al Machinery, Apparatus And Appliances, N.E.S. And Electric al Parts Thereof	1,010	1,035	1,249	1,026	1,223	1,273	1,191	1,343	1,197	1,180	1,263	1,177

Figure 4:1 Original E&E Export Dataset

4.3 Dataset Collecting and Loading

The dataset was collected from multiple data sources and imported into the Python environment for subsequent data preprocessing, transformation, integration, and analysis.

```
import pandas as pd

# Import ER.csv
er_df = pd.read_csv('/content/ER.csv')

# Import CPI Inflation.csv
cpi_inflation_df = pd.read_csv('/content/CPI Inflation.csv')

# Import IPI.csv
ipi_df = pd.read_csv('/content/IPI.csv')

# Import E&E Exports.xlsx
ee_exports_df = pd.read_excel('/content/E&E Exports.xlsx')

print("Files imported successfully!")
```

Files imported successfully!

Figure 4:2 Importing files

4.3.1 Initial Inspection for Each Variables

An initial inspection was conducted for each variable to assess the structure, data type, time coverage, frequency, and completeness of the collected datasets before further preprocessing. The inspection involved examining the date column and recorded values for the E&E export series and the selected macroeconomic variables, namely inflation, exchange rate, and U.S. Industrial Production Index. The number of observations, and consistency of the monthly time-series structure were also checked. This initial inspection helped any inconsistencies in the data structure that required further preprocessing before the datasets could be integrated.

```

print("\n--- E&E Exports DataFrame Initial Inspection ---")
print(ee_exports_df.head())
print(ee_exports_df.columns)
print("-----")

```

```

--- E&E Exports DataFrame Initial Inspection ---
SITC 3 DIGIT      DESCRIPTIONS \
0      NaN      NaN
1      77.0  Electrical Machinery, Apparatus And Appliances...

      1990  Unnamed: 3  Unnamed: 4  Unnamed: 5  Unnamed: 6  Unnamed: 7 \
0      Jan      Feb      Mar      Apr      May      Jun
1  1010.235516  1034.73632  1249.33347  1025.820057  1223.404799  1273.317912

      Unnamed: 8  Unnamed: 9  ...  Unnamed: 415  Unnamed: 416  Unnamed: 417 \
0      Jul      Aug      ...      Jun      Jul      Aug
1  1190.580015  1342.993456  ...  41083.016724  39357.590921  39863.748126

      Unnamed: 418  Unnamed: 419  Unnamed: 420  Unnamed: 421      2025 \
0      Sep      Oct      Nov      Dec      Jan
1  44006.856395  39226.126364  39795.67259  47002.306925  40494.054767

      Unnamed: 423  Unnamed: 424
0      Feb      Mar
1  36504.760024  45813.712073

[2 rows x 425 columns]
Index(['SITC 3 DIGIT', 'DESCRIPTIONS',      1990, 'Unnamed: 3',
      'Unnamed: 4', 'Unnamed: 5', 'Unnamed: 6', 'Unnamed: 7',
      'Unnamed: 8', 'Unnamed: 9',
      ...,
      'Unnamed: 415', 'Unnamed: 416', 'Unnamed: 417', 'Unnamed: 418',
      'Unnamed: 419', 'Unnamed: 420', 'Unnamed: 421',      2025,
      'Unnamed: 423', 'Unnamed: 424'],
      dtype='object', length=425)
-----

```

Figure 4:3 Initial inspection of the E&E export dataset showing its original wide-format structure and column names

The CPI inflation, exchange rate and U.S. IPI datasets were inspected using the same procedure; each was confirmed to contain a single date column and a single value column at monthly frequency, with no structural inconsistencies requiring correction before transformation.

4.4 Data Preprocessing Results

The table below summarises how the number of observations changed across each preprocessing stage, from the four raw source files through to the final integrated dataset:

Table 4:2 Dataset Observations Before and After Pre-processing

Stage	Observations	Notes
Raw source files (each)	423	Before transformation
After individual transformation	423 each	No observations lost; format-only changes
After merging/integration	423	No observations lost during merge
After missing-value and duplicate checks	423	0 missing values, 0 duplicates found

4.4.1 Data Cleaning Results

Table 4.x below summarises the cleaning outcome for each of the four datasets. In every case, the cleaning process altered only the format and structure of the data (reshaping, renaming, date standardisation) rather than the underlying numeric values themselves no values were modified, imputed or removed during this stage.

Table 4:3 Data Cleaning Results

Dataset	Observations Before	Observations After	Values Changed?	Format Changed?
E&E Exports	423	423	No	Yes, reshaped from wide to long format
CPI Inflation	423	423	No	Yes, irrelevant columns dropped, date standardised

Exchange Rate	423	423	No	Yes, column names and date field standardised
U.S. IPI	423	423	No	Yes, date field standardised

4.4.1.1 Data Transformation

Data transformation was performed to standardise the variables into a consistent monthly time-series format before integrating them for further analysis and modelling. The transformed dataset contains monthly observations of E&E exports, CPI inflation, exchange rate, and the U.S. Industrial Production Index, with the observation period extending from January 1990 to March 2025. The resulting integrated dataset contains 423 monthly observations. The variables were also renamed to consistent format for easy readability.

The E&E export series was maintained at a monthly frequency, with each observation representing the corresponding monthly export value. The date variable was converted into a standard datetime format to ensure that the observations could be correctly ordered chronologically and used for subsequent time-series feature construction.

```

import pandas as pd
data_start_idx = 2
data_columns_headers = ee_exports_df.columns[data_start_idx:]

years_list = []
current_year = None
for col_header in data_columns_headers:
    # Check if the column header is a year (e.g., '1990', '2025')
    if isinstance(col_header, int) or (isinstance(col_header, str) and col_header.isdigit() and len(col_header) == 4):
        current_year = str(col_header)
        years_list.append(current_year)
years_filled = pd.Series(years_list).ffill().tolist()
months_row_data = ee_exports_df.iloc[0, data_start_idx:]
export_values_data = ee_exports_df.iloc[1, data_start_idx:]
combined_dates = []
for year, month in zip(years_filled, months_row_data):
    if pd.notna(year) and pd.notna(month):
        combined_dates.append(f"{year}-{str(month).strip()}")
    else:
        combined_dates.append(None)
valid_indices = [i for i, date_str in enumerate(combined_dates) if date_str is not None]

ee_exports_transformed_df = pd.DataFrame({
    'Date': [combined_dates[i] for i in valid_indices],
    'E&E Exports': [export_values_data.iloc[i] for i in valid_indices]
})
ee_exports_transformed_df['Date'] = pd.to_datetime(ee_exports_transformed_df['Date'], format='%Y-%b')
ee_exports_transformed_df = ee_exports_transformed_df.sort_values('Date').reset_index(drop=True)
ee_exports_df = ee_exports_transformed_df

```

Figure 4:4 Transformation of the E&E export dataset from the original wide format into a chronological monthly time-series format

The CPI Inflation, Exchange Rate, and U.S. Industrial Production Index were incorporated into the same monthly structure.

```

import pandas as pd
cpi_inflation_transformed_df = cpi_inflation_df.copy()
cpi_inflation_transformed_df = cpi_inflation_transformed_df.rename(columns={'date': 'Date'})
cpi_inflation_transformed_df['Date'] = pd.to_datetime(cpi_inflation_transformed_df['Date'])
if 'division' in cpi_inflation_transformed_df.columns:
    cpi_inflation_transformed_df = cpi_inflation_transformed_df[cpi_inflation_transformed_df['division'] == 'overall']
start_date = '1990-01-01'
end_date = '2025-03-01'
cpi_inflation_transformed_df = cpi_inflation_transformed_df[
    (cpi_inflation_transformed_df['Date'] >= start_date) &
    (cpi_inflation_transformed_df['Date'] <= end_date)
].reset_index(drop=True)
cpi_inflation_transformed_df = cpi_inflation_transformed_df.drop_duplicates(subset=['Date']).reset_index(drop=True)
cpi_inflation_transformed_df = cpi_inflation_transformed_df.rename(columns={'inflation_mom': 'CPI Inflation'})
cpi_inflation_transformed_df = cpi_inflation_transformed_df.drop(columns=['division', 'inflation_yoy'], errors='ignore')
cpi_inflation_df = cpi_inflation_transformed_df
print("Transformed 'CPI Inflation' DataFrame head:")
display(cpi_inflation_df.assign(Date=cpi_inflation_df['Date'].dt.strftime('%Y-%m')).head())

```

Figure 4:5 Transformed CPI inflation dataset after selecting the overall division, retaining the monthly inflation measure and standardising the date variable.

```

import pandas as pd
er_transformed_df = er_df.copy()
er_transformed_df = er_transformed_df.rename(columns={'observation_date': 'Date'})
er_transformed_df = er_transformed_df.rename(columns={'EXMAUS': 'Exchange Rate'})
er_transformed_df['Date'] = pd.to_datetime(er_transformed_df['Date'])
er_df = er_transformed_df
print("Transformed 'Exchange Rate' DataFrame head:")
display(er_df.head())

```

Figure 4:6 Transformed exchange rate dataset after standardising the date and exchange rate variable names.

```
import pandas as pd
ipi_transformed_df = ipi_df.copy()
ipi_transformed_df = ipi_transformed_df.rename(columns={'observation_date': 'Date'})
ipi_transformed_df = ipi_transformed_df.rename(columns={'INDPRO': 'US Industrial Production Index'})
ipi_transformed_df['Date'] = pd.to_datetime(ipi_transformed_df['Date'])
ipi_df = ipi_transformed_df
print("Transformed 'US Industrial Production Index' DataFrame head:")
display(ipi_df.head())
```

Figure 4:7 Transformed U.S. Industrial Production Index dataset after standardising the date and variable names

Four specific transformations were applied during this stage:

1. Reshaping (E&E Exports only): the original dataset was provided in wide format, with separate columns for each month within a year; this was reshaped ("melted") into a long format with a single Date column and a single export-value column, enabling it to be treated as a standard monthly time series.
2. Column selection (CPI Inflation): the original file contained inflation broken down by expenditure division (e.g. food, transport, housing); only the overall/composite CPI inflation column was retained for this study.
3. Renaming: all four datasets had their date and value columns renamed to a consistent naming convention (e.g. Date, E&E_Exports, CPI_Inflation) to simplify subsequent merging.
4. Date standardisation: all date fields, which were originally represented in differing formats across the four sources, were converted to a common datetime type at monthly resolution to allow consistent chronological ordering and merging.

4.4.1.2 Missing Value Treatment

Missing-value checks were performed after the individual dataset transformations.

Table 4:4 Missing Values in the Integrated Dataset

Variable	Observations	Missing Values
E&E Exports	423	0
CPI Inflation	423	0
Exchange Rate	423	0
U.S. Industrial Production Index	423	0

```

null_values_in_ee_exports = ee_exports_df.isnull().sum()
print("\nNull values in the transformed 'E&E Exports' DataFrame:")
print(null_values_in_ee_exports)

```

Figure 4:8 Missing Value Assessment of the E and E Exports

```

null_values_in_cpi_inflation = cpi_inflation_df.isnull().sum()
print("\nNull values in the transformed 'CPI Inflation' DataFrame:")
print(null_values_in_cpi_inflation)

```

Figure 4:9 Missing Value Assessment of the CPI Inflation

```

null_values_in_er_ = er_df.isnull().sum()
print("\nNull values in the transformed 'CPI Inflation' DataFrame:")
print(null_values_in_er_)

```

Figure 4:10 Missing Value Assessment of Exchange Rate

```

null_values_in_ipi_ = ipi_df.isnull().sum()
print("\nNull values in the transformed 'CPI Inflation' DataFrame:")
print(null_values_in_ipi_)

```

Figure 4:11 Missing Value Assessment of the Industrial Production Index

The results showed that the four prepared datasets contained no missing values in their respective analytical variables.

Therefore, no missing-value imputation was required at the dataset integration stage. This provided a complete monthly dataset for the subsequent exploratory analysis and feature engineering procedures.

4.4.1.3 Duplicate Assessment

A duplicate-value check was performed on the individual analytical variables.

```
duplicates_in_ee_exports = ee_exports_df.duplicated().sum()
print(f"\nNumber of duplicate rows in 'E&E Exports' DataFrame: {duplicates_in_ee_exports}")
if duplicates_in_ee_exports > 0:
    print("Duplicate rows found in 'E&E Exports' DataFrame:")
    display(ee_exports_df[ee_exports_df.duplicated()])
duplicates_in_date_column = ee_exports_df.duplicated(subset=['Date']).sum()
print(f"\nNumber of duplicate 'Date' entries in 'E&E Exports' DataFrame: {duplicates_in_date_column}")
if duplicates_in_date_column > 0:
    print("Rows with duplicate 'Date' entries in 'E&E Exports' DataFrame:")
    display(ee_exports_df[ee_exports_df.duplicated(subset=['Date'])])
```

Figure 4:12 Duplicate check for E and E exports

```
duplicates_in_cpi_inflation = cpi_inflation_df.duplicated().sum()
print(f"\nNumber of duplicate rows in 'CPI Inflation' DataFrame: {duplicates_in_cpi_inflation}")
if duplicates_in_cpi_inflation > 0:
    print("Duplicate rows found in 'CPI Inflation' DataFrame:")
    display(cpi_inflation_df[cpi_inflation_df.duplicated()])
duplicates_in_date_column_cpi = cpi_inflation_df.duplicated(subset=['Date']).sum()
print(f"\nNumber of duplicate 'Date' entries in 'CPI Inflation' DataFrame: {duplicates_in_date_column_cpi}")
if duplicates_in_date_column_cpi > 0:
    print("Rows with duplicate 'Date' entries in 'CPI Inflation' DataFrame:")
    display(cpi_inflation_df[cpi_inflation_df.duplicated(subset=['Date'])])
```

Figure 4:13 Duplicate check for CPI Inflation

```
duplicates_in_er_df = er_df.duplicated().sum()
print(f"\nNumber of duplicate rows in 'Exchange Rate' DataFrame: {duplicates_in_er_df}")
if duplicates_in_er_df > 0:
    print("Duplicate rows found in 'Exchange Rate' DataFrame:")
    display(er_df[er_df.duplicated()])
duplicates_in_date_column_er = er_df.duplicated(subset=['Date']).sum()
print(f"\nNumber of duplicate 'Date' entries in 'Exchange Rate' DataFrame: {duplicates_in_date_column_er}")
if duplicates_in_date_column_er > 0:
    print("Rows with duplicate 'Date' entries in 'Exchange Rate' DataFrame:")
    display(er_df[er_df.duplicated(subset=['Date'])])
```

Figure 4:14 Duplicate check for Exchange Rate

```

duplicates_in_ipi_df = ipi_df.duplicated().sum()
print(f"\nNumber of duplicate rows in 'US Industrial Production Index' DataFrame: {duplicates_in_ipi_df}")
if duplicates_in_ipi_df > 0:
    print("Duplicate rows found in 'US Industrial Production Index' DataFrame:")
    display(ipi_df[ipi_df.duplicated()])
duplicates_in_date_column_ipi = ipi_df.duplicated(subset=['Date']).sum()
print(f"\nNumber of duplicate 'Date' entries in 'US Industrial Production Index' DataFrame: {duplicates_in_date_column_ipi}")
if duplicates_in_date_column_ipi > 0:
    print("Rows with duplicate 'Date' entries in 'US Industrial Production Index' DataFrame:")
    display(ipi_df[ipi_df.duplicated(subset=['Date'])])

```

Figure 4:15 Duplicate check for U.S. Industrial Production Index

The results indicated that E&E Exports, CPI inflation, exchange rate and U.S. IPI contained no duplicated values.

4.4.2 Data Integration Results

After the individual datasets had been cleaned and standardised, they were integrated using the monthly date as the common reference key. A temporary Date_YM variable in YYYY-MM format was created for each dataset to ensure that observations from the same month were correctly aligned.

The E&E export dataset was used as the starting dataset and subsequently combined with CPI inflation, exchange rate and U.S. IPI data. After integration, the temporary Date_YM variable was removed, the date was converted into a datetime index, and the observations were sorted chronologically.

```

# Perform merges
# Start with E&E Exports
merged_df = ee_exports_df.copy()
# Merge with CPI Inflation
merged_df = pd.merge(merged_df, cpi_inflation_df[['Date_YM', 'CPI Inflation']], on='Date_YM', how='outer')
# Merge with Exchange Rate
merged_df = pd.merge(merged_df, er_df[['Date_YM', 'Exchange Rate']], on='Date_YM', how='outer')
# Merge with Industrial Production Index
merged_df = pd.merge(merged_df, ipi_df[['Date_YM', 'US Industrial Production Index']], on='Date_YM', how='outer')
merged_df['Date'] = pd.to_datetime(merged_df['Date_YM'])
merged_df = merged_df.drop(columns=['Date_YM'])
merged_df = merged_df.sort_values('Date').set_index('Date')
print("Merged DataFrame Head:")
# Create a temporary DataFrame for display with the index formatted as 'YYYY-MM'
display_df_head = merged_df.head().copy()
display_df_head.index = display_df_head.index.strftime('%Y-%m')
display(display_df_head)

```

Figure 4:16 Merging Dataset

Also, it was checked if the merged dataset contained any missing values and checked for duplicates to ensure if any inconsistencies was resulted but there was no missing values and duplicates found.

```

null_values_in_merged_=merged_df.isnull().sum()
print(null_values_in_merged_)

```

Figure 4:17 Null value checking in merged dataset

Missing-value assessment of the integrated dataset showing zero missing values across all analytical variables.

```

for column in merged_df.columns:
    if merged_df[column].duplicated().any():
        print(f" Column '{column}' has duplicate values.")
    else:
        print(f" Column '{column}' has no duplicate values.")

```

Figure 4:18 Duplicate checking in merged dataset

Duplicate observation assessment of the integrated dataset showing no duplicated rows or monthly date entries.

The resulting integrated dataset contained 423 monthly observations from January 1990 to March 2025 and four analytical variables.

Table 4:5 Integrated Dataset

Variable	Description	Observations
E&E Exports	Malaysia's monthly E&E export value	423
CPI Inflation	Monthly CPI inflation	423
Exchange Rate	MYR/USD exchange rate	423
U.S. Industrial Production Index	U.S. industrial production index	423

Each of the four datasets contained 423 monthly observations immediately prior to merging. After the successive left-join operations on the common Date_YM key, the resulting integrated dataset also contained 423 monthly observations — confirming that no observations were lost due to unmatched dates during integration, since all four sources shared an identical January 1990 – March 2025 coverage period at monthly frequency.

The integration results confirmed that all four variables were successfully aligned at a monthly frequency and that no missing values were introduced during the integration process. The resulting dataset therefore provided the basis for the exploratory data analysis and subsequent feature construction.

4.4.3 Original Integrated Dataset

The original integrated dataset consisted of 423 monthly observations covering the period from January 1990 to March 2025. The dataset was obtained by integrating the E&E export series with the three selected macroeconomic indicators using the monthly date as the common reference key.

The E&E export variable represents the target variable of the study, while inflation, the MYR/USD exchange rate and the U.S. Industrial Production Index represent the initial explanatory variables. At this stage, the dataset represented the raw integrated data before the construction of lagged, rolling, growth-rate and time-based features required for the machine-learning models.

The 35-year observation period provided a broad historical representation of Malaysia's E&E export performance across different economic conditions. It included periods of economic expansion and contraction, the 2008 Global Financial Crisis, the COVID-19 pandemic, and the subsequent economic recovery. These periods provided variation in the E&E export series for subsequent exploratory analysis and forecasting model development.

Table 4:6 Original Integrated Dataset

Item	Description
Target variable	E&E Exports
Macroeconomic variables	Inflation, MYR/USD Exchange Rate, U.S. Industrial Production Index
Frequency	Monthly
Start date	January 1990
End date	March 2025
Number of observations	423
Initial dataset type	Multivariate time-series dataset

4.4.4 Dataset Structure and Variables

The original integrated dataset was structured as a multivariate monthly time-series dataset, where each row represented one monthly observation and each column represented the target variable or an explanatory variable. The Date variable served as the common temporal index for integrating the datasets and maintaining the chronological order of observations.

The integrated dataset has dimensions of 423 rows $\times$ 5 columns (one Date column plus four analytical variables), covering January 1990 to March 2025. A sample of the first and last three rows is shown below:

Table 4:7 First and Last 3 Row of the Dataset

Date	E&E Exports	CPI Inflation	Exchange Rate	US Industrial Production Index
01-01-1990	1010.235516	0.7	2.7041	61.729

01-02-1990	1034.73632	0.5	2.7032	62.2896
01-03-1990	1249.33347	-0.2	2.717	62.5999
01-01-2025	40494.05477	0.1	4.462	100.0647
01-02-2025	36504.76002	0.4	4.4408	101.0993
01-03-2025	45813.71207	0	4.4325	101.0404

Because all four source variables were merged on a common Date_YM (year-month) key rather than an exact calendar date, every row of the integrated dataset is confirmed to represent the same calendar month across all four variables — i.e., the E&E export, inflation, exchange-rate and IPI values in any given row all correspond to the identical month.

The target variable was Malaysia's E&E export value, while the initial predictor variables consisted of CPI inflation, the MYR/USD exchange rate and the U.S. Industrial Production Index. These variables provided information on domestic price conditions, exchange-rate movements and external industrial activity, respectively.

Before feature engineering, the integrated dataset therefore contained the original target and macroeconomic variables. Additional predictors were subsequently generated from the historical E&E export series and the selected macroeconomic variables through lag construction, rolling statistics, growth-rate calculation and time-based feature construction. The resulting feature set was then used for the supervised machine-learning forecasting models.

Table 4:8 Structure of Original Dataset

Variable	Description	Unit	Data Type	Role	Source
Date	Monthly observation date	YYYY-MM	Date/Time	Time index	—
E&E Exports	Malaysia's monthly E&E export value	RM million	Numerical	Target	Ministry of Economy Malaysia
CPI Inflation	Monthly inflation indicator	% change	Numerical	Predictor	DOSM
MYR/USD Exchange Rate	Exchange rate between MYR and USD	MYR per USD	Numerical	Predictor	FRED
U.S. Industrial Production Index	Indicator of U.S. industrial activity	Index (2017=100)	Numerical	Predictor	FRED

4.4.5 Dataset Transformation and Modelling Sequence

The original integrated dataset underwent a series of transformations before it was used for supervised machine-learning modelling. The transformation process was designed to preserve the chronological structure of the monthly time-series data while generating predictors required for different forecasting horizons.

First, historical E&E export values were transformed into lagged and rolling features to capture short-term and seasonal historical patterns. The lag features included one-month, three-month and twelve-month lags, while rolling statistics included three-month and twelve-month rolling means and standard deviations. The monthly growth rate of E&E exports was also calculated to capture changes in export

momentum. In addition, month and quarter variables were included to represent recurring seasonal patterns.

The forecasting targets were then constructed for the 1-month-, 2-month- and 3-month-ahead horizons. This transformation resulted in some observations being unavailable because the required historical lag values, rolling statistics or future target values could not be calculated. Consequently, the number of usable observations decreased from the original 423 observations.

The resulting supervised dataset was subsequently arranged in chronological order and divided into training and testing subsets using an 80:20 chronological split. The training portion was used for model training and hyperparameter optimisation using five-fold time-series cross-validation, while the most recent observations were retained as a hold-out test set. The same chronological evaluation procedure was applied across SVR, Random Forest and XGBoost to ensure a consistent comparison.

Following model evaluation, the best-performing model for each forecasting horizon was identified based primarily on RMSE. The selected final model was then retrained using the complete historical modelling data and used to generate forecasts for April, May and June 2025, based on information available up to March 2025.

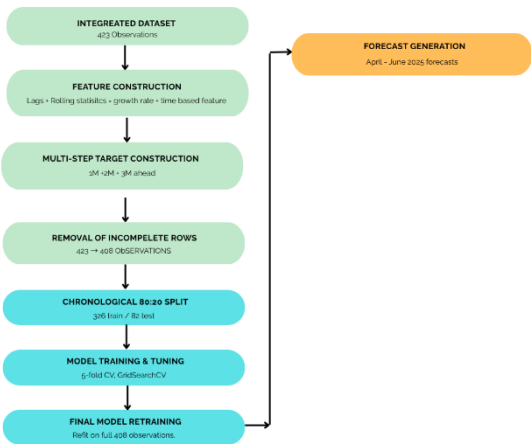


Figure 4:19 Dataset Transformation and Modelling Sequence

4.5 Exploratory Data Analysis

4.5.1 E&E Export Trends

To examine the long-run behaviour of Malaysia's E&E exports, the monthly export series was plotted over the full January 1990 – March 2025 study period. This visualisation provides the basis for identifying the overall trend, structural shifts and periods of heightened volatility discussed below.

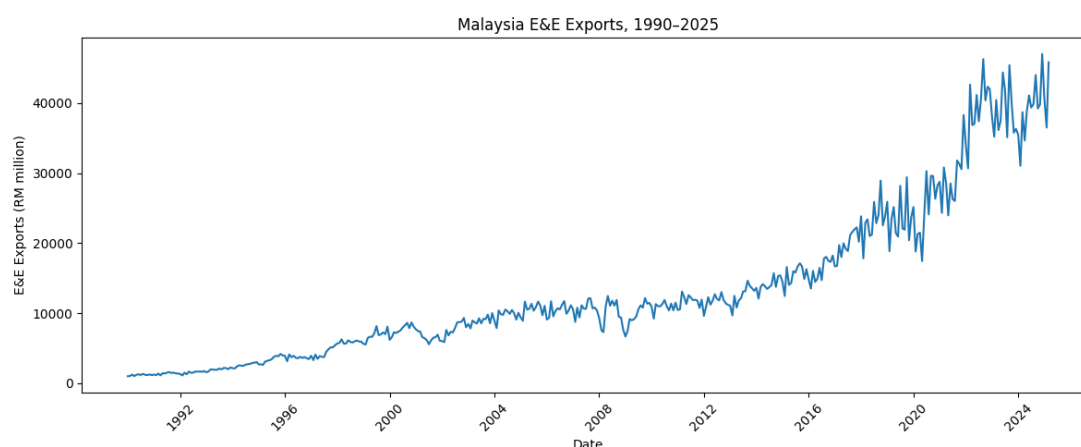


Figure 4:20 Monthly E&E Export Values, January 1990 – March 2025

The line chart shows E&E Exports rising from just above RM1,000 million in 1990 to over RM45,000 million by 2025 a long, steady climb rather than a flat or cyclical series. Between January 1990 and March 2025, E&E exports grew at a compound annual growth rate of approximately 11.1%, rising from RM1,142.30 million to RM45,813.71 million. Growth is gradual and low-volatility through the 1990s and 2000s, then the line becomes visibly "noisier" (sharper up-and-down swings) from around 2016 onward, even as the overall upward trend continues.

Specifically, February recorded the lowest average export value (~RM11,500 million), while March and September recorded the highest (~RM14,000–14,300 million), with secondary peaks in October and December. The steepest, most volatile stretch is the final few years of the series, where the line oscillates strongly between ~RM35,000–47,000 million.

4.5.2 E&E Export Seasonality and Patterns

With the date index fixed, the monthly average line chart now shows a genuine seasonal pattern. Exports dip to their lowest point in February (~RM11,500 million) and are consistently higher in March and September (~RM14,000–14,300 million), with a secondary lift in October and December. The shape is a jagged zig-zag rather than a smooth curve no single long stretch of the year is uniformly high or low, but Feb/Apr/May stand out as the weaker months and Mar/Sep/Oct/Dec as the stronger ones.

To assess whether E&E exports exhibit recurring seasonal patterns, average export values were calculated for each calendar month across the full study period and plotted below.

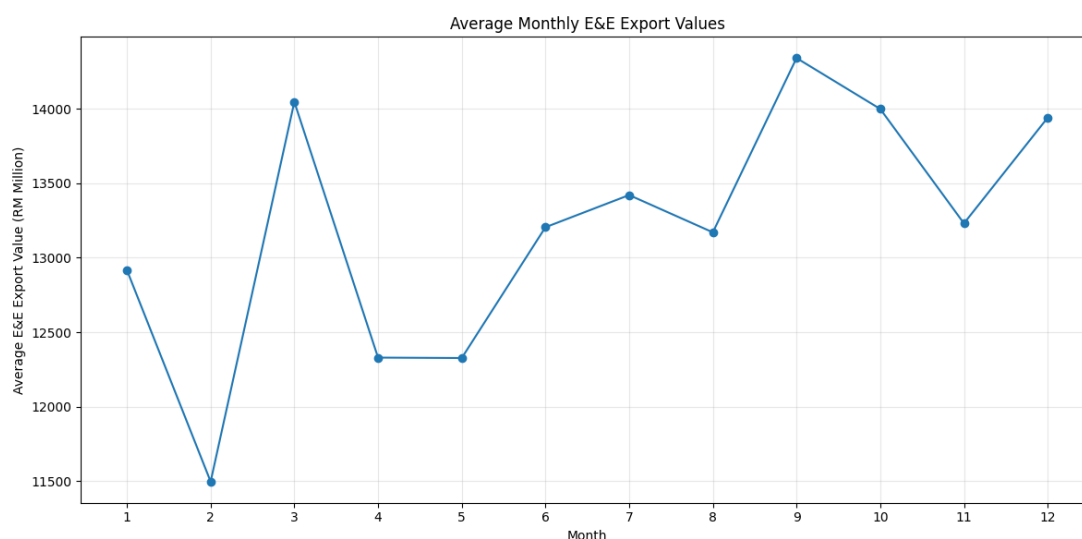


Figure 4:21 Average Monthly E&E Export Values

The monthly boxplot tells a complementary story: every month's box sits at a similar height (medians cluster around RM9,000–11,500 million) and every month shows a long upper whisker with scattered high outliers stretching to RM40,000+ million. This means the seasonal effect is modest relative to the *trend* effect — the difference between a strong month and a weak month is small compared to how much

a given month's exports have grown across the 35-year span. February visibly has the lowest median and a noticeably shorter box than the rest, confirming it as the weakest month overall.

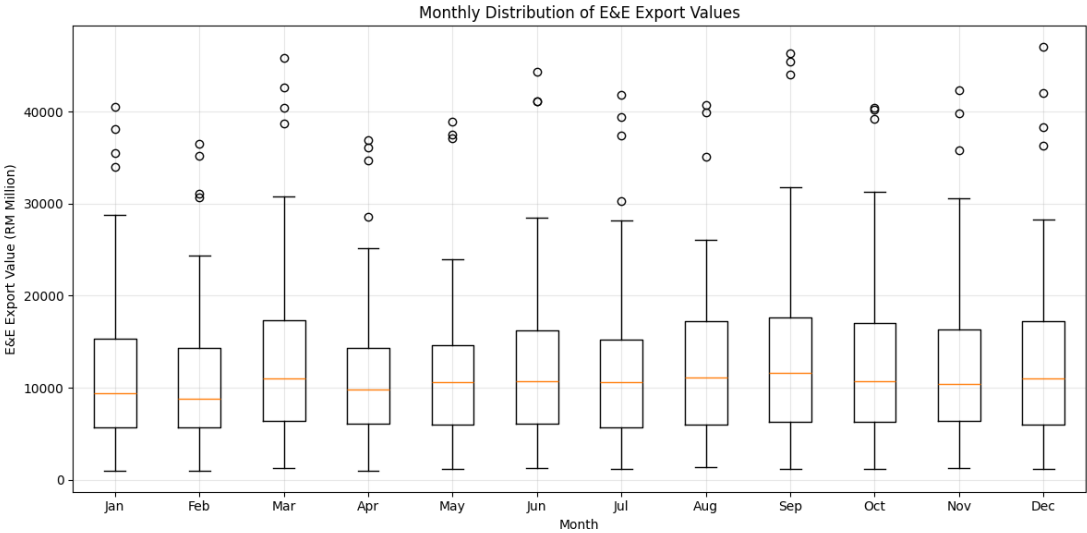


Figure 4:22 Monthly Distribution of E&E Export Values

4.5.3 Impact of Major Economic Events

To assess how E&E exports behaved during major periods of economic disruption, the export series was plotted with the 2008–09 Global Financial Crisis and 2020–21 COVID-19 pandemic periods highlighted.

The shaded-region chart now renders correctly on a proper 1990–2025 timeline. Visually:

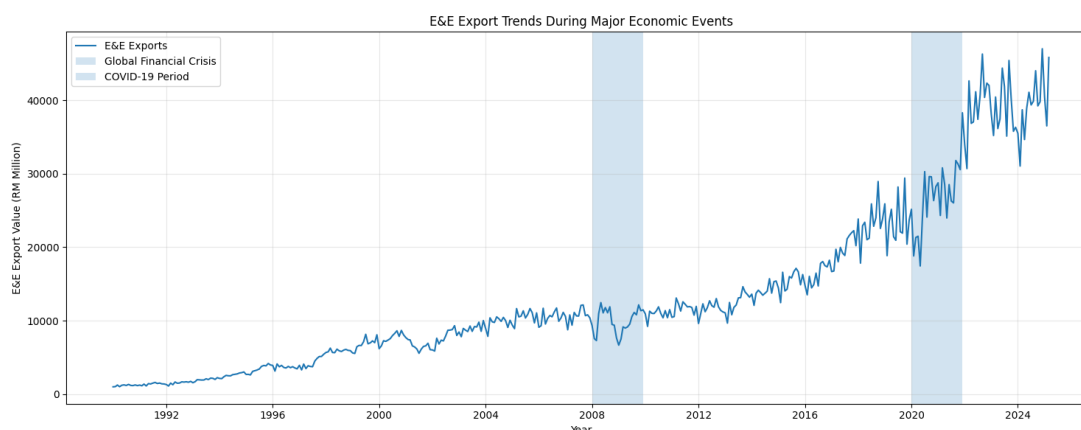


Figure 4:23 E&E Export Trends During Major Economic Events

- (f) The 2008–09 Global Financial Crisis band sits on a relatively flat, low part of the curve — exports were still in their gradual-growth phase, and the shaded window shows little visible disruption to the trend; if anything the line looks flat-to-slightly-dipping rather than sharply falling.
- (g) The 2020–21 COVID-19 band sits much higher up the curve, right where exports were already trending strongly upward. The line continues climbing through the shaded region rather than crashing, and it's followed immediately by the steepest, most volatile growth segment of the entire chart in 2022–2024.

This visual impression is backed by the statistics: mean exports during the COVID period (~RM26,910 million) were more than double the mean during the GFC period (~RM9,942 million) although this difference largely reflects the substantially higher structural level of E&E exports by 2020 compared with 2008 rather than the direct effect of COVID-19. In growth-rate terms, GFC-period YoY growth averaged -2.4% (the only negative average among the three periods), while COVID-period growth averaged +12.8%, close to the +13.1% seen in normal periods which is visually consistent with the relatively flat GFC segment and the continued upward movement during the COVID period.

These structural shifts have direct implications for model performance: because the post-2016 segment of the series exhibits substantially higher volatility than the 1990–2015 period, models trained predominantly on the calmer earlier years may

under-estimate the magnitude of recent fluctuations, a pattern consistent with the systematic under-prediction of sharp increases discussed in Chapter 6.

4.5.4 Descriptive Statistics

Table 4:9 Descriptive statistics table and interpretation

Variable	Mean	Median	Std. Dev.	Min	Max	Skewness	CV
E&E Exports (RM million)	13,199 .49	10,528.5 0	10,892.02	1,010 .24	47,002.3 1	1.31	0.825
CPI Inflation (%)	0.21	0.20	0.48	-2.70	4.00	1.92	—
Exchange Rate	3.54	3.76	0.64	2.44	4.77	-0.20	0.18
U.S. IPI	90.09	94.08	12.92	60.33	104.10	-1.08	0.14

The coefficient of variation is not reported for CPI Inflation. CV is calculated as the standard deviation divided by the mean, and is only meaningful for variables where the mean is a meaningful, non-zero reference point. CPI Inflation has a mean of just 0.21% a value very close to zero so dividing its standard deviation (0.48) by this near-zero mean produces a CV of approximately 2.3, an artificially inflated figure that reflects the smallness of the denominator rather than genuine instability in the series. Because CPI Inflation also takes both positive and negative values (ranging from -2.70% to 4.00%), a ratio-based measure like CV is not statistically appropriate for it in the first place. For this reason, only the exchange rate, U.S. IPI and E&E exports — all strictly positive, ratio-scale variables are reported with a CV, while CPI Inflation is better characterised by its standard deviation and range alone.

The mean E&E export value of RM13,199.49 million, well above the median of RM10,528.50 million, together with a skewness of 1.31, confirms that the distribution is right-skewed consistent with a series that spent most of its 35-year history at lower export levels before growing substantially in its later years. The standard deviation of RM10,892.02 million relative to the mean produces a coefficient of variation (CV) of 0.825. This CV describes the dispersion of monthly export values relative to their mean across the entire 35-year series it does not mean that exports change by 82.5% from one month to the next. A CV of this magnitude for a strongly trending series simply reflects the fact that export levels near the start of the series (RM1,010.24 million) are vastly different from levels near the end (RM45,813.71 million) an overall increase of 4,434.95% across the full sample rather than being a measure of month-to-month change.

To assess the distributional characteristics of each variable including skewness and the presence of extreme values histograms were plotted for E&E exports and each of the three macroeconomic indicators.

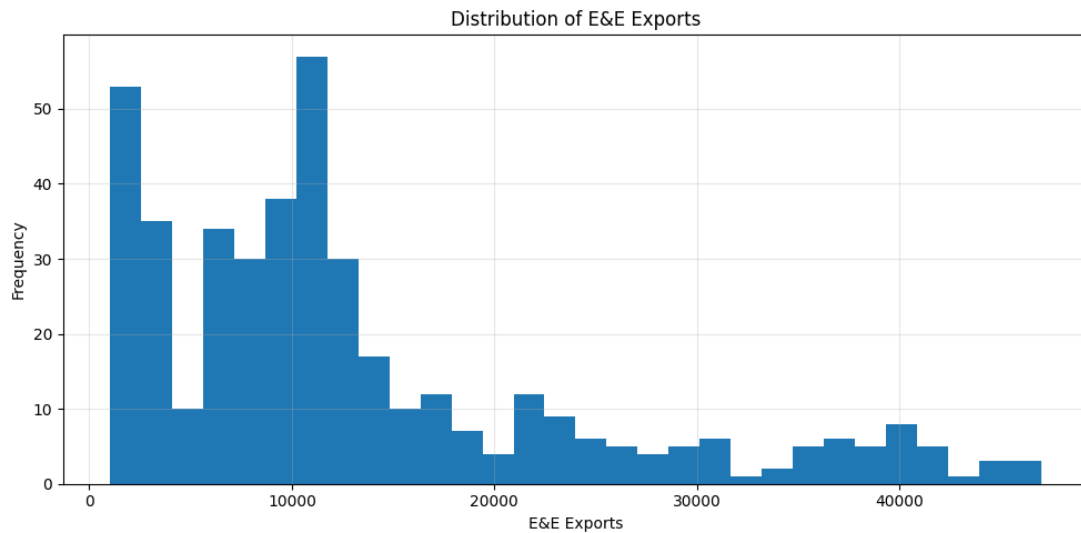


Figure 4:24 Distribution of E&E Exports

E&E Exports histogram: strongly right-skewed — a tall cluster of bars sits in the RM0–15,000 million range, with a long thinning tail stretching to RM47,000 million. This visually matches the skewness value of 1.31 and reflects the trend structure: many years of lower values, fewer years of very high values.

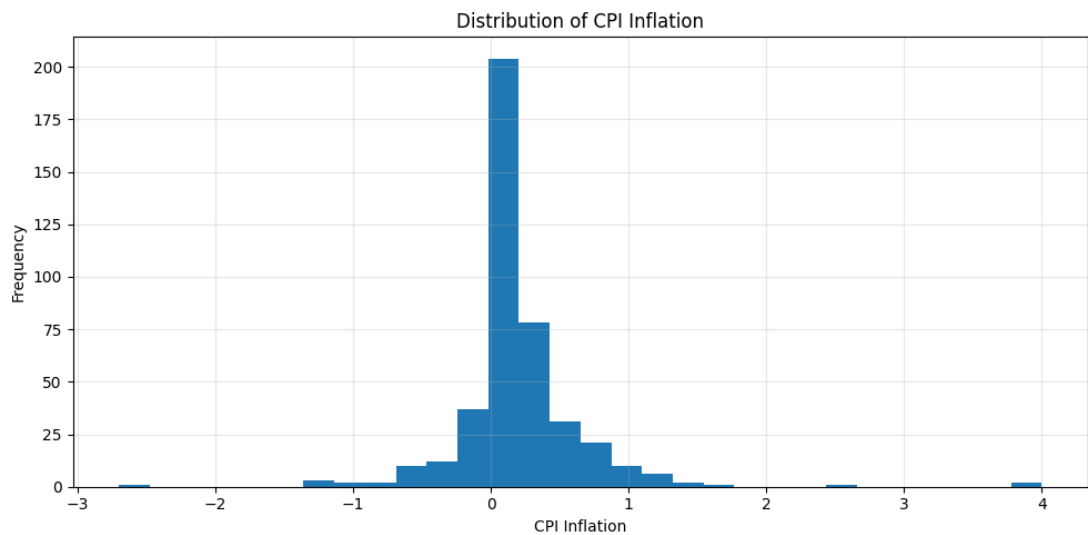


Figure 4:25 Distribution of CPI Inflation

CPI Inflation histogram: tightly clustered near 0–1%, with a handful of extreme outlier bars pulling the shape rightward — these correspond to the inflation spikes visible around 2002, 2008, and 2022 in the earlier line chart.

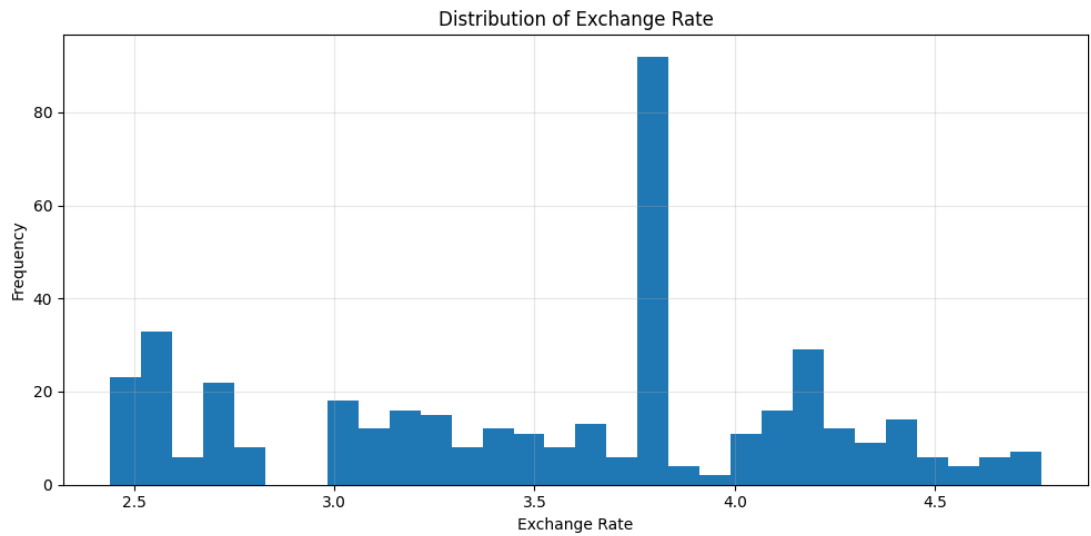


Figure 4:26 Distribution of Exchange Rate

Exchange Rate histogram: loosely bimodal — one cluster around 2.5–3.0 (the pre-1998 peg era) and another around 3.5–4.7 (the post-float era), rather than one smooth bell curve.

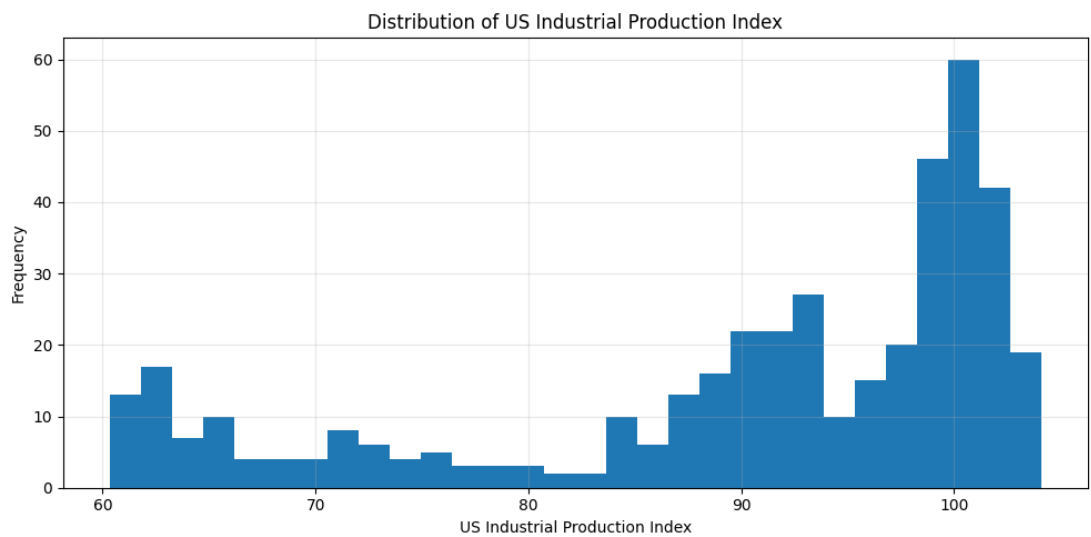


Figure 4:27 Distribution of U.S. Industrial Production Index

US Industrial Production Index histogram: left-skewed, with most values bunched at the higher end (85–104) and a thinner tail of lower values, reflecting the index spending most of its time near historical highs and dipping only briefly during 2008 and 2020.

E&E Exports (skewness = 1.31) and CPI Inflation are both right-skewed, while U.S. IPI is left-skewed. Because none of the three machine-learning models used in this study (Random Forest, SVR, XGBoost) assume normally distributed features, and because the primary purpose of these variables is prediction rather than statistical inference, no distributional transformation (e.g. log transformation) was considered necessary. Feature scaling was applied only for SVR, as described in Section 4.6.5.

4.5.5 Correlation Analysis

To examine the linear relationships between E&E exports and the selected macroeconomic indicators, a correlation matrix was constructed and visualised as a heatmap.

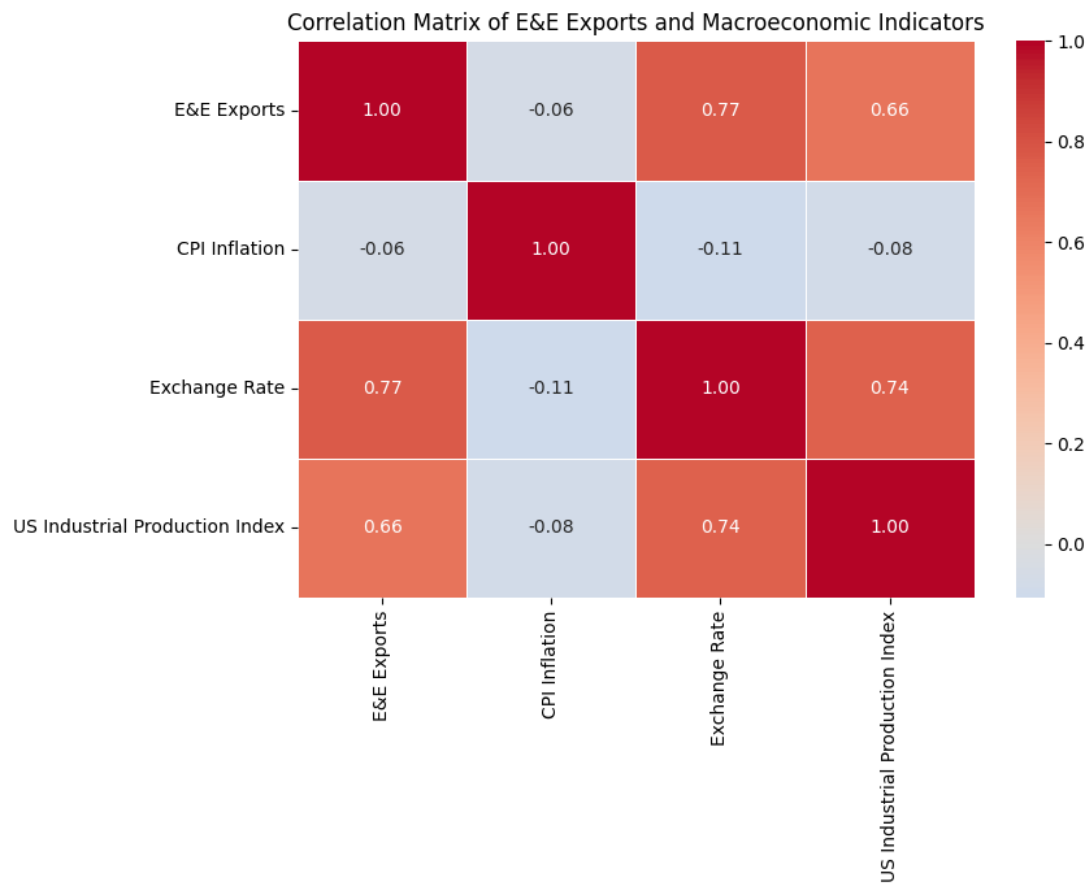


Figure 4:28 Correlation Matrix of E&E Exports and Macroeconomic Indicators

The heatmap makes the relationships easy to scan: the darkest/warmest cells are E&E Exports–Exchange Rate (0.77) and Exchange Rate–U.S. Industrial Production (0.74), both showing strong positive correlations. E&E Exports–U.S. Industrial Production (0.66) is a slightly lighter red, indicating a moderately strong positive correlation. By contrast, the entire CPI Inflation row/column is pale, cool-toned near zero, visually standing out as having relatively weak linear relationships with the other variables, while the exchange rate, U.S. Industrial Production and E&E exports show stronger positive associations.

4.6 Feature Engineering Results

Feature engineering was performed to transform the cleaned monthly E&E export and macroeconomic data into predictor variables suitable for the forecasting

models. Since the study focuses on time-series forecasting, the feature construction process was designed to capture historical export behaviour, short- and long-term trends, changes in export momentum, seasonality, and the influence of the selected macroeconomic indicators.

The three targets were constructed using the following formulas, where $E\&E_t$ is the export value observed at month t :

$$Target_1M(t) = E\&E(t + 1)$$

$$Target_2M(t) = E\&E(t + 2)$$

$$Target_3M(t) = E\&E(t + 3)$$

Worked example: for a predictor row observed at December 2024 (t), the three targets are matched as follows: $Target_1M = E\&E$ value for January 2025, $Target_2M = E\&E$ value for February 2025, and $Target_3M = E\&E$ value for March 2025. Because March 2025 is the last available month in the dataset, December 2024 is the last row for which all three targets can be constructed this is the reason the final supervised modelling period ends in December 2024 even though raw data exists up to March 2025 (see also Section 4.7.3 below).

After removing rows with unavailable historical predictors or future targets, 408 observations remained in the supervised dataset. This count is identical across the three horizons because a single combined supervised dataset (with three target columns) was constructed and rows lacking *any* of the three targets were dropped together, rather than dropping rows independently per horizon.

The feature engineering process resulted in lag, rolling statistical, growth-rate, and time-based features.

4.6.1 Target Variable Creation

To formulate the forecasting problem as a supervised learning task, three future $E\&E$ export targets were created using the $E\&E$ export series. $Target_1M$, $Target_2M$,

and Target_3M represent E&E export values one, two, and three months ahead, respectively. These targets allow the machine-learning models to be evaluated across the three-month forecasting horizon.

```
df['Target_1M'] = df['E&E Exports'].shift(-1)
df['Target_2M'] = df['E&E Exports'].shift(-2)
df['Target_3M'] = df['E&E Exports'].shift(-3)
```

Figure 4:29 E and E Export Target Creation

4.6.2 Lag Features

Lag features were constructed from the historical E&E export series to provide the machine-learning models with information about previous export values. Three lag periods were created: one-month lag, three-month lag, and twelve-month lag.

The one-month lag represents the E&E export value recorded in the immediately preceding month and captures short-term persistence in export behaviour. The three-month lag was included to capture slightly longer-term movements in export performance, while the twelve-month lag represents the E&E export value from the same month of the previous year and was included to capture annual patterns and potential seasonality.

The lag features were generated by shifting the E&E export series backward by the respective number of months. As a result, the earliest observations for which sufficient historical information was unavailable contained missing values. These observations were subsequently excluded from the supervised modelling dataset during the feature preparation process.

The 1-month lag was selected to capture the most recent short-term persistence in exports; the 3-month lag was selected to capture the influence of the most recent quarter, a horizon commonly used in trade and business-cycle analysis; and the 12-month lag was selected to capture annual, same-month-last-year patterns, given the seasonal effects identified in Section 4.5.2 (e.g. the recurring February trough).

Table 4:10 Lag Features

Feature	Description
E&E Lag-1	E&E export value from the previous month
E&E Lag-2	E&E export value from two months earlier
E&E Lag-3	E&E export value from three months earlier

```
df['E&E Lag 1'] = df['E&E Exports'].shift(1)
df['E&E Lag 3'] = df['E&E Exports'].shift(3)
df['E&E Lag 12'] = df['E&E Exports'].shift(12)
```

Figure 4:30 Creation of Lag Features

4.6.3 Rolling Statistics

Rolling statistical features were constructed to capture the underlying trend and variability of E&E exports over different time windows. Two rolling windows were used: 3 months and 12 months.

For each window, both the rolling mean and rolling standard deviation were calculated. The three-month rolling mean captures short-term export trends, while the twelve-month rolling mean provides a broader representation of the annual movement of E&E exports. The corresponding rolling standard deviations measure the degree of variation in export values within each window and therefore provide information about short-term and longer-term volatility.

The 3-month window was chosen to represent short-term trend and volatility consistent with the 3-month lag, while the 12-month window was chosen to represent the annual trend and volatility level, consistent with the 12-month lag — together these two windows allow the models to distinguish between recent momentum and the underlying annual baseline.

Table 4:11 Rolling Statistical Features

Feature	Description
3-month Rolling Mean	Average E&E export value over the previous three months
12-month Rolling Mean	Average E&E export value over the previous twelve months
3-month Rolling Standard Deviation	Short-term variability of E&E exports over three months
12-month Rolling Standard Deviation	Longer-term variability of E&E exports over twelve months

These features allow the models to consider not only individual historical observations but also the recent trend and volatility of E&E export performance.

The current month's own E&E export value (i.e., $E\&E(t)$) is **not** included as a predictor for any horizon; only lagged values ($t-1$, $t-3$, $t-12$) and rolling statistics computed over past months ($t-1$ through $t-w$) are used as predictors. This design deliberately excludes the current-period target value from the feature set to prevent data leakage, since including $E\&E(t)$ as a predictor would allow the model implicit access to information it should not have at forecast time.

```

df['E&E Rolling Mean 3M'] = (
    df['E&E Exports']
    .shift(1)
    .rolling(window=3)
    .mean()
)
df['E&E Rolling Mean 12M'] = (
    df['E&E Exports']
    .shift(1)
    .rolling(window=12)
    .mean()
)
df['E&E Rolling Std 3M'] = (
    df['E&E Exports']
    .shift(1)
    .rolling(window=3)
    .std()
)
df['E&E Rolling Std 12M'] = (
    df['E&E Exports']
    .shift(1)
    .rolling(window=12)
    .std()
)

```

Figure 4:31 Creation of Rolling Features

4.6.4 Growth Rate Features

A monthly growth-rate feature was created to represent changes in E&E export performance over time. The monthly percentage change was calculated from the E&E export series to capture the rate at which exports increased or decreased relative to the previous month.

Unlike the lag and rolling features, which primarily represent the level and historical pattern of exports, the growth-rate feature provides information about the momentum and direction of recent export movements. Positive values indicate an increase in exports compared with the previous month, while negative values indicate a decline.

```
df['E&E Growth Rate'] = df['E&E Exports'].pct_change()
df['Month'] = df['Date'].dt.month
df['Quarter'] = df['Date'].dt.quarter
df['Year'] = df['Date'].dt.year
```

Figure 4:32 Creation of Growth Rate Features

The first observation does not have a previous month from which growth can be calculated and therefore produces a missing value. This was handled during the subsequent preparation of the modelling dataset.

4.6.5 Time-Based Features

Time-based features were incorporated to allow the machine-learning models to learn recurring temporal patterns in the monthly E&E export series. Two calendar-based variables, month and quarter, were extracted from the date index.

The month feature identifies the month of the year from 1 to 12, while the quarter feature identifies the corresponding quarter from 1 to 4. These variables provide the models with explicit information about the position of each observation within the annual calendar and can therefore help capture recurring seasonal patterns in E&E export activity.

The use of time-based features complements the Lag-12 feature because both provide information related to annual patterns but represent the information differently. The Lag-12 feature provides the actual export value from the corresponding month in the previous year, whereas month and quarter provide calendar information.

```
df['CPI Inflation Lag 1'] = df['CPI Inflation'].shift(1)
df['Exchange Rate Lag 1'] = df['Exchange Rate'].shift(1)
df['US Industrial Production Index Lag 1'] = (
    df['US Industrial Production Index'].shift(1)
)
```

Figure 4:33 Creation of Time-Based Features

4.6.6 Feature Scaling

Feature scaling was applied to the predictor variables used by the Support Vector Regression (SVR) model to ensure that variables with different numerical ranges contributed appropriately to the model. The scaling parameters were fitted using the training data and subsequently applied to the testing data to prevent information from the test set from influencing the model development process. The same scaling procedure was maintained consistently when generating forecasts from the selected model.

Feature scaling was performed using scikit-learn's StandardScaler, which transforms each feature using

$$z(x) = \frac{(x - \mu)}{\sigma} \quad (4.1)$$

,where μ and σ are the mean and standard deviation of that feature. Critically, μ and σ were calculated using only the 326 training-set observations; the resulting fitted scaler was then applied, without refitting, to transform the 82 test-set observations. This ensures the test set's own statistical properties never influence the scaling parameters, avoiding information leakage. Scaling was applied only for SVR; Random Forest and XGBoost, being tree-based, do not require feature scaling.

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Figure 4:34 Feature Scaling

4.6.7 Final Feature Set

Following the feature construction process, the final predictor set consisted of historical E&E export features, rolling statistical features, growth information, temporal features, and lagged macroeconomic indicators.

The final machine-learning feature set consisted of 13 predictor variables, comprising:

Historical E&E export features

5. E&E Lag-1
6. E&E Lag-3
7. E&E Lag-12
8. 3-month rolling mean
9. 12-month rolling mean
10. 3-month rolling standard deviation
11. 12-month rolling standard deviation
12. E&E monthly growth rate

Time-based features

13. Month
14. Quarter

Lagged macroeconomic features

15. One-month lagged CPI inflation

16. One-month lagged MYR/USD exchange rate
17. One-month lagged U.S. Industrial Production Index

Table 4:12 Final Feature Set

Feature	Formula	Purpose
E&E Lag-1	$E\&E(t - 1)$	Captures immediate short-term persistence
E&E Lag-3	$E\&E(t - 3)$	Captures quarterly-scale historical behaviour
E&E Lag-12	$E\&E(t - 12)$	Captures annual/seasonal patterns
3-month Rolling Mean	$\frac{1}{3} \sum E\&E(t - 1 \dots t - 3)$	Captures short-term trend level
12-month Rolling Mean	$\frac{1}{12} \sum E\&E(t - 1 \dots t - 12)$	Captures longer-term/annual trend level
3-month Rolling Std	$std\{E\&E(t - 1 \dots t - 3)\}$	Captures short-term volatility
12-month Rolling Std	$std\{E\&E(t - 1 \dots t - 12)\}$	Captures longer-term volatility
Growth Rate	$\frac{E\&E_t - E\&E_{t-1}}{E\&E_{t-1}} \times 100$	Captures export momentum/direction
Month	Calendar month (1–12) of t	Captures monthly seasonality
Quarter	Calendar quarter (1–4) of t	Captures quarterly seasonality
CPI Inflation Lag-1	Inflation $t - 1$	Captures lagged domestic price effects
Exchange Rate Lag-1	ExchangeRate $t - 1$	Captures lagged currency effects
U.S. IPI Lag-1	IPI $t - 1$	Captures lagged external industrial demand

After feature construction, observations without the required historical or future target information were removed from the supervised learning dataset. The resulting dataset was subsequently used for chronological training and testing of the forecasting models.

4.7 Final Prepared Dataset

After the data preprocessing and feature engineering stages, the resulting dataset was prepared for supervised machine-learning modelling. The final preparation process involved removing observations affected by unavailable historical feature values and unavailable future target values, while maintaining the chronological structure of the monthly time-series data.

4.7.1 Dataset Reduction After Feature Construction

The original integrated dataset consisted of 423 monthly observations covering January 1990 to March 2025. The construction of lagged and rolling features required historical observations that were not available at the beginning of the series. In addition, the multi-step forecasting setup required future E&E export values to be available for the corresponding forecasting horizons.

As a result, observations that could not provide the required historical predictors or future target values were excluded from the supervised learning dataset. Following this process, the final supervised dataset contained 408 observations.

```
model_df = df[
    features + ['Target_1M', 'Target_2M', 'Target_3M', 'Date']
].copy()
model_df = model_df.dropna(subset=features).reset_index(drop=True)
```

Figure 4:35 Construction of the Supervised Modelling Dataset with Forecasting Targets

```

historical_df = model_df.dropna(
    subset=['Target_1M', 'Target_2M', 'Target_3M']
).copy()
historical_df = historical_df.reset_index(drop=True)

```

Figure 4:36 Removal of Observations with Unavailable Future Forecasting Targets

The reduction in observations was therefore a consequence of the time-series feature construction and multi-step forecasting formulation rather than missing data in the original integrated dataset.

In addition to the historical predictor features, future E&E export targets were constructed for the three forecasting horizons. For each observation, the target represented the E&E export value one month, two months, or three months ahead. Therefore, observations for which the required future target value was unavailable were excluded from the corresponding supervised learning formulation.

This process resulted in a reduction from the original 423 monthly observations to 408 observations in the final supervised learning dataset. The reduction was therefore attributable to the requirements of lagged and rolling feature construction and future target construction, rather than missing values in the original integrated dataset.

4.7.2 Training and Testing Dataset

The final supervised dataset was divided chronologically into training and testing subsets using an 80:20 time-series split. The earliest 326 observations were assigned to the training set, while the most recent 82 observations were reserved as the testing set.

The training data were used for model development and hyperparameter optimisation, while the test data were kept separate and used only for the final evaluation of forecasting performance. This chronological splitting approach prevents

future observations from being used during model training and provides a more realistic representation of forecasting conditions.

```
split_index = int(len(historical_df) * 0.80)
train_df = historical_df.iloc[:split_index].copy()
test_df = historical_df.iloc[split_index:].copy()
```

Figure 4:37 Chronological 80:20 Train-Test Split of the Supervised Dataset

Table 4:13 Training and Testing Dataset

Dataset	Observations	Proportion	Purpose
Training set	326	80%	Model training and hyperparameter optimisation
Testing set	82	20%	Final out-of-sample model evaluation
Total	408	100%	Final supervised dataset

The training period was from January 1991 to February 2018 and the testing period was from March 2018 to December 2024.

The same chronological partition was applied consistently to the machine-learning models to ensure a fair comparison of SVR, Random Forest and XGBoost. The ARIMA baseline was evaluated against the corresponding test-period observations using its own univariate forecasting procedure.

4.7.3 Final Modelling Dataset

The final supervised modelling dataset the set of rows with a complete predictor set and all three future targets available spans January 1991 to December 2024 (408 observations). This is because, as illustrated in the worked example in Section 4.6.1, a row can only have a valid Target_3M value if data exists three months later; since the dataset ends in March 2025, December 2024 is the last month for which a 3-month-ahead target (March 2025) is available. Rows for January, February and

March 2025 therefore have complete predictor information but incomplete target information (their future targets fall beyond the data's end date) and are consequently excluded from model training. However, information up to March 2025 is still used: the predictor values from the March 2025 row (Table 5.3) are retained separately and used as the forecast origin from which the operational April, May and June 2025 forecasts are generated in Chapter 5. In short, December 2024 marks the end of the trainable dataset, while March 2025 marks the end of the available data used to generate forward-looking forecasts.

The final predictor set consisted of 13 features, including E&E export lag features, rolling statistics, monthly growth rate, time-based variables, and one-month lagged values of CPI inflation, the MYR/USD exchange rate, and the U.S. Industrial Production Index.

Table 4:14 Final Dataset Summary

Dataset Stage	Period	Observations
Original integrated dataset	January 1990 – March 2025	423
Final supervised modelling dataset	January 1991 – December 2024	408

The final dataset maintained the chronological ordering of observations to ensure that the temporal relationship between historical predictors and future E&E export values was preserved.

4.8 Chapter Summary

This chapter presented the results of the data preparation, exploratory data analysis and feature engineering processes conducted for the Malaysia E&E export forecasting study. The original integrated dataset consisted of 423 monthly observations from January 1990 to March 2025, containing E&E export values, CPI inflation, MYR/USD exchange rate and the U.S. Industrial Production Index.

The data preprocessing results demonstrated that the datasets were successfully cleaned, standardised and integrated using a common monthly date index. The exploratory data analysis examined the historical trends, seasonal patterns, major economic events, descriptive statistics and relationships between E&E exports and the selected macroeconomic indicators. The analysis provided an understanding of the characteristics and behaviour of Malaysia's E&E export series before model development.

Feature engineering was subsequently performed to transform the historical time-series data into features suitable for machine-learning forecasting. The constructed features included 1-, 3- and 12-month lag features, 3- and 12-month rolling means and standard deviations, monthly growth rate, month and quarter features, and lagged macroeconomic variables. These features were designed to capture historical dependence, trend, volatility, seasonality and the potential delayed effects of macroeconomic conditions on E&E exports.

After feature construction and the removal of observations for which the required historical or future target values were unavailable, the dataset was prepared for supervised learning. The resulting modelling dataset was maintained in chronological order and subsequently divided into training and testing periods without randomisation to preserve the temporal structure of the forecasting problem.

Overall, the results presented in this chapter established the prepared dataset and feature set required for the subsequent modelling stage.

CHAPTER 5

IMPLEMENTATION

5.1 Introduction

This chapter presents the implementation of the proposed machine-learning-based forecasting system for Malaysia's Electrical and Electronics (E&E) exports. The implementation follows the methodology described in Chapter 3 and uses the final prepared dataset obtained after the data preprocessing and feature engineering procedures presented in Chapter 4.

The forecasting system implements three machine-learning models, namely Support Vector Regression (SVR), Random Forest and XGBoost. ARIMA is additionally implemented as a statistical baseline to provide a benchmark for evaluating the performance of the machine-learning models. The models are developed to forecast Malaysia's E&E export values at three forecasting horizons: one-month-, two-month- and three-month-ahead.

The implementation begins with the preparation of the forecasting targets and forecasting horizons, followed by the implementation of the selected models. A chronological train-test split is used to preserve the temporal order of the observations, while five-fold time-series cross-validation is applied during hyperparameter optimisation. The models are subsequently trained using the selected configurations and evaluated using the procedure defined in Chapter 3.

Following model development and evaluation, the selected machine-learning model was refitted using the complete supervised modelling dataset covering January 1991 to December 2024. Information available up to March 2025 was then used as the forecasting origin to generate forecasts for April, May and June 2025. Finally, the

forecasting outputs and model evaluation results are prepared for visualisation through an interactive Power BI dashboard.

The implementation described in this chapter provides the technical foundation for the comparative evaluation of the forecasting models and the interpretation of the forecasting results presented in Chapter 6.

5.2 Forecasting System Architecture

The forecasting system was implemented as an end-to-end machine-learning forecasting workflow consisting of forecasting target construction, model development, validation, hyperparameter optimisation, model evaluation, forecast generation and result visualisation. The architecture follows the CRISP-DM methodology established in Chapter 3 and uses the final prepared dataset described in Chapter 4.

The system begins with the final monthly E&E export dataset containing the historical E&E export values, selected macroeconomic indicators and engineered time-series features. The prepared data are arranged in chronological order and used to construct separate forecasting targets for the one-month-, two-month- and three-month-ahead horizons. The resulting datasets are then divided chronologically into training and testing sets to prevent future observations from being used during model training.

Four forecasting approaches are implemented in the system. ARIMA is used as a univariate statistical baseline and relies only on the historical E&E export series. The three machine-learning approaches, namely Support Vector Regression (SVR), Random Forest and XGBoost, use the engineered time-series features and selected macroeconomic indicators as predictors. For the machine-learning models, time-series cross-validation is applied during hyperparameter optimisation to identify suitable model configurations while preserving the temporal order of the observations.

After optimisation, the selected configurations are used to train the respective models and generate predictions for the held-out test observations. Model performance is subsequently assessed using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and R^2 . The evaluation results are then used to determine the best-performing forecasting approach for each forecasting horizon.

For the final forecasting task, information available up to March 2025 is used to construct the required forecasting inputs. The best final machine learning model is then used to generate forecasts for April, May and June 2025. Feature importance analysis is performed to examine the contribution of the input features, while the resulting historical data, model performance and forecast outputs are prepared for presentation through the Power BI dashboard. Figure 5.1 illustrates the overall architecture of the implemented forecasting system, showing the flow from the prepared dataset through model development and validation to final forecasting and visualisation.

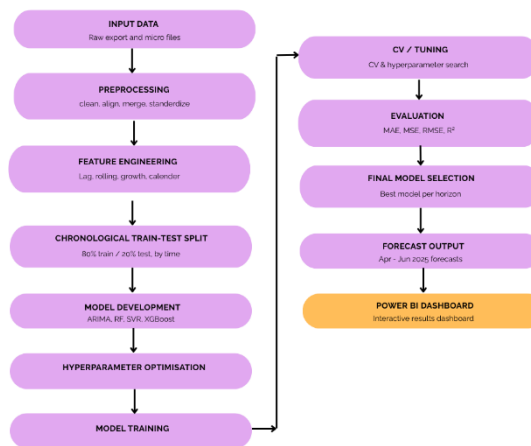


Figure 5:1 Architecture of the Implemented E&E Export Forecasting

5.3 Forecasting Target and Horizons

The forecasting task focuses on predicting the monthly value of Malaysia's Electrical and Electronics (E&E) exports at three different forecasting horizons. The

forecasting horizons were defined to assess the capability of the developed models to predict E&E export values over short-term periods of one, two and three months ahead. This multi-horizon approach allows the forecasting performance of the models to be examined as the forecasting period increases.

The forecasting targets were constructed from the historical E&E export series by shifting the target variable forward according to the required forecasting horizon. The predictor variables consisted of the historical E&E export lag and rolling features, growth rate and time-based features, together with the selected macroeconomic indicators. The same predictor framework was applied to the three machine-learning models to ensure a consistent comparison.

Separate forecasting models were developed for each horizon. Therefore, the one-month-ahead, two-month-ahead and three-month-ahead forecasting tasks were treated as individual supervised learning problems. This allows the models to learn the relationship between the amount of historical information and the E&E export value at the future horizon.

Example: If the observed predictor row is for the month of December 2024, then the three predictor horizon targets are: 1-month target = export value, month of January 2025, 2-month target = export value, month of February 2025, 3-month target = export value, month of March 2025. All predictor values (lags, rolling statistics, lagged macroeconomic indicators) used for this row are calculated using information available at or before December 2024 only.

The three forecasting horizons implemented in the system are described in the following subsections.

5.3.1 One-Month-Ahead Forecasting

The one-month-ahead forecasting task predicts the E&E export value for the month immediately following the latest observation available to the model. For each

observation, the target variable represents the E&E export value at $t+1$, while the predictor variables contain information available at time t and earlier.

The one-month-ahead models were developed separately for SVR, Random Forest and XGBoost. ARIMA was also used to produce a corresponding one-month-ahead baseline forecast using only the historical E&E export series. The one-month-ahead horizon represents the shortest forecasting period and provides an assessment of the models' ability to capture short-term movements in E&E exports.

At prediction time t , the information available to the one-month-ahead model consists of: the E&E export lag features ($t-1$, $t-3$, $t-12$), the 3- and 12-month rolling mean and standard deviation (computed using data up to t), the monthly growth rate up to t , the calendar month/quarter of t , and the one-month-lagged macroeconomic indicators (inflation, exchange rate, U.S. IPI at $t-1$). No information from $t+1$ (the target period) is available to or used by the model.

```
rf_1M = RandomForestRegressor(  
    n_estimators=300,  
    random_state=42,  
    n_jobs=-1  
)  
  
rf_1M.fit(X_train, y_train_1M)  
  
rf_pred_1M = rf_1M.predict(X_test)  
  
print("1M predictions:", rf_pred_1M.shape)
```

Figure 5:2 Forecast One-Month Using Random Forest

```
svr_1M = SVR(  
    kernel='rbf',  
    C=100,  
    gamma='scale',  
    epsilon=0.1  
)  
  
svr_1M.fit(X_train_scaled, y_train_1M)  
  
svr_pred_1M = svr_1M.predict(X_test_scaled)  
  
print("1M predictions:", svr_pred_1M.shape)
```

Figure 5:3 Forecast One-Month Using SVR

```

xgb_1M = XGBRegressor(
    n_estimators=300,
    learning_rate=0.05,
    max_depth=6,
    random_state=42,
    n_jobs=-1,
    objective='reg:squarederror'
)

xgb_1M.fit(X_train, y_train_1M)

xgb_pred_1M = xgb_1M.predict(X_test)

print("1M predictions:", xgb_pred_1M.shape)

```

Figure 5:4 Forecast One-Month Using XGBoost

5.3.2 Two-Month-Ahead Forecasting

The two-month-ahead forecasting task predicts the E&E export value two months after the available observation. The target variable is therefore defined as the E&E export value at $t+2$, while the predictor variables are constructed using information available up to time t .

Separate SVR, Random Forest and XGBoost models were trained for this forecasting horizon, together with a corresponding ARIMA baseline. The two-month-ahead horizon provides an intermediate forecasting period for examining whether model performance changes when the forecasting distance is increased.

The two-month-ahead model does not require future ($t+1$ or $t+2$) macroeconomic values as inputs like the one-month model, it uses only information available up to and including time t (i.e., macroeconomic indicators lagged by one month relative to t , not relative to the target period). A separate two-month-ahead model is trained independently from the one-month and three-month models: each horizon has its own dedicated XGBoost, Random Forest and SVR model, tuned and trained specifically on that horizon's target variable, rather than a single model producing all three horizons simultaneously.

```

rf_2M = RandomForestRegressor(
    n_estimators=300,
    random_state=42,
    n_jobs=-1
)

rf_2M.fit(X_train, y_train_2M)

rf_pred_2M = rf_2M.predict(X_test)

print("2M predictions:", rf_pred_2M.shape)

```

Figure 5:5 Forecast Two-Month Using Random Forest

```

svr_2M = SVR(
    kernel='rbf',
    C=100,
    gamma='scale',
    epsilon=0.1
)

svr_2M.fit(X_train_scaled, y_train_2M)

svr_pred_2M = svr_2M.predict(X_test_scaled)

print("2M predictions:", svr_pred_2M.shape)

```

Figure 5:6 Forecast Two-Month Using SVR

```

xgb_2M = XGBRegressor(
    n_estimators=300,
    learning_rate=0.05,
    max_depth=6,
    random_state=42,
    n_jobs=-1,
    objective='reg:squarederror'
)

xgb_2M.fit(X_train, y_train_2M)

xgb_pred_2M = xgb_2M.predict(X_test)

print("2M predictions:", xgb_pred_2M.shape)

```

Figure 5:7 Forecast Two-Month Using XGBoost

5.3.3 Three-Month-Ahead Forecasting

The three-month-ahead forecasting task predicts the E&E export value three months after the available observation. The target variable is defined as the E&E export value at $t+3$, using predictor information available at time t and earlier.

Separate SVR, Random Forest and XGBoost models were implemented for the three-month-ahead horizon and compared with the ARIMA baseline. This horizon is particularly important to the proposed forecasting system because the final forecasting application generates three consecutive monthly forecasts using information available up to March 2025. The resulting forecasts correspond to April, May and June 2025.

```
rf_3M = RandomForestRegressor(  
    n_estimators=300,  
    random_state=42,  
    n_jobs=-1  
)  
  
rf_3M.fit(X_train, y_train_3M)  
  
rf_pred_3M = rf_3M.predict(X_test)  
  
print("3M predictions:", rf_pred_3M.shape)
```

Figure 5:8 Forecast Three-Month Using Random Forest

```
svr_3M = SVR(  
    kernel='rbf',  
    C=100,  
    gamma='scale',  
    epsilon=0.1  
)  
  
svr_3M.fit(X_train_scaled, y_train_3M)  
  
svr_pred_3M = svr_3M.predict(X_test_scaled)  
  
print("3M predictions:", svr_pred_3M.shape)
```

Figure 5:9 Forecast Three-Month Using SVR

```

xgb_1M = XGBRegressor(
    n_estimators=300,
    learning_rate=0.05,
    max_depth=6,
    random_state=42,
    n_jobs=-1,
    objective='reg:squarederror'
)

xgb_1M.fit(X_train, y_train_1M)

xgb_pred_1M = xgb_1M.predict(X_test)

print("1M predictions:", xgb_pred_1M.shape)

```

Figure 5:10 Forecast Three-Month Using XGBoost

```

arma_order = (2, 1, 2)
arma_predictions = []
history = arima_train.copy()
for i in range(len(test_df)):
    model = ARIMA(
        history,
        order=arma_order
    )
    fitted_model = model.fit()
    forecast = fitted_model.forecast(steps=3)
    arma_predictions.append({
        'Date': test_df.iloc[i]['Date'],
        'Actual_1M': test_df.iloc[i]['Target_1M'],
        'Actual_2M': test_df.iloc[i]['Target_2M'],
        'Actual_3M': test_df.iloc[i]['Target_3M'],
        'Predicted_1M': forecast.iloc[0],
        'Predicted_2M': forecast.iloc[1],
        'Predicted_3M': forecast.iloc[2]
    })

```

Figure 5:11 Forecast One, Two, Three Month Ahead Forecasting Using ARIMA

5.4 Model Implementation

Four candidate forecasting approaches were configured for each of the three horizons: an ARIMA statistical baseline fitted on the univariate E&E Exports series, and three supervised machine learning regressors Support Vector Regression, Random Forest, and XGBoost fitted on the thirteen-feature engineered matrix. This section describes the libraries, configuration, and initial (untuned) settings used to establish each model family before hyperparameter optimisation (Section 5.5.3) and final training (Section 5.6).

5.4.1 ARIMA Baseline Implementation

The ARIMA baseline was implemented using `statsmodels.tsa.arima.model`. ARIMA on the univariate E&E Exports series restricted to the training period (326 observations, January 1991 – February 2018). Six candidate (p, d, q) orders were fitted directly to the training series and ranked by Akaike Information Criterion (AIC): (1,1,0), (1,1,1), (2,1,0), (2,1,1), (2,1,2), and (3,1,1). The order (2, 1, 2) achieved the lowest AIC (5,354.80), marginally ahead of (2,1,1) (AIC = 5,356.88) and (3,1,1) (AIC = 5,357.15), and was selected as the baseline specification. The procedure by which this specification was subsequently used to generate walk-forward forecasts is described in Section 5.6.4.

```
import warnings
warnings.filterwarnings('ignore')
orders = [
    (1, 1, 0),
    (1, 1, 1),
    (2, 1, 0),
    (2, 1, 1),
    (2, 1, 2),
    (3, 1, 1)
]
arima_results = []
for order in orders:
    try:
        model = ARIMA(arima_train, order=order)
        fitted_model = model.fit()

        arima_results.append([
            'Order': order,
            'AIC': fitted_model.aic
        ])
    except Exception as e:
        print(f"ARIMA{order} failed: {e}")
arima_results_df = pd.DataFrame(arima_results)
arima_results_df = arima_results_df.sort_values(
    'AIC'
).reset_index(drop=True)
```

Figure 5:12 Choosing the lowest AIC Value

5.4.2 Support Vector Regression Implementation

The implementation was done using Support Vector Regression (SVR) from `scikit-learn`'s `svm`. The three targets were treated separately with an SVR class with a radial basis function (RBF) kernel. The thirteen features were standardized using `StandardScaler`, which was fitted only on the training partition, and applied to the training and test partitions to prevent information leakage since the features are sensitive to scale. The initial, untuned configuration used `C = 100`, `gamma = 'scale'`,

and $\epsilon = 0.1$. This configuration was not very successful with the held-out test set (the R^2 was around -8.6 on all horizons, and the range of predicted values was extremely small compared to the range of actual values, as shown in Section 5.5.3).

5.4.3 Random Forest Implementation

Random Forest regression was performed with the help of scikit-learn's `ensemble.RandomForestRegressor` that is trained independently for each horizon on the unscaled feature matrix. The first configuration is: 300 trees (`n_estimators = 300`), random state = 42 (for reproducibility) and `n_jobs = -1` (for parallelisation). In this untuned setting, predictions were within a plausible numerical range, but the magnitude and volatility of exports in the test period 2018–2024 were under-predicted (R^2 negative at all three horizons), again providing a motivation for hyperparameter tuning.

5.4.4 XGBoost Implementation

The `XGBRegressor` class from the `xgboost` package was used to implement XGBoost regression. The first one is with 300 boosting rounds (`n_estimators=300`), learning rate of 0.05, max tree depth of 6, objective of squared-error (`reg:squarederror`), and `random_state = 42`. The three untuned machine learning models performed similarly with both outperforming the untuned SVR configuration, making XGBoost a good candidate for the initial stage of the process outlined in Section 5.5.3.5.5.3.

5.5 Data Splitting and Validation Implementation

Because the dataset is a monthly time series, standard random k-fold cross-validation was avoided, as it would allow future information to inform predictions of earlier periods. A strictly chronological validation strategy was used throughout: a single chronological train-test split for final model comparison, and time-series cross-validation for hyperparameter tuning.

5.5.1 Chronological Train-Test Split

This resulted in a training partition of 326 observations (January 1991 – February 2018) and a testing partition of 82 observations (March 2018 – December 2024) which were divided using a fixed partition rather than a random one, so that every test observation was made after every training observation. The period of the COVID-19 pandemic and its aftermath (2020–2021) is used as the test period, giving a meaningful and challenging out-of-sample evaluation period. All four model families were trained with the same X_{train} , X_{test} , and target vectors for each horizon.

5.5.2 Time-Series Cross-Validation

To perform hyperparameter tuning, the `TimeSeriesSplit` was employed from `scikit-learn` with 5 splits (`n_splits = 5`) instead of the usual k -fold cross validation. The `TimeSeriesSplit` method creates successive training folds of growing size followed by a validation fold that is completely drawn from the period after the training fold, maintaining the forward-looking nature of the forecasting problem while tuning. To guarantee the same validation regime, the same `TimeSeriesSplit` object was used for the hyperparameter optimization for the Random Forest, SVR and XGBoost.

```
tscv = TimeSeriesSplit(n_splits=5)
```

Figure 5:13 Time-Series Split

Table 5:1 Time Series 5 Split Cross Validation

Fold	Training Window	Validation Window
1	Jan 1991 – Aug 1995 (56 obs)	Sep 1995 – Feb 2000 (54 obs)
2	Jan 1991 – Feb 2000 (110 obs)	Mar 2000 – Aug 2004 (54 obs)
3	Jan 1991 – Aug 2004 (164 obs)	Sep 2004 – Feb 2009 (54 obs)
4	Jan 1991 – Feb 2009 (218 obs)	Mar 2009 – Aug 2013 (54 obs)
5	Jan 1991 – Aug 2013 (272 obs)	Sep 2013 – Feb 2018 (54 obs)

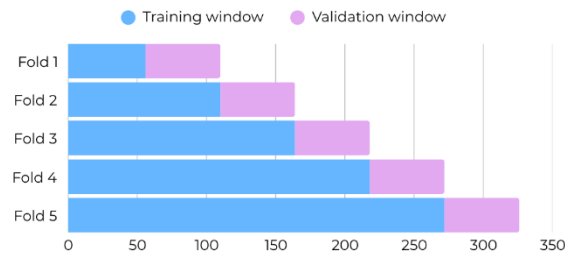


Figure 5:14 Time Series 5 Split Cross Validation

5.5.3 Hyperparameter Optimisation

For each machine learning family and each horizon (nine grid searches), the hyperparameters were optimized using GridSearchCV with the five-fold TimeSeriesSplit and the negative root mean squared error as the scoring function. The Random Forest grid included 36 combinations of number of trees (200, 300, 500), maximum depth (None, 10, 20), minimum samples to split (2, 5) and minimum samples per leaf (1, 2) per horizon. The SVR grid covered C (10, 100, 500, 1000), gamma ('scale', 0.001, 0.01, 0.1), and epsilon (0.01, 0.1, 0.5) 48 combinations per horizon. The XGBoost grid covered the number of boosting rounds (200, 300, 500),

maximum depth (3, 5, 7), learning rate (0.03, 0.05, 0.1), row subsampling (0.8, 1.0), and column subsampling (0.8, 1.0) 108 combinations per horizon. The selected configuration and cross-validated RMSE for each model and each horizon are presented in Table 5.1.

Table 5:2 : Best hyperparameter configuration and cross-validated RMSE per model and forecast horizon

Model	Horizon	Best Hyperparameters	CV RMSE
Random Forest	1M	n_estimators=300, max_depth=None, min_samples_split=2, min_samples_leaf=1	2,772.12
Random Forest	2M	n_estimators=200, max_depth=10, min_samples_split=2, min_samples_leaf=1	2,905.75
Random Forest	3M	n_estimators=500, max_depth=10, min_samples_split=5, min_samples_leaf=2	3,116.99
SVR	1M	C=1000, epsilon=0.5, gamma=0.01, kernel=rbf	2,542.88
SVR	2M	C=1000, epsilon=0.5, gamma=0.01, kernel=rbf	2,531.79
SVR	3M	C=1000, epsilon=0.5, gamma=0.01, kernel=rbf	2,594.00
XGBoost	1M	n_estimators=200, max_depth=3, learning_rate=0.1, subsample=0.8, colsample_bytree=0.8	2,454.83
XGBoost	2M	n_estimators=200, max_depth=3, learning_rate=0.1, subsample=0.8, colsample_bytree=0.8	2,670.91
XGBoost	3M	n_estimators=500, max_depth=3, learning_rate=0.05, subsample=1.0, colsample_bytree=0.8	2,756.33

5.6 Model Training and Forecast Generation

This section describes how each tuned model family was fitted using the configurations identified in Table 5.1, how the ARIMA baseline's walk-forward

forecasts were generated, how the final production model was selected, and how the operational April–June 2025 forecasts were produced.

5.6.1 ARIMA Forecasting Procedure

Rather than fitting a single ARIMA(2,1,2) model once and forecasting three steps ahead from a fixed origin, the baseline was implemented as a walk-forward (expanding-window) forecast: for each of the 82 test-set months, an ARIMA(2,1,2) model was refitted on all data observed up to and including that point and used to forecast three steps ahead simultaneously (forecast(steps=3)), yielding 1M, 2M and 3M predictions for that origin. After each step, the actual E&E Exports value for that month was appended to the training history before the model was refitted for the next origin, so that ARIMA always had access to the same information a real-time forecaster would have had. This design required 82 separate re-estimations but avoids any artificial accuracy advantage from fitting only once, producing a like-for-like comparison against the machine learning models.

5.6.2 SVR Training

The best SVR configuration identified for each horizon ($C = 1000$, $\epsilon = 0.5$, $\gamma = 0.01$, RBF kernel) was refitted on the scaled training partition ($X_{\text{train_scaled}}$) using GridSearchCV's `best_estimator_`, and used to generate predictions on $X_{\text{test_scaled}}$ for all three horizons. Test-set predictions and the resulting error metrics are reported in Section 5.7.

```

svr_grid_1M = GridSearchCV(
    estimator=SVR(),
    param_grid=svr_param_grid,
    cv=tscv,
    scoring='neg_root_mean_squared_error',
    n_jobs=-1,
    verbose=1
)

svr_grid_1M.fit(X_train_scaled, y_train_1M)

print("Best parameters for SVR 1M:")
print(svr_grid_1M.best_params_)

print("Best CV RMSE for SVR 1M:")
print(-svr_grid_1M.best_score_)

```

Figure 5:15 Best Configuration Identification For SVR 1-month ahead Forecasting

5.6.3 Random Forest Training

For each horizon (Table 5.1), the best fit Random Forest model was retrained on the unscaled training partition (X_{train}), and applied to the unscaled test partition (X_{test}) to make predictions. The test-set metrics for the selected 1M configuration are the same as the initial untuned configuration ($n_{\text{estimators}} = 300$, default depth) as it was essentially the same configuration, and the 2M and 3M configurations, which used shallower trees ($\text{max_depth} = 10$), showed small increases in RMSE compared to their untuned versions.

```

rf_grid_1M = GridSearchCV(
    estimator=RandomForestRegressor(
        random_state=42,
        n_jobs=-1
    ),
    param_grid=rf_param_grid,
    cv=tscv,
    scoring='neg_root_mean_squared_error',
    n_jobs=-1,
    verbose=1
)

rf_grid_1M.fit(X_train, y_train_1M)

print("Best parameters for 1M:")
print(rf_grid_1M.best_params_)

print("Best CV RMSE for 1M:")
print(-rf_grid_1M.best_score_)

```

Figure 5:16 Best Configuration Identification For Random Forest 1-month ahead Forecasting

5.6.4 XGBoost Training

The best XGBoost configuration for each horizon (Table 5.1) was refitted on X_{train} and used to generate predictions on X_{test} . The tuned XGBoost models performed better across all three horizons than the tuned Random Forest and tuned SVR (section 5.8.1), and were the best machine learning performer.

```

xgb_grid_1M = GridSearchCV(
    estimator=XGBRegressor(
        objective='reg:squarederror',
        random_state=42,
        n_jobs=-1
    ),
    param_grid=xgb_param_grid,
    cv=tscv,
    scoring='neg_root_mean_squared_error',
    n_jobs=-1,
    verbose=1
)

xgb_grid_1M.fit(X_train, y_train_1M)

print("Best parameters for XGBoost 1M:")
print(xgb_grid_1M.best_params_)

print("Best CV RMSE for XGBoost 1M:")
print(-xgb_grid_1M.best_score_)

```

Figure 5:17 Best Configuration Identification For XGBoost 1-month ahead Forecasting

5.6.5 Final Machine Learning Model Selection

On this test set, ARIMA outperformed every machine learning model at every horizon and was the only model to achieve a positive R^2 at any horizon. Nonetheless, XGBoost was selected as the model carried forward into production forecasting, for two reasons: first, ARIMA is univariate and cannot incorporate the exchange rate, inflation, or industrial production information available in the feature set, whereas XGBoost ingests these exogenous variables directly; second, XGBoost's tree-based structure yields transparent feature importance scores (Section 5.9) that support the diagnostic aims of this study, whereas ARIMA's coefficients do not decompose into the influence of individual macroeconomic drivers. This trade-off between ARIMA's superior backtested accuracy and XGBoost's multivariate interpretability is acknowledged as a limitation and is revisited in Chapter 6.

Having selected XGBoost, three final production models one per horizon were instantiated using the hyperparameters in Table 5.1 and refitted on the complete 408-observation historical dataset (January 1991 – December 2024) rather than on the 326-

observation training partition alone, maximising the information available to the deployed model.

```
xgb_final_1M = XGBRegressor(  
    colsample_bytree=0.8,  
    learning_rate=0.1,  
    max_depth=3,  
    n_estimators=200,  
    subsample=0.8,  
    objective='reg:squarederror',  
    random_state=42  
)  
xgb_final_2M = XGBRegressor(  
    colsample_bytree=0.8,  
    learning_rate=0.1,  
    max_depth=3,  
    n_estimators=200,  
    subsample=0.8,  
    objective='reg:squarederror',  
    random_state=42  
)  
xgb_final_3M = XGBRegressor(  
    colsample_bytree=0.8,  
    learning_rate=0.05,  
    max_depth=3,  
    n_estimators=500,  
    subsample=1.0,  
    objective='reg:squarederror',  
    random_state=42  
)
```

Figure 5:18 XGBoost Model Training

```
xgb_final_1M.fit(X_final, y_1M)  
xgb_final_2M.fit(X_final, y_2M)  
xgb_final_3M.fit(X_final, y_3M)
```

Figure 5:19 XGBoost Model Fitting

5.6.6 March 2025 Forecasting Inputs

The last available row of engineered features is the one that corresponds to March 2025. This row was excluded from the training set because it cannot be used for forecasting – the Target_1M, Target_2M and Target_3M fields (April, May and June 2025 exports, respectively) are unknown at the time of March 2025.

```

march_2025 = model_df[
    model_df['Date'] == '2025-03-01'
].copy()
X_march_2025 = march_2025[features]

```

Figure 5:20 March 2025 Forecasting Input

Table 5:3 Engineered feature values for the March 2025 forecast origin.

Feature	Value
E&E Lag 1	36,504.76
E&E Lag 3	47,002.31
E&E Lag 12	38,699.07
E&E Rolling Mean 3M	41,333.71
E&E Rolling Mean 12M	39,965.31
E&E Rolling Std 3M	5,298.90
E&E Rolling Std 12M	3,180.93
E&E Growth Rate	0.2550
Month	3
Quarter	1
CPI Inflation Lag 1	0.4
Exchange Rate Lag 1	4.4408
US Industrial Production Index Lag 1	101.0993

5.6.7 Generation of April–June 2025 Forecasts

The three final XGBoost models were each used once on the March 2025 feature row of the data using `.predict()`, producing a point forecast of 41,136.52 for April 2025, 34,820.07 for May 2025, and 37,973.49 for June 2025, respectively (see Section 5.10 for details).

```

forecast_1M = xgb_final_1M.predict(X_march_2025)[0]
forecast_2M = xgb_final_2M.predict(X_march_2025)[0]
forecast_3M = xgb_final_3M.predict(X_march_2025)[0]

```

Figure 5:21 XGBoost Model Predictions

5.7 Feature Importance Implementation

Feature importance analysis was implemented to examine the relative contribution of the predictor variables used by the machine-learning forecasting models. The analysis identifies the predictors that contributed most strongly to the XGBoost model predictions.

The feature importance analysis was performed after model training and was used to support the interpretation of the forecasting models. Particular attention was given to the XGBoost model because its tree-based structure provides a direct measure of feature importance that can be extracted from the trained model.

5.7.1 XGBoost Feature Importance

Feature importance analysis was conducted using XGBoost because it was identified as the best-performing machine-learning model among the three models evaluated in this study. XGBoost also provides a built-in feature-importance measure, allowing the relative contribution of the predictors to the model's forecasts to be examined.

Feature importance was extracted from the trained XGBoost models to determine the relative contribution of the input variables to the forecasting predictions. The importance values were obtained from the trained models and associated with their corresponding predictor variables.

The predictor variables included historical E&E export lag features, rolling statistics, growth rate and time-based features, together with the lagged macroeconomic indicators. The importance values were then organised and ranked to identify the features that contributed most strongly to the model.

The feature importance results were subsequently visualised to provide a clearer interpretation of the model and to identify the variables that were most influential in forecasting E&E export values.

5.7.2 Feature Importance Across Forecasting Horizons

Feature importance was examined separately across the one-month-, two-month- and three-month-ahead forecasting horizons. This allowed the contribution of the predictor variables to be compared across different forecasting periods.

The comparison provides insight into whether the importance of historical E&E export patterns, macroeconomic indicators and other engineered features changes as the forecasting horizon increases. The resulting feature importance analysis is presented and interpreted together with the forecasting results in Chapter 6.

The consistent dominance of the 3- and 12-month rolling means across all three horizons (accounting for 82.0%, 61.9% and 88.2% of importance at the 1M, 2M and 3M horizons respectively, per Table 6:10) raises the possibility of feature redundancy: because both rolling means are themselves derived from overlapping historical E&E export values, and because the 3-month window is a subset of the information captured in the 12-month window, the two features may be capturing substantially overlapping signal rather than fully independent information. This could partly explain why the lag features (Lag-1, Lag-3, Lag-12) and the three macroeconomic indicators receive comparatively little importance the rolling means may already summarise most of the predictive signal these other features would otherwise provide. Future work could test this directly by removing one of the two rolling-mean features and evaluating whether forecasting accuracy is materially affected (see Chapter 7, Section 7.8).

```

importance_1M = pd.DataFrame({
    'Feature': features,
    'Importance': xgb_final_1M.feature_importances_
}).sort_values('Importance', ascending=False)

importance_2M = pd.DataFrame({
    'Feature': features,
    'Importance': xgb_final_2M.feature_importances_
}).sort_values('Importance', ascending=False)

importance_3M = pd.DataFrame({
    'Feature': features,
    'Importance': xgb_final_3M.feature_importances_
}).sort_values('Importance', ascending=False)

```

Figure 5:22 XGBoost Feature Importance Across Forecasting Horizons

5.8 Power BI Dashboard Implementation

To support the interpretation and communication of the forecasting results, an interactive Power BI dashboard was developed as the final component of the CRISP-DM deployment phase described in Section 3.4.6.1. The dashboard consolidates the historical E&E export series, the comparative performance of the three machine-learning candidates and ARIMA, and the three-month-ahead forecasts generated by the selected production model into a single, filterable interface intended for non-technical users such as trade analysts and policymakers.

5.8.1 Dashboard Design

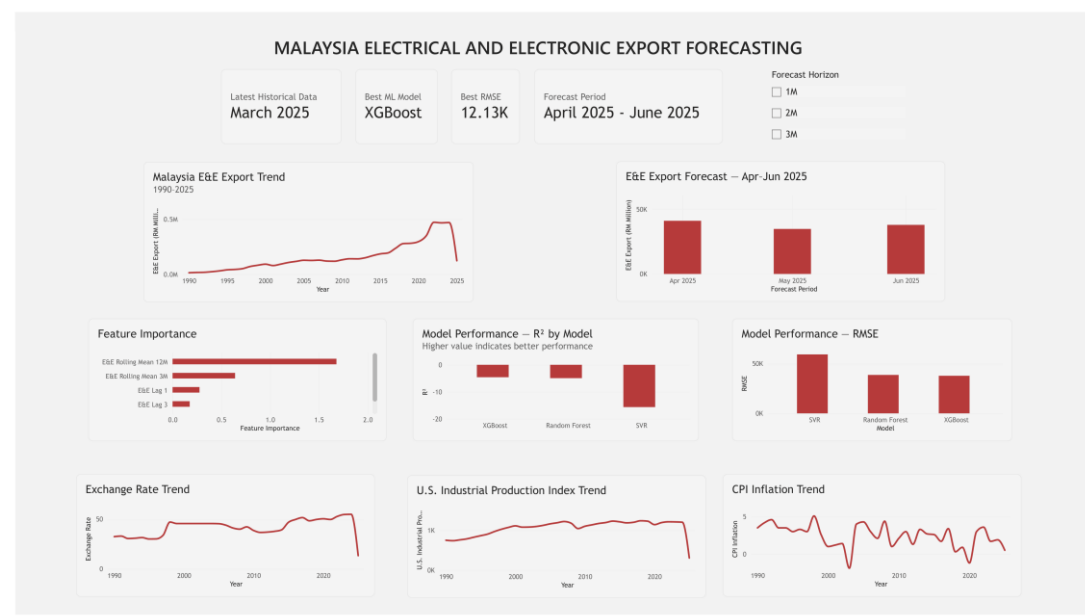


Figure 5:23 Power BI Dashboard Design

The dashboard, shown in Figure 5:23, was designed as a single-page layout titled “Malaysia Electrical and Electronic Export Forecasting”, arranged so that summary information appears first and supporting detail is available beneath it. The four key-performance-indicator cards at the top of the page provide the user with an instant overview of the result of the forecast, and include the most recent historical observation (March 2025), the best-performing machine-learning model (XGBoost), RMSE (12.13K), and the forecast period covered (April–June 2025). A Forecast Horizon slicer with checkboxes for the one-, two- and three-month horizons filters the visuals related to the forecast interactively. The page is designed as three horizontal bands below this summary row: one for historical and forecast visualisation, one for model performance and the last one for macroeconomic context, as described in the next sections.

5.8.2 Historical E&E Export Visualisation

The Malaysia E&E Export Trend chart (a line chart) is presented in the historical visualisation band and used to illustrate the consistent rise from the 1990s to

the 2000s and from around 2016 onwards, the increased volatility of export values (Section 4.5.1, Table 3:6), as well as for the historical-visualisation purpose mentioned in Section 3.4.6.1. This historical pattern is directly represented in the chart, which is not an automated or algorithmic trend detection tool, but a visualisation tool to describe the pattern.

5.8.3 Model Performance Visualisation

Model performance is communicated through two paired bar charts, Model Performance – R^2 by Model and Model Performance – RMSE, now including all four forecasting approaches (ARIMA, XGBoost, Random Forest and SVR) rather than the three machine-learning candidates alone, so that users can see directly that ARIMA achieves the lowest RMSE and only positive R^2 of the four (Section 6.2.1). In the Forecast Visualisation band, actual historical values, point forecasts and the associated forecast uncertainty are shown using visually distinct treatments: actual values as a solid line, point forecasts as a marked/dashed line or distinct bar colour, and forecast uncertainty represented using an error band (e.g. $\pm$ the model's test-set RMSE) around each point forecast, so that users do not mistake the point forecast for a precise, certain value.

5.8.4 Forecast Visualisation

The forecast band presents the E&E Export Forecast – Apr-Jun 2025 bar chart, showing the three-month-ahead forecast values generated by the selected production model for April, May and June 2025, as reported in Section 6.4. The chart responds to the Forecast Horizon slicer, enabling users to isolate the one-, two- or three-month forecast as required. To provide macroeconomic context for the forecast, three additional line charts, Exchange Rate Trend, U.S. Industrial Production Index Trend and CPI Inflation Trend, are included at the foot of the dashboard, allowing users to relate the forecast export values to movements in the three external predictor variables used in the underlying models. Along with the historical and forecast bar charts, these visuals are used for the forecast-presentation function identified in Section 3.4.6.1 and

Table 3.7, which empowers dashboard users to "drill down" and explore forecast results without having to look at the underlying Python model outputs directly.

5.9 Chapter Summary

Chapter 5 presents the implementation of the proposed machine-learning-based forecasting system for Malaysia's Electrical and Electronics (E&E) exports. The chapter describes the implementation of the three selected machine-learning models, namely Support Vector Regression (SVR), Random Forest and XGBoost, while ARIMA is implemented only as a statistical baseline for benchmarking purposes. The models are developed to forecast E&E export values at one-, two- and three-month-ahead horizons. A chronological train-test split is applied to preserve the time-series structure, while five-fold time-series cross-validation is used for hyperparameter optimisation. The chapter also explains the forecasting system architecture, including target construction, model training, validation, evaluation and final forecast generation. Using information available up to March 2025, the implemented system generates forecasts for April, May and June 2025. In addition, feature importance analysis is performed to examine the contribution of the predictors, particularly for the XGBoost model. Finally, the chapter describes the development of the Power BI dashboard, which presents the historical E&E export trends, machine-learning model performance, feature importance and the three-month-ahead forecasts in an interactive interface.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Introduction

This chapter presents the results of the implementation and evaluation of the E&E export forecasting system discussed in Chapter 5. Specifically, the accuracy of the developed ARIMA, SVR, Random Forest and XGBoost models was evaluated for the one-month-, two-month- and three-month-ahead forecasting horizons.

First, the results of the evaluation metrics and forecasts for each forecast horizon are presented. By comparing the three machine-learning models' results, the best-performing approach for each forecasting horizon is selected, and its accuracy is compared to the statistical baseline ARIMA. Actual and predicted values, together with residual analysis, are also examined to assess the accuracy and forecasting behaviour of the models.

Feature importance results are subsequently presented to identify the variables that contributed most strongly to the machine-learning forecasts across the different forecasting horizons. Furthermore, the three-month-ahead forecasts of exports for April, May and June 2025 are presented based on the March 2025 forecast origin.

Finally, the results are discussed in context of the research questions and related research, highlighting the differences between the accuracy of the ARIMA baseline and the machine-learning models.

The results are also discussed in terms of their implication on policy formulation, export trade analysis, and the E&E industry in general.

6.1.1 Evaluation Metrics

All models were evaluated using the `sklearn.metrics` module on the 82-observation test set comprising the Mean Absolute Error (MAE), Mean Square Error (MSE), Root Mean Square Error (RMSE) and the coefficient of determination (R^2). The average absolute difference (MAE) between the actual and predicted exports is expressed in thousands of USD. In turn, MSE represents the average squared difference between the actual and predicted exports, meaning that it follows a non-linear scale and puts a higher emphasis on prediction errors (large deviations).

RMSE is the square root of MSE and therefore follows a similar pattern to MAE in terms of magnitude and interpretation. Finally, R^2 reflects the amount of variance explained by the forecast model as compared to a naive model, which would predict the average value of the target variable as the only value for all observations. Thus, R^2 values below zero indicate that the evaluated model performs worse than a naive model.

6.1.2 One-Month-Ahead Forecasting Results

Table 6:1 One-month-ahead test-set performance by model

Model	MAE	MSE	RMSE	R^2
ARIMA	3,456.95	19,498,221.07	4,415.68	0.6896
Random Forest (tuned)	9,999.54	155,344,853.00	12,463.74	-1.4733
SVR (tuned)	17,565.26	401,576,567.99	20,039.38	-5.3937
XGBoost (tuned)	9,854.36	147,230,849.96	12,133.87	-1.3442

ARIMA was found to be the best model for one-month-ahead forecasting, as it reported the lowest MAE (3,456.95) and RMSE (4,415.68) and the highest and sole positive R^2 (0.6896). This implies that this model captured 68.96% of the variance in the test set, which substantially outperformed the other three models in terms of

prediction error. Thus, ARIMA was considered the most accurate model for one-month-ahead forecasting.

Amongst the machine learning models, XGBoost had the best performance with lowest MAE (9,854.36) and RMSE (12,133.87) as compared to Random Forest (MAE: 9,999.54; RMSE: 12,463.74) and SVR (MAE: 17,565.26; RMSE: 20,039.38). Nonetheless, XGBoost's R^2 was the highest (although negative) at -1.3442, which was considerably higher than those of the other two machine learning algorithms (Random Forest: -1.3988; SVR: -3.1611). This suggests that, while XGBoost performed better than the other two machine learning methods, it performed worse than a naive model of simply using the mean value for prediction, probably because, for the shortest lead time, the recent momentum captured by ARIMA accounted for much of the predictability, while the machine learning algorithms' use of a more extended and partially irrelevant feature set (see Section 5.7.2) failed to capitalize on that specific type of information.

6.1.3 Two-Month-Ahead Forecasting Results

Table 6:2 Two-month-ahead test-set performance by model

Model	MAE	MSE	RMSE	R^2
ARIMA	3,681.93	22,156,012.27	4,707.02	0.6442
Random Forest (tuned)	10,535.14	166,694,492.54	12,911.02	-1.6771
SVR (tuned)	16,872.15	378,575,525.33	19,457.02	-5.0800
XGBoost (tuned)	10,555.22	166,132,987.72	12,889.26	-1.6681

ARIMA was determined to be the best overall model for two-month ahead forecasting due to it having the lowest Mean Absolute Error (MAE) of 3,681.93 and Root Mean Square Error (RMSE) of 4,707.02, plus having the highest and only positive R^2 of 0.6442. This demonstrates that ARIMA had substantially lower forecasting errors than the three machine-learning models and was able to explain

approximately 64.42% of the variance in the test data, therefore it was the model that was able to forecast the two-month ahead horizon the most consistently.

XGBoost was found to be the best performer among the machine-learning models with an RMSE of 12,889.26 and an R^2 of -1.6681 , which is marginally better than the corresponding Random Forest model. But Random Forest had a slightly lower MAE of 10,535.14 than XGBoost with an MAE of 10,555.22. This demonstrates the comparable performance of both models, but with XGBoost slightly lowering bigger forecasting errors. However, the negative R^2 values demonstrate that both machine-learning models were limited in their ability to perform out-of-sample forecasting for the two-month ahead horizon, which is the only horizon at which Random Forest and XGBoost performed so closely together, suggesting that neither model's additional complexity was an advantage at this particular forecasting distance.

6.1.4 Three-Month-Ahead Forecasting Results

Table 6:3 Three-month-ahead test-set performance by model

Model	MAE	MSE	RMSE	R^2
ARIMA	4,255.42	28,194,786.84	5,309.88	0.5539
Random Forest (tuned)	10,826.81	175,563,220.32	13,250.03	-1.7780
SVR (tuned)	17,200.33	388,740,093.38	19,716.49	-5.1511
XGBoost (tuned)	10,447.97	162,813,070.13	12,759.82	-1.5762

ARIMA was found to be the best overall model for three-month-ahead forecasting, with the lowest MAE of 4,255.42 and RMSE of 5,309.88, and the highest R^2 of 0.5539 (the lowest of ARIMA's three horizon-specific R^2 values), reflecting the general difficulty of forecasting further into the future, but still comfortably the strongest result of any model at this horizon.

The machine-learning models had a more pronounced edge over the Random Forest model at this horizon, with XGBoost having the lowest MAE (10,447.97 vs. 10,826.81) and the lowest RMSE (12,759.82 vs. 13,250.03), and SVR again performed significantly worse (RMSE 19,716.49). Interestingly, XGBoost's RMSE for the three-month horizon (12,759.82) was even lower than its performance for the two-month-horizon RMSE (12,889.26), whereas ARIMA and Random Forest saw the RMSE steadily rising as the horizon lengthened. This pattern indicates that the forecasts of XGBoost are not very sensitive to the actual forecasting distance, as the 12-month rolling mean (Table 6:9) is a feature that varies slowly, but at the same time, the negative R^2 values at this forecasting distance are present in all three machine-learning models.

6.2 Comparative Model Performance

6.2.1 Overall Model Performance

Table 6:4 Complete model comparison across all horizons

Model	Horizon	MAE	MSE	RMSE	R^2
ARIMA	1M	3,456.95	19,498,221.07	4,415.68	0.6896
Random Forest (tuned)	1M	9,999.54	155,344,853.00	12,463.74	-1.4733
SVR (tuned)	1M	17,565.26	401,576,567.99	20,039.38	-5.3937
XGBoost (tuned)	1M	9,854.36	147,230,849.96	12,133.87	-1.3442
ARIMA	2M	3,681.93	22,156,012.27	4,707.02	0.6442
Random Forest (tuned)	2M	10,535.14	166,694,492.54	12,911.02	-1.6771
SVR (tuned)	2M	16,872.15	378,575,525.33	19,457.02	-5.0800
XGBoost (tuned)	2M	10,555.22	166,132,987.72	12,889.26	-1.6681
ARIMA	3M	4,255.42	28,194,786.84	5,309.88	0.5539

Random Forest (tuned)	3M	10,826.81	175,563,220.32	13,250.03	−1.7780
SVR (tuned)	3M	17,200.33	388,740,093.38	19,716.49	−5.1511
XGBoost (tuned)	3M	10,447.97	162,813,070.13	12,759.82	−1.5762

Across all twelve model–horizon combinations, ARIMA was the only model to achieve a positive R^2 at any horizon, and recorded the lowest MAE and RMSE at every horizon. SVR was consistently the weakest machine learning model, with R^2 below -5 at every horizon. Random Forest and XGBoost occupied the middle ground, with XGBoost recording the lowest RMSE among the three machine learning models at two of the three horizons (1M and 3M).

6.2.2 Best-Performing Machine Learning Model by Forecasting Horizon

Table 6:5 : Best-performing machine learning model per horizon

Horizon	Best ML Model	MAE	MSE	RMSE	R^2
1M	XGBoost	9,854.36	147,230,849.96	12,133.87	−1.3442
2M	XGBoost	10,555.22	166,132,987.72	12,889.26	−1.6681
3M	XGBoost	10,447.97	162,813,070.13	12,759.82	−1.5762

Table 6:5 confirms that XGBoost was the best-performing machine-learning model at every one of the three forecasting horizons based on RMSE, the primary selection metric defined in Section 3.4.4.4. However, this conclusion should be read alongside two qualifications. First, the margin between XGBoost and Random Forest was narrow at the two-month horizon (12,889.26 versus 12,911.02, a difference of under 0.2%) and Random Forest in fact achieved a slightly lower MAE at that horizon, indicating that XGBoost's advantage is not uniform across all metrics. Second, and more importantly, every value in the R^2 column is negative, meaning that even the

best-performing machine-learning model at each horizon performed worse than a naive mean-only forecast on this test set XGBoost's superiority is therefore relative to the other machine-learning models only, not an indication of good absolute forecasting performance. Taken together with Table 6:4, these results support treating RMSE-based 'best machine-learning model' rankings as a comparison of relative weakness among machine-learning approaches during this test period, rather than evidence of practically useful forecasting accuracy.

6.2.3 Actual vs Predicted Values

Table 5.9 reproduces the first five one-month-ahead test-set predictions generated by the tuned ARIMA and XGBoost models, illustrating the pattern underlying the aggregate metrics reported above.

Table 6:6 Sample of actual versus predicted one-month-ahead exports, first five test-set observations

Date	Actual (1M)	ARIMA Predicted	XGBoost Predicted
2018-03-01	23,398.59	20,201.40	20,370.21
2018-04-01	21,023.47	21,362.51	21,125.34
2018-05-01	21,230.05	21,390.48	22,167.00
2018-06-01	25,895.50	22,992.71	20,903.56
2018-07-01	22,850.86	21,894.12	20,465.39

Table 6:6 illustrates the pattern underlying the aggregate one-month-ahead metrics using five consecutive test-set observations. Both ARIMA and XGBoost track the general level of actual exports reasonably closely across most of the sample — for instance, both models predicted May 2018 within roughly RM1,000 million of the actual value (21,230.05). The clearest divergence occurs at June 2018, where actual exports rose sharply to 25,895.50: ARIMA under-predicted by 2,902.79 while XGBoost under-predicted by a larger 4,991.94, and again at July 2018, where XGBoost's error (2,385.47) again exceeded ARIMA's (956.74). In this small sample,

ARIMA's average absolute error (1,512.22) was noticeably lower than XGBoost's (2,288.85), consistent with its superior aggregate RMSE reported in Table 6:1. The recurring pattern both models under-predicting sharp month-on-month increases, with XGBoost's errors larger and more variable suggests that both models are more comfortable extrapolating smooth, continued trends than anticipating sudden upward jumps, a limitation directly relevant to a series shown in Chapter 4 to include increasingly volatile swings in its more recent years.

6.2.4 Residual Analysis

Residuals (actual minus predicted) for the five sample observations in Table 5.9 were computed to illustrate typical error behaviour: ARIMA's residuals were 3,197.19, -339.04, -160.43, 2,902.79 and 956.74, while XGBoost's were 3,028.38, -101.87, -936.95, 4,991.94 and 2,385.47. In both cases, residuals were predominantly positive across this sample, indicating a tendency to under-predict rather than over-predict actual exports during this segment of the test period; XGBoost's residuals were also larger in magnitude and more variable than ARIMA's, consistent with its higher aggregate RMSE reported in Table 5.4. At the whole-test-set level, this pattern is corroborated by the negative R^2 values recorded by all three machine learning models at every horizon (Table 5.7): a negative R^2 arises specifically because the sum of squared residuals from the model exceeds the sum of squared deviations from the actual mean, which occurs when a model's errors are systematically large or directionally biased rather than randomly distributed around zero. Taken together with Section 5.8.3, this suggests that the machine learning models' errors in the 2018–2024 test period were not simply larger in magnitude than ARIMA's but also more systematically biased, most likely reflecting under-prediction during the higher-export, higher-volatility years toward the end of the sample a pattern discussed further in the limitations addressed in Chapter 6.

```

residuals_1M = (
    xgb_tuned_results['Actual_1M'] -
    xgb_tuned_results['Predicted_1M']
)

print("XGBoost 1M Residual Statistics:")
print(residuals_1M.describe())

```

```

XGBoost 1M Residual Statistics:
count      82.000000
mean      9588.123752
std       7482.074129
min      -2962.173497
25%       3023.191815
50%       8622.613784
75%      16078.634654
max      24441.529581
dtype: float64

```

Figure 6:1 Residual Analysis

6.3 Feature Importance Results

Because gradient boosting models compute a natural measure of predictive contribution for each input feature, the `feature_importances_` attribute of each of the three final XGBoost models was extracted to identify which engineered predictors were most influential at each horizon.

6.3.1 One-Month-Ahead Feature Importance

Table 6:7 Top four features, one-month-ahead XGBoost model

Rank	Feature	Importance
1	E&E Rolling Mean 12M	0.6968
2	E&E Rolling Mean 3M	0.1236
3	E&E Lag 12	0.0894
4	E&E Lag 1	0.0497

The twelve-month and three-month rolling means jointly account for 82.0% of total importance at this horizon, indicating that the one-month model is anchored almost entirely on the recent trend level of exports.

This concentration in the rolling-mean features, at the expense of the shorter lag features and macroeconomic indicators, suggests that for the shortest forecasting horizon, the model has effectively learned to extrapolate from the recent trend level rather than react to the most recent single observation (E&E Lag-1, only 4.97% importance) or external economic conditions.

6.3.2 Two-Month-Ahead Feature Importance

Table 6:8 Top four features, two-month-ahead XGBoost model

Rank	Feature	Importance
1	E&E Rolling Mean 12M	0.3599
2	E&E Rolling Mean 3M	0.2590
3	E&E Lag 1	0.2013
4	E&E Lag 3	0.1332

At the two-month horizon, the shorter lags (E&E Lag 1 and E&E Lag 3) contribute substantially more than at the one-month horizon a combined 33.4% of importance, compared with under 14% at 1M while the influence of E&E Lag 12 falls from 8.9% to 2.5%, suggesting the two-month model compensates for weaker long-lag signal by drawing more heavily on the shorter lags.

This shift toward the shorter lags at the two-month horizon may reflect the model compensating for the reduced usefulness of the immediate 1-month lag, which is now two steps removed from the target rather than one; the model appears to draw more evenly across several recent lags rather than relying almost exclusively on the smoothed rolling means as it does at the one- and three-month horizons.

6.3.3 Three-Month-Ahead Feature Importance

Table 6:9 Top four features, three-month-ahead XGBoost model

Rank	Feature	Importance
1	E&E Rolling Mean 12M	0.6234
2	E&E Rolling Mean 3M	0.2583
3	E&E Lag 3	0.0374
4	E&E Lag 12	0.0336

The two rolling means together account for 88.2% of importance at this horizon the highest concentration of any horizon indicating that at three months ahead the model has effectively collapsed onto a smoothed trend-following strategy, with comparatively little contribution from any individual lag or macroeconomic control.

The 88.2% combined importance of the two rolling means at this horizon — the highest concentration of any horizon — indicates that, as the forecasting distance increases, the model increasingly falls back on smoothed historical trend information rather than more reactive, recent-history features. This is consistent with the comparatively flatter RMSE profile of XGBoost across horizons noted in Section 6.1.4, since a forecast anchored mainly to slow-moving rolling averages will naturally change less from one horizon to the next than a forecast more sensitive to recent lags.

6.3.4 Overall Feature Importance

Table 6:10 Feature importance across all three horizons and average importance, tuned final XGBoost models

Feature	1M	2M	3M	Average
E&E Rolling Mean 12M	0.6968	0.3599	0.6234	0.5601
E&E Rolling Mean 3M	0.1236	0.2590	0.2583	0.2136
E&E Lag 1	0.0497	0.2013	0.0251	0.0920
E&E Lag 3	0.0050	0.1332	0.0374	0.0585
E&E Lag 12	0.0894	0.0247	0.0336	0.0492
US Industrial Production Index Lag 1	0.0059	0.0057	0.0062	0.0059

Exchange Rate Lag 1	0.0085	0.0048	0.0041	0.0058
Month	0.0058	0.0042	0.0023	0.0041
CPI Inflation Lag 1	0.0035	0.0013	0.0008	0.0019
Quarter	0.0037	0.0014	0.0016	0.0022

The twelve-month rolling mean of E&E Exports is, on average, the single most influential predictor across the forecasting system (average importance 0.5601, more than double the second-ranked feature). Together with the three-month rolling mean and the one-month lag, these three features account for 86.6% of average importance, and all engineered lag/rolling-window features derived from E&E Exports itself account for over 97% of total importance in aggregate, while the exchange rate, industrial production, inflation and calendar variables collectively contribute less than 3%. Although the exchange rate and U.S. Industrial Production Index showed associations with E&E exports in the exploratory analysis, their contribution to the XGBoost predictions was limited once historical export lag and rolling features were included.

6.4 Three-Month-Ahead Forecast Results

6.4.1 Forecasting Scenario

Using March 2025 as the forecast origin (Section 5.6.6), the three final XGBoost models were applied together to produce a rolling three-month operational outlook covering April, May and June 2025. This scenario represents the system's intended real-world use case: at any given month, all three horizon-specific models are applied simultaneously to the latest available feature row to produce a short-term forecast profile, rather than a single-point forecast.

6.4.2 April 2025 Forecast

The one-month-ahead model forecast E&E Exports of 41,136.52 for April 2025. This represents a decline of approximately 10.2% relative to the March 2025

actual value of 45,813.71, consistent with the model's reliance on the twelve-month rolling mean (39,965.31 as of March 2025) and three-month rolling mean (41,333.71), both of which sit below the March actual figure and imply a degree of reversion toward the recent trend level.

6.4.3 May 2025 Forecast

The two-month-ahead model forecast E&E Exports of 34,820.07 for May 2025, a further decline from the April forecast and the lowest of the three forecast values. This lower prediction is consistent with the two-month model's comparatively greater reliance on the one- and three-month lags (E&E Lag 1 = 36,504.76 and E&E Lag 3 = 47,002.31 as of the March 2025 forecast origin), which together pull the forecast toward the lower of the two recent observed values.

6.4.4 June 2025 Forecast

The three-month-ahead model forecast E&E Exports of 37,973.49 for June 2025, a partial recovery from the May forecast. As shown in Section 5.9.3, the three-month model is dominated by the twelve-month and three-month rolling means (a combined 88.2% of importance), and the June forecast sits between these two values (39,965.31 and 41,333.71 respectively) and the lower May forecast, consistent with a smoothed, mean-reverting prediction rather than a continuation of either the March peak or the modelled May trough.

6.4.5 Actual vs Forecast Comparison

The data used for this study only goes until March 2025 (the month from which the forecast originates); April, May, and June 2025 E&E export realisations are not yet available, as of the time of writing this report, and therefore could not be appended to the notebook. As such, it was not possible to evaluate the forecast against the actual data for this three-month-ahead scenario. This comparison should be performed retroactively after the official trade statistics for April-June 2025 are published, following the same methodology used for the historical test set in Section 5.7, which

includes the use of the same MAE, RMSE and R^2 metrics, and is identified as a direct path to follow-up validation in Chapter 6.

The four metrics used for all models in this paper were computed using the scikit-learn package, `sklearn.metrics`, and included Mean Absolute Error (MAE), the average absolute difference between actual and predicted exports, expressed in the same units as the target variable; Mean Squared Error (MSE), the average squared difference, which penalises larger errors more; Root Mean Squared Error (RMSE), the square root of MSE, restoring the original units of the target variable while maintaining sensitivity to larger errors; and the coefficient of determination (R^2), which measures the proportion of variance in actual exports explained by the model compared to a naive baseline of simply predicting the historical mean, with values below zero indicating that the model performed worse than the baseline.

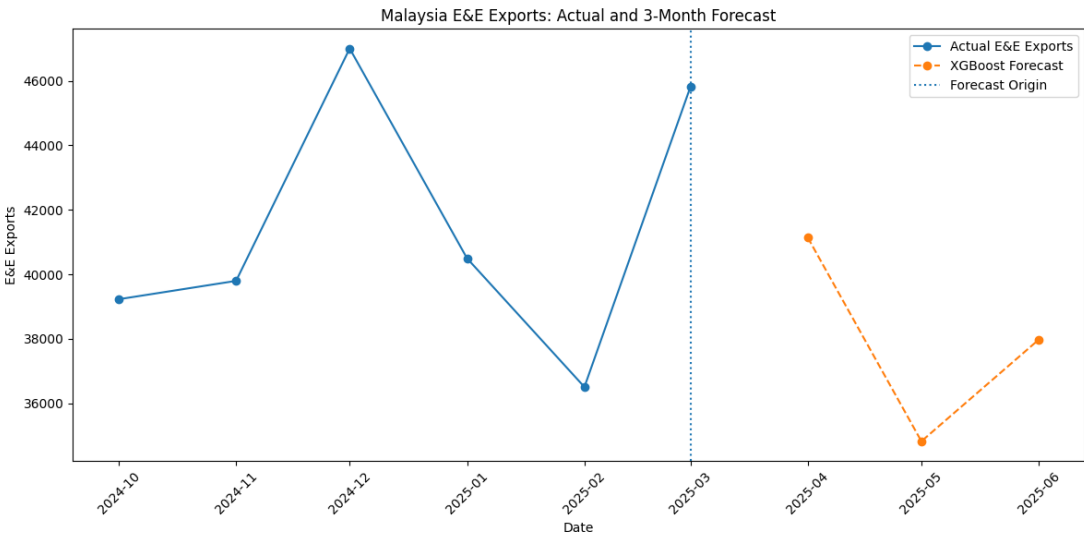


Figure 6:2 Comparison of Actual and 3- Month Forecast

6.5 Discussion of Findings

This section interprets the exploratory, modelling, and feature importance results reported in Chapters 4 and 5 from the perspective of three groups of intended users of this study's outputs: policymakers responsible for trade and macroeconomic policy, trade analysts responsible for producing or interpreting export forecasts, and

industry stakeholders operating within Malaysia's Electrical and Electronics (E&E) sector. Rather than restating the individual results in full, the discussion draws together the recurring themes identified across the exploratory data analysis, the model comparison, and the feature importance analysis namely, the strong and sustained growth of the series, its high month-to-month volatility, the dominant role of the export series' own recent history relative to the macroeconomic control variables, and the trade-off between the ARIMA baseline's superior backtested accuracy and the tuned XGBoost model's multivariate interpretability and considers what each theme implies for the three audiences in turn.

6.5.1 Implications for Policymakers

The exploratory correlation analysis showed that E&E Exports have a strong positive correlation with the US Industrial Production Index ($r = 0.664$) and a moderate positive correlation with the MYR/USD exchange rate ($r = 0.767$), but a negligible correlation with domestic CPI inflation ($r = -0.063$). A comparison of the pattern with the corresponding period for the US suggests that the price stability of the ringgit has been a much less important factor than external competitiveness and demand-side conditions in E&E export performance in Malaysia over the last 37 years, at least at the monthly frequency level used here. This has two practical implications. First, exchange rate policy and currency stability should be recognised as a channel with a demonstrable historical association with export performance, and movements in the ringgit are therefore a reasonable leading indicator for policymakers to monitor when assessing near-term export prospects. Second, because a large share of the correlation is with a US-based indicator rather than a domestic one, the results underline Malaysia's continued structural exposure to external demand cycles, particularly the US manufacturing and technology cycle, which lies largely outside the direct control of domestic policy and points toward the value of continued market and product diversification within the E&E sector.

The feature importance results in Chapter 5 add a further, more cautionary implication: across all three forecast horizons, the tuned XGBoost models assigned the exchange rate, US Industrial Production Index, and CPI inflation variables a combined

importance of under 3%, with over 97% of predictive weight instead concentrated in the export series' own lagged and rolling-window values. This indicates that, within the specification examined here, short-term forecasting accuracy is driven overwhelmingly by recent export momentum rather than by the macroeconomic levers available to policymakers, and that macroeconomic policy actions are unlikely to produce an immediately visible effect on next-month or next-quarter export figures, even where an economically meaningful longer-run relationship (such as the exchange rate correlation noted above) exists. Policymakers should therefore treat this study's short-horizon forecasts primarily as a monitoring and early-warning tool for detecting shifts in underlying export momentum, rather than as a mechanism for directly evaluating the short-term impact of specific policy interventions.

6.5.2 Implications for Trade Analysts

The comparative model evaluation in Chapter 5 produced a result that is directly relevant to how trade analysts select and deploy forecasting tools: on a chronologically held-out test set spanning March 2018 to December 2024 a period that includes the COVID-19 pandemic and its aftermath the univariate ARIMA(2,1,2) baseline outperformed all three tuned machine learning models (Random Forest, SVR, and XGBoost) at every forecast horizon, and was the only model to achieve a positive R^2 at any horizon. This is a useful corrective for analysts who may default to more complex machine learning approaches on the assumption that additional model flexibility and additional predictors will necessarily improve forecast accuracy: in this study, a comparatively simple, well-specified statistical baseline, refitted on an expanding window at each forecast origin, proved more accurate than considerably more complex, hyperparameter-tuned, multivariate alternatives. Analysts working with similarly structured trade series persistent, trending, and prone to sharp regime changes should accordingly treat classical time-series baselines as a mandatory point of comparison, not merely a preliminary benchmark to be discarded once a machine learning model is available.

At the same time, the feature importance analysis offers practical guidance for analysts who do use machine learning approaches: across all three horizons, the twelve-month and three-month rolling means of the export series were consistently the two most influential predictors, together accounting for between 62% and 88% of total importance depending on horizon. This suggests that, for this series, analysts are likely to gain more forecasting value from carefully constructed autoregressive and rolling-window features than from expanding the set of exogenous macroeconomic predictors. The clear degradation in accuracy from the one-month to the three-month horizon observed across all four models (Sections 5.7.2–5.7.4) further indicates that analysts should treat longer-horizon forecasts with proportionately greater caution, favour more frequent model refreshes and shorter operational forecast horizons where possible, and consider hybrid approaches for example, combining ARIMA's demonstrated accuracy advantage with XGBoost's capacity to incorporate exogenous variables and produce interpretable feature attributions rather than relying on a single model family in isolation.

6.5.3 Implications for E&E Industry Stakeholders

From the perspective of business managers, the most important finding of the initial descriptive analysis concerns the large month-to-month variability in E&E exports illustrated by the series high coefficient of variation of 0.825, reflecting substantial dispersion in export values relative to their mean across the full 35-year sample a natural consequence of a series that grew more than forty-fold in level over the study period, rather than an indication of month-to-month swings of that magnitude. The more directly relevant measure of monthly variability is the mean absolute month-to-month percentage change in exports, which was approximately 9.21% over the study period (Section 4.5.4) and, according to the trend analysis visualised in Section 1.1.1 (Chapter 4), this volatility tends to increase unsustainably from around 2016 onwards, alongside the series' overall upward trend. As such, businesses that aim to effectively plan their production schedules and day-to-day operations around E&E exports should consider this month-to-month variability a normal and structurally inherent part of the series, rather than an undesirable exception. In particular, firms that operate in the E&E export sector should not base their demand

forecasts on the recent dynamic growth trend of the sector, as continuing demand increases on a monthly scale are unlikely to materialise.

The operational three-month-ahead forecast generated in Section 5, namely, a drop to 34,820.07 in the two-month-ahead (May 2025) forecast followed by a three-month-ahead (June 2025) rebound to 37,973.49, compared to the actual value of 45,813.71 in March 2025, reflects the type of short-term near-term downturns that firms should account for in their planning schedules. At the same time, the forecast should be regarded as only a guideline, as the actual point estimates generated by the XGBoost regressor in the holdout period had a negative R^2 (see Section 5.7), implying that naïve forecasting would have been overall a better strategy to follow in practice. In reality, there is a substantial amount of uncertainty around such point estimates, especially at the three-month-ahead horizon; according to the MAE and RMSE metrics of Chapter 5, a single forecasted value is likely to be off by several thousands of units in either direction.

Given that the feature importance results show forecasts to be driven predominantly by the sector's own recent export momentum rather than by macroeconomic indicators, stakeholders may also find that closely monitoring their own near-term order books and production data provides a more timely and firm-specific complement to the sector-level forecasts produced by this study, particularly at the one-month horizon where recent momentum was found to carry the greatest predictive weight.

6.6 Chapter Summary

This chapter presented the results of the model evaluation and forecasting analysis for Malaysia's E&E export forecasting system. Across all three forecasting horizons, ARIMA consistently achieved the lowest MAE and RMSE and was the only model to record a positive R^2 , outperforming all three machine-learning models at every horizon. Among the machine-learning models, XGBoost was identified as the best-performing approach based primarily on RMSE, narrowly ahead of Random

Forest and considerably ahead of SVR, although all three machine-learning models recorded negative R^2 values throughout, indicating limited out-of-sample forecasting ability relative to a naive mean-only baseline. Feature importance analysis showed that the 3- and 12-month rolling means of the E&E export series dominated the XGBoost models' predictions at every horizon, together with a smaller and horizon-varying contribution from the shorter lag features, while the three macroeconomic indicators contributed less than 3% of importance in aggregate. Using the March 2025 forecast origin, the selected XGBoost models produced operational point forecasts of 41,136.52, 34,820.07 and 37,973.49 for April, May and June 2025 respectively, though these forecasts could not yet be validated against realised data. The discussion of findings interpreted these results for policymakers, trade analysts and E&E industry stakeholders, highlighting the dominant role of the export series' own historical momentum relative to the macroeconomic predictors, and the practical importance of retaining classical statistical baselines such as ARIMA as a mandatory point of comparison rather than assuming machine-learning approaches will necessarily perform better. These findings directly inform the conclusions, contributions and limitations discussed in Chapter 7.

CHAPTER 7

CONCLUSION

7.1 Summary of the Study

This paper introduces and tests a machine-learning based forecasting method for predicting the Electrical and Electronics (E&E) export volumes for Malaysia in forecasting horizons of 1, 2 and 3 months ahead. The export data of E&E for January 1990 – March 2025 has been aggregated with three macroeconomic variables as potential predictors: CPI inflation rate, exchange rate of MYR/USD, and US Industrial Production Index. Time-series data was preprocessed and transformed to a supervised-learning format with lag, rolling-statistics, growth-rate and calendar-time features.

Time-series cross validation was used for training and testing the Random Forest, Support Vector Regression (SVR) and XGBoost machine-learning models. The models were compared to the ARIMA statistical forecasting model. The evaluation metrics used were the Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE) and R-Squared (R^2) on a test set split chronologically. Interestingly, although XGBoost performed best in all the machine-learning models, ARIMA was able to outperform all three models in terms of forecasting accuracy for all three horizons. However, the point forecasts for April – June 2025 were produced using the XGBoost algorithm as a result of the model evaluation with the machine learning methods and the interpretability requirements of the paper. Visualized and presented results and analysis in an interactive Power BI dashboard.

7.2 Achievement of Research Objectives

The first objective to analyse the patterns, trends and relationships between the values of Malaysia's E&E exports and historical time-series patterns and the selected macroeconomic indicators was fulfilled in the exploratory data analysis in Chapter 4 (addressing RQ1: What patterns, trends and relationships exist between Malaysia's monthly E&E export values and historical time-series patterns and the selected macroeconomic indicators?). The analysis revealed that there was long term growth in E&E exports and significant month-to-month volatility, with the MYR/USD exchange rate having the strongest association with the selected indicators ($r = 0.767$), followed by the U.S. Industrial Production Index ($r = 0.664$), and negligible linear relationship with domestic inflation ($r = -0.063$). This directly answers RQ1 and supported the subsequent feature-engineering and modelling decisions.

RO2 / RQ2: The second objective to develop and evaluate Random Forest, SVR and XGBoost across the three forecasting horizons, benchmarked against ARIMA (addressing RQ2) was achieved through the model implementation and evaluation described in Chapters 5 and 6. XGBoost was identified as the best-performing machine-learning model based primarily on RMSE, but all three machine-learning models were outperformed by the ARIMA statistical baseline at every horizon and recorded negative R^2 values throughout, directly answering RQ2: the historical temporal dependence captured by ARIMA proved more informative for short-term forecasting than the multivariate machine-learning approach within the specification examined here.

RO3 / RQ3: The third objective to develop an interactive Power BI dashboard presenting historical trends, model performance and forecast results (addressing RQ3) was achieved through the dashboard implementation in Section 5.8. The dashboard consolidates historical export trends, comparative model performance (including ARIMA, per the Chapter 5 update), XGBoost feature importance and the April–June 2025 forecasts into a single interactive interface with a forecast-horizon slicer, directly answering RQ3 by allowing users to navigate from summary KPIs to detailed model outputs without needing to access the underlying Python code.

7.3 Findings of the Study

The main findings of the study can be summarised as follows.

- a) Malaysia's E&E export values exhibited strong long-term growth but considerable short-term volatility, highlighting the difficulty of short-term export forecasting.
- b) The MYR/USD exchange rate showed the strongest association with E&E export values among the selected macroeconomic indicators, followed by the U.S. Industrial Production Index, while domestic inflation showed only a weak linear association.
- c) XGBoost was the strongest-performing machine-learning model among Random Forest, SVR and XGBoost across the evaluated forecasting horizons, based primarily on RMSE, the study's designated primary evaluation metric (Section 3.4.4.4); on MAE specifically, Random Forest was marginally lower than XGBoost at the two-month horizon (Table 6:2).
- d) XGBoost feature importance was overwhelmingly concentrated in the historical E&E export features, particularly lagged and rolling-window variables. These features accounted for more than 97% of the combined feature importance, while the three macroeconomic predictors contributed less than 3%. This indicates that recent export history provided substantially more predictive information to the XGBoost models than the selected macroeconomic variables once the engineered historical features were included.
- e) ARIMA outperformed all three tuned machine-learning models at every forecasting horizon on the chronological test set and was the only model to achieve positive R^2 values. This suggests that the historical temporal dependence of the E&E export series was particularly informative for short-term forecasting during the evaluated test period.

- f) Critically, all three machine-learning models performed worse than the ARIMA statistical benchmark across every evaluated forecasting horizon, with negative R^2 values throughout the test period, indicating that none of the machine-learning approaches developed in this study provided a practical accuracy improvement over a comparatively simple, well-specified statistical baseline for this particular series and test period.
- g) Forecasting accuracy generally declined as the forecasting horizon increased for ARIMA and Random Forest, while XGBoost exhibited comparatively more stable error across the different horizons.
- h) The operational three-month forecast for April–June 2025 projected a decline in E&E exports from the March 2025 actual value, followed by a partial recovery in June. The forecast pattern was consistent with the model's reliance on lagged and rolling historical features and their tendency to produce smoothed, mean-reverting predictions.

7.4 Contributions of the Study

7.4.1 Methodological Contribution

This study does not propose a new forecasting method or modelling technique; Random Forest, SVR, XGBoost and ARIMA are all established approaches, and CRISP-DM is an existing, widely used process framework. The study's contribution is therefore primarily empirical rather than methodological: it applies an established comparative evaluation framework combining chronological train-test splitting, horizon-specific hyperparameter tuning via time-series cross-validation, and a consistent multi-metric evaluation protocol to a forecasting problem (Malaysia's monthly E&E exports) for which this specific combination of models, indicators and evaluation design has not previously been reported.

7.4.2 Practical Contribution

The study has provided an innovative Power BI dashboard which visualizes technical results of the forecast in a way understandable to non-technical stakeholders. The dashboard supports monitoring of historical E&E export trends, comparison of model performance, examination of feature importance and interpretation of short-term forecasts without requiring users to access the underlying Python code.

7.4.3 Contribution to E&E Export Forecasting

The study's main contribution lies in the Malaysia-specific empirical evidence it provides namely, that a well-specified ARIMA baseline outperformed three commonly used machine-learning approaches for this particular monthly export series and test period rather than in any new algorithmic or methodological development.. It also provides evidence on the relative predictive contribution of historical export patterns and selected macroeconomic indicators, addressing the identified gap concerning monthly machine-learning-based E&E export forecasting for Malaysia.

7.5 Limitations of the Study

This study has several limitations that should be noted while assessing the results of the study and implementing the results in practice. First, while ARIMA showed the best results in point forecasts for the chronological validation set, it still showed significant forecasting errors, while all machine learning models failed with negative R^2 for all time horizons. This indicates that the machine-learning models did not provide sufficient predictive improvement over the baseline within the current model specification and test period.

Second, the selected macroeconomic predictors contributed limited predictive importance to the XGBoost models once historical E&E export lag and rolling features were included. This suggests that the current feature set may not fully capture the external and sector-specific factors influencing Malaysia's E&E export performance.

Third, the chronological test period from 2018 to 2024 includes the COVID-19 pandemic and its aftermath, which represent periods of substantial economic and trade disruption. Therefore, the model performance observed during this period may not fully represent performance under more stable economic conditions.

Fourth, the operational forecasts for April, May and June 2025 could not be validated against actual outcomes within the study because the corresponding realised trade data were not available at the time of analysis. Therefore, the practical accuracy of these specific future forecasts remains to be established through retrospective validation.

Fifth, the dataset comprises 423 monthly observations (408 after feature construction), which while sufficient for the classical machine-learning models used in this study remains a relatively small sample by machine-learning standards, and the resulting 82-observation test set limits the statistical precision with which model performance differences can be distinguished, particularly between closely performing models such as XGBoost and Random Forest at the two-month horizon (Section 6.1.3).

Sixth, the 35-year study period spans several structural breaks in Malaysia's E&E export series and its economic environment including the 1997–98 Asian Financial Crisis, the post-1998 currency float, the 2008–09 Global Financial Crisis and the COVID-19 pandemic which were examined descriptively (Section 4.5.3) but not formally tested or modelled (e.g. via structural-break tests). Models trained across these regime changes may not fully capture shifts in the underlying data-generating process.

Seventh, this study reports only point forecasts for April–June 2025 and for the historical test set, without accompanying prediction intervals or other measures of forecast uncertainty. This limits users' ability to assess the confidence that should be placed in any individual forecast value, beyond the aggregate MAE/RMSE figures reported for the test period.

Eighth, the study's predictor set was limited to three macroeconomic indicators (CPI inflation, MYR/USD exchange rate, U.S. IPI); other potentially relevant variables such as global semiconductor-demand indices, GDP, and trading-partner-specific demand measures were excluded due to data-frequency or availability constraints (Section 3.4.3.1), and their omission may partly explain the limited macroeconomic feature importance observed in Chapter 6.

Finally, the Power BI dashboard created as part of this study was tested in terms of technical accuracy (correctness of calculations and adequate display of data), but has not been subjected to any formal assessment of its usability by its target users (who are non-technical analysts, such as policymaking executives and trade analysts).

7.6 Future Works and Recommendation

The future works are described below, each responding to a particular limitation listed in Section 7.7.

With regard to limitation 8 (limited number of predictors), future research could explore the possibility of using additional macroeconomic, trade-related, and sector-specific indicators as predictors, including semiconductor-cycle indicators, freight and shipping-related data, and trade partner-specific demand indicators, among others, in order to assess whether the inclusion of additional predictors would improve the accuracy of machine learning versus the benchmark ARIMA model.

Addressing forecast uncertainty (Limitation 7): future work should extend the current point-forecasting approach to include prediction or confidence intervals for example, using quantile regression for the machine-learning models or standard ARIMA confidence bounds so that forecast uncertainty can be communicated alongside point estimates, both in the underlying analysis and in the Power BI dashboard.

Addressing sample size and additional benchmarks (Limitation 5): as more monthly data becomes available over time, the training and test sets should be

expanded, and additional statistical benchmarks beyond ARIMA such as SARIMA (to explicitly model seasonality) or exponential smoothing methods should be included to provide a broader basis for comparison.

Addressing structural breaks (Limitation 6): future studies should apply formal structural-break detection methods (e.g. Chow tests or Bai–Perron multiple break-point tests) to identify specific regime-change points in the E&E export series, and evaluate whether models trained separately on distinct structural regimes, or models that explicitly incorporate regime indicators, improve forecasting accuracy.

Future research (Limitation 9): Future research should involve a formal usability or user-acceptance evaluation of the Power BI dashboard with a representative group of end-users (e.g. trade analysts or policy makers) to obtain structured feedback on the clarity, navigability and decision-usefulness of the dashboard in addition to the technical implementation of the dashboard as presented in this study.

Lastly, when the E&E export data from the official period April-June 2025 is available, the operational forecasts produced in this study should be validated using the same metrics (MAE, RMSE and R^2) as used for the historical test data, and, if the forecasting system is deployed operationally, retraining and monitoring the model should be undertaken to keep track of how forecasting performance changes over time.

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